COURSE OUTLINE

Naval Architecture I (Ship Geometry and Hydrostatics)
BME 203

TOPICS

1. Definitions/Dimensions/Coefficients
    a. Hull; Superstructure; Parallel middle body; Fore and aft ends; Entrance and run
    b. Forward and aft perpendiculars; Length overall and between perpendiculars; Keel; Baseline; Midship; Centreline
    c. Loadline; Draught; Freeboard; Depth
    d. Sheer; Camber; Rise of floor; Tumblehome; Flare; Stem; Stern; Bilge
    e. Mass and volume displacement; Tonnage
    f. Block, prismatic, vertical prismatic, waterplane area and midship section coefficients
    g. Deadweight coefficient
2. Ship Geometry and Calculations
    a. Lines plan; Table of offsets
    b. Approximate methods of integration
    c. Calculation of areas, volumes,
    d. Determination of centroids of flotation and buoyancy
    e. Calculation of second moments of area
    f. Displacement sheet; Hydrostatic data and curves
    g. Deadweight scale
3. Flotation/Reserve Buoyancy/TPC
    a. Archimedes principle and principle of flotation
    b. Effects of external and internal load changes on buoyancy
    c. Need for reserve buoyancy and Loadline markings
    d. Tonne per centimeter immersion (TPC)
    e. Fresh Water Allowance (FWA)
    f. Assignment of freeboard
4. Loaded Centre of Gravity/Heel/List
    a. Effects of addition, removal, shifting and suspension of masses on centre of gravity
    b. Determination of centre of gravity after loading
    c. Definitions of list and trim
5. Transverse Statical Stability – Initial
    a. Definition of heel and heeling moment
    b. Definitions of righting moment (Moment of statical stability), righting lever (GZ), transverse metacentre and metacentric height (GM_T)
    c. Relationship between GZ and GM_T
    d. Stable, unstable and neutral equilibrium
    e. Angle of loll
    f. Relationship between KG, KB, BM_T and GM_T
    g. Metacentric diagram

00-INTRODUCTION

0.1 MARINE VEHICLES

Marine vehicles are man-made structures which operate on or in water for the purposes of transportation, recovery of natural resources (such as oil, fish and minerals), research, warfare or leisure. They may be classified as shown in Figure 0.1.1

Vehicles that operate on the surface are usually referred to as ships, while those underwater are called submarine vehicles. When submarine vehicles surface, and when offshore structures are moving in the floating mode in between locations, they may be considered as ships.

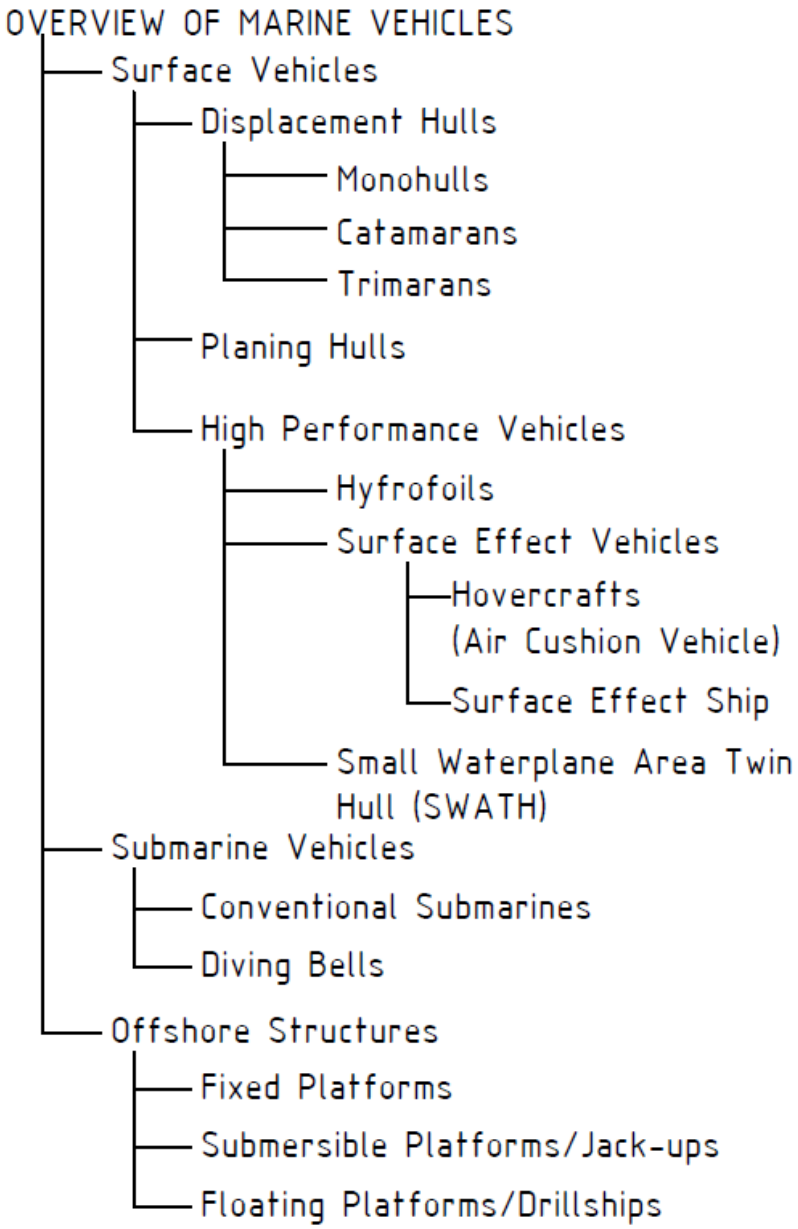


Figure 0.1.1 – Marine Vehicles

0.2 SURFACE VEHICLES OR SHIPS

Ships are surface vehicles. They are usually categorized into merchant ships, industrial, fishing, governmental and private vessels as indicated in Figure 1.2.1. It must be stated that these classifications are not to be considered as finite. Different classifications may be adopted by different individuals in accordance to their own perspectives, while on a regular basis new types are being added as and when they get developed.

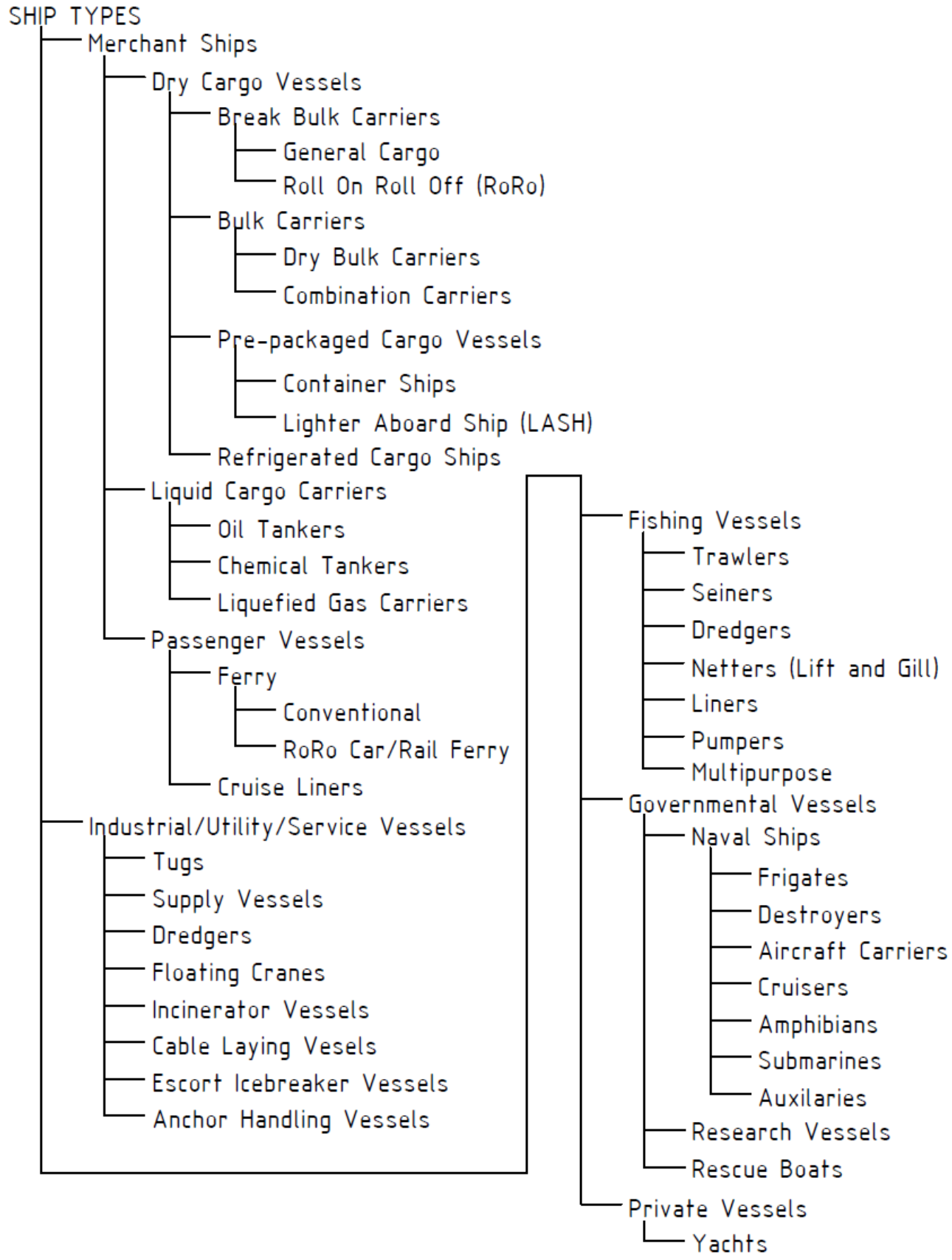


Figure 0.2.1 – Ship Types

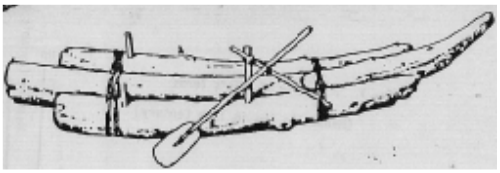
0.3 PICTORIAL HISTORY OF SHIPS



Reed Raft (Circa 4000 BC)



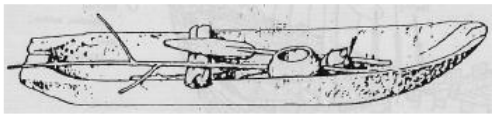
Cog (Circa 200-1600 AD)



Log Raft (Circa 3500 BC)



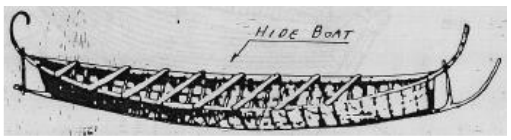
1700s



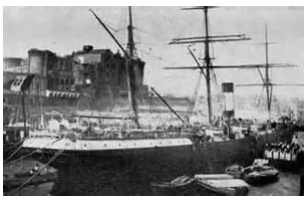
Dugout Boat (Circa 3000 BC)



1700-1900 AD



Hide Boat (Circa 3000 BC)



1800s



Sailing Ships – (Circa 100 AD)



Early 1900s

0.4 PICTURES OF MODERN SHIP TYPES

 River Boat	 Ferry-Captain Zaman	 Heavy Load Carrier-Explorer	 Reefer-Arctic	 Yacht-Callisto
 Car Carrier-Brussel	 Fishing Vessel-Alboniga	 High Speed Craft-Blau de Formentera	 Rescue Boat-Salvamar Aldemaran	 Bulk Carrier-Bled
 Cement Carrier-Frida	 Fishing Vessel-Cidade de Albufeira	 High Speed Craft-Patricia Olivia	 Research Vessel-Coopernaut Franer	 Water Tanker
 Chemical Tanker-Anette Theresa	 Gas Carrier-Artic Princess	 Naval Vessel-Albion	 RoRo Vessel-Don Fernando	 Dredger
 Container Ship-Parsifal	 General Cargo Ship-Aficaborg	 Naval Vessel-Kearsarge	 Cruise Liner	 Supply Vessel
 Cruise Liner-Celebrity Century	 General Cargo Ship-Anna K	 Naval Vessel-Montrose	 Supply Vessel-Vos Simpathy	 Hydrofoil
 Cruise Liner-Voyager	 Harbour Ferry-Cala Moreya	 Crude Oil Tanker-Tropic Brilliance	 Training Ship-Empire State	 Submarine
 Ferry-Abel Matutes	 Harbour Ferry-N. Cala Castell	 Pilot Boat-Lancha Mahon	 Tug-Allesandro Neri	 Submarine

1–DEFINITIONS AND DIMENSIONS

1. HULL: The main body of a ship.

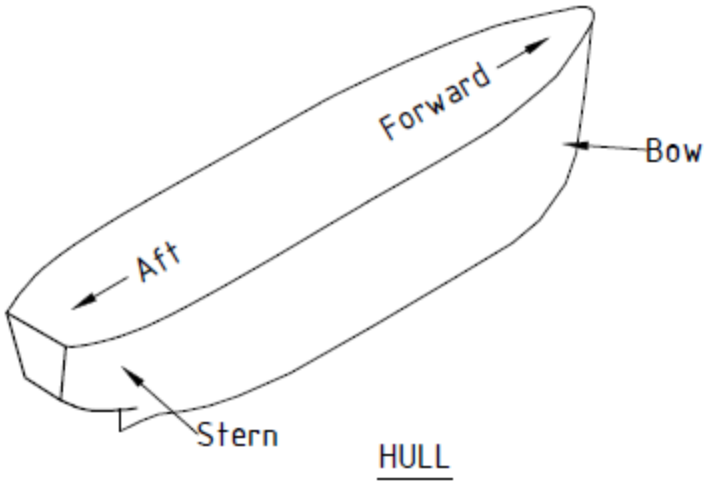


Figure 1.1

2. BOW: The front part of a ship.

3. STERN: The rear part of a ship.

4. FORWARD: Toward, at or near the front of a ship.

5. AFT: Toward, at or near the stern.

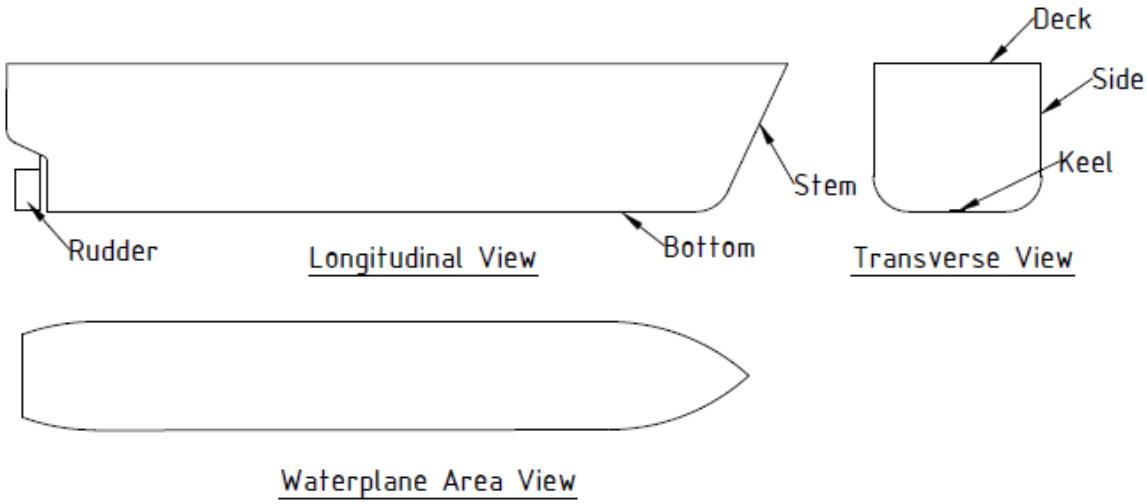


Figure 1.2

6. LONGITUDINAL VIEW: Any view of a ship's hull when looked at lengthwise from fore to aft.

7. TRANSVERSE VIEW: Any view of a ship's hull when looked at in cross-section from port to starboard.

8. WATERPLANE AREA: Any view of a ship's hull when cut across a waterline.

9. BOTTOM: The bottom most enclosure of the hull.

10. KEEL: The primary structural member which occupies the central part of a ship's hull at the bottom.

11. DECK: Any horizontal platform built into a vessel. The Main Deck being the uppermost deck exposed to the weather.

12. SHIPSIDES: The vertical sides that joins the bottom to the main deck to complete the hull.

13. STEM: The foremost structural member at the bow of a vessel.

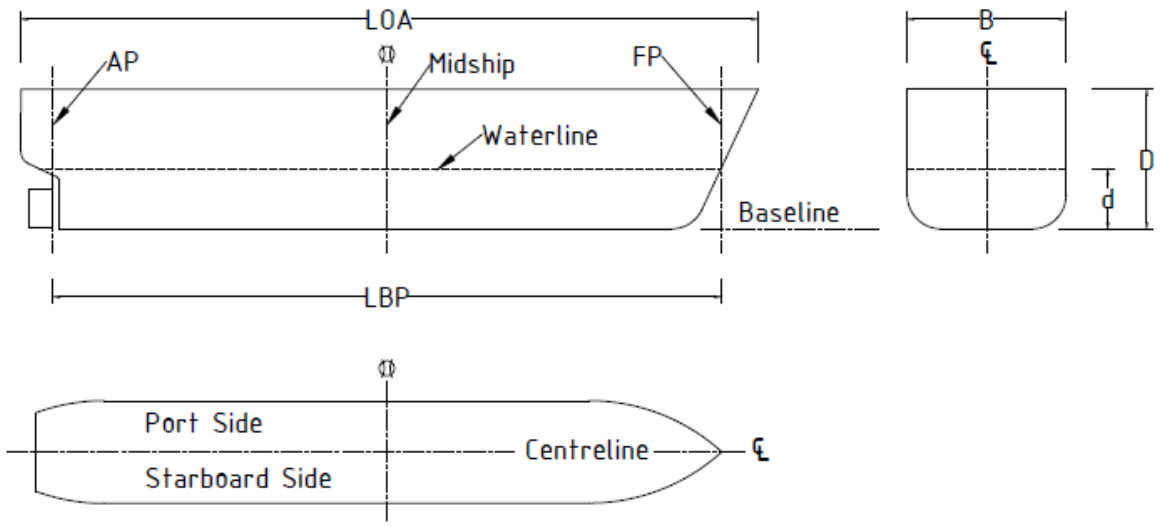


Figure 1.3

14. PORT SIDE: The left side of a ship when looking from behind.

15. STARBOARD SIDE: The right side of a ship when looking from behind.

16. WATERLINE: The line at a ship's side formed by the surface of water at a specific draught.

17. BASELINE (BL): A horizontal line drawn at the top of the keel plate.

18. CENTRELINE (CL): An imaginary plane drawn to divide a transverse view of a vessel symmetrically into two halves.

19. FORWARD PERPENDICULAR (F.P.): Perpendicular drawn to the waterline at the point of intersection of the stem and the Summer Load Line.

20. AFT PERPENDICULAR (A.P.): Perpendicular drawn to the waterline at the point where the aft sides of the rudder post meet the Summer Load Line. Where no rudder post is fitted the centreline of the rudder stock is used as reference.

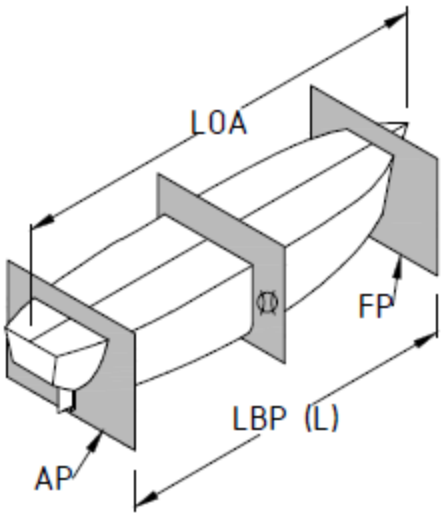


Figure 1.4

21. LENGTH OVERALL (L.O.A.): Length of a vessel taken over all extremities.

22. LENGTH BETWEEN PERPENDICULARS (L.B.P. or L): Length between the forward and aft perpendiculars measured along the Summer Load line.

23. MIDHIP (MS): An imaginary plane drawn midway to divide the Length Between Perpendiculars into two.

24. DISPLACEMENT (Δ): The mass of a vessel and all it contains measured in (Tonne). One Tonne = 1000 kg.

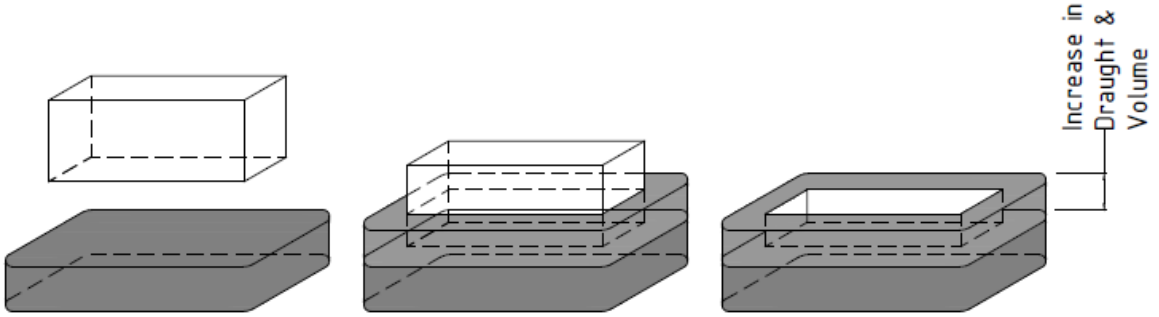


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25. VOLUME DISPLACEMENT (▽): The volume of water displaced by a vessel measured in (m^3).

26. LIGHTSHIP DISPLACEMENT (Δ_L): The mass of a ship as built including: mass of water in the boiler, mass of jacket cooling water in the engine, and mass of lubricating oil in the sump of the main engine all to the working levels.

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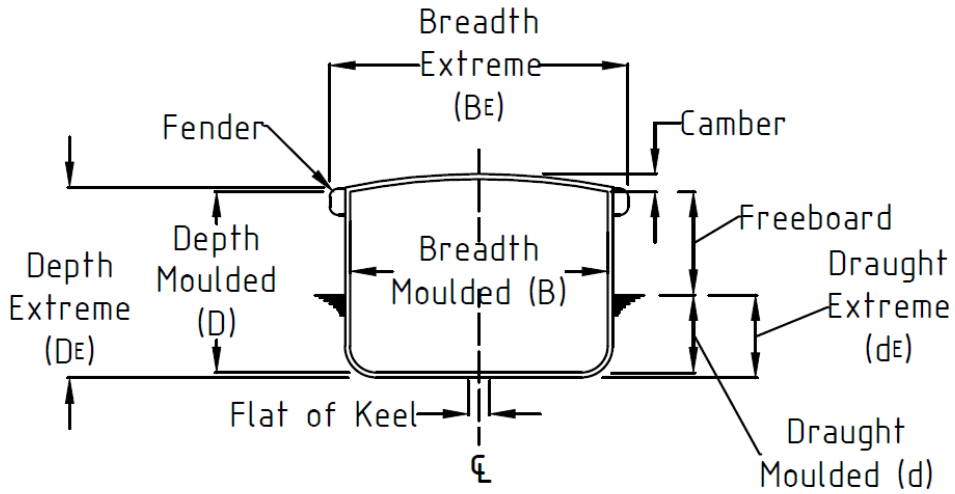


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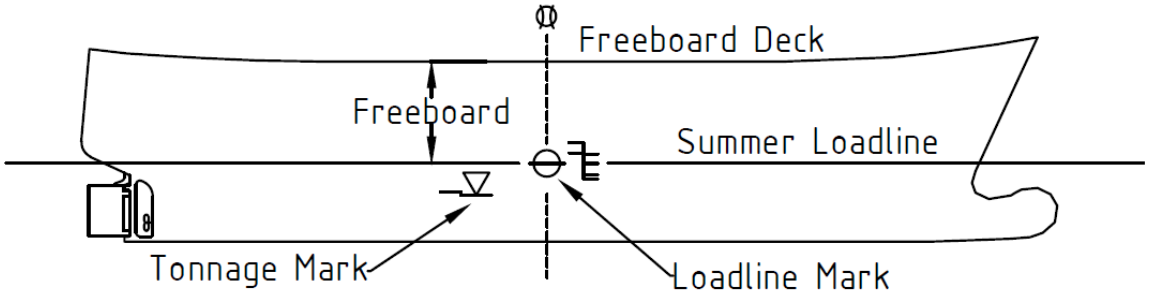


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37. TONNAGE: The capacity of a merchant ship expressed in Ton. One Ton = 40 ft^3 (1.13 m^3) of freight, or 100 ft^3 (2.83 m^3) of bulk cargo.

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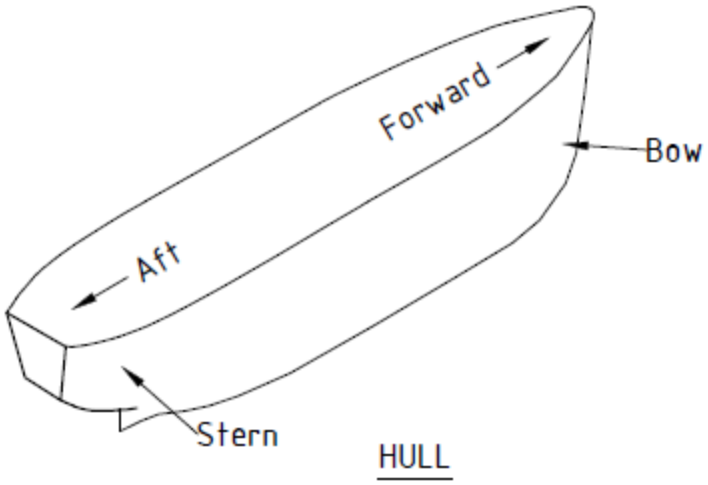


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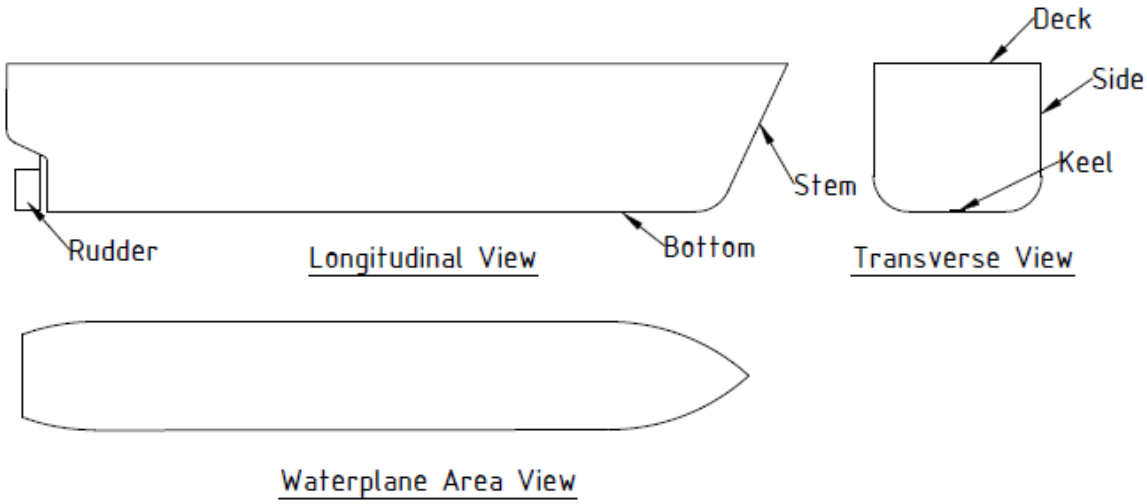


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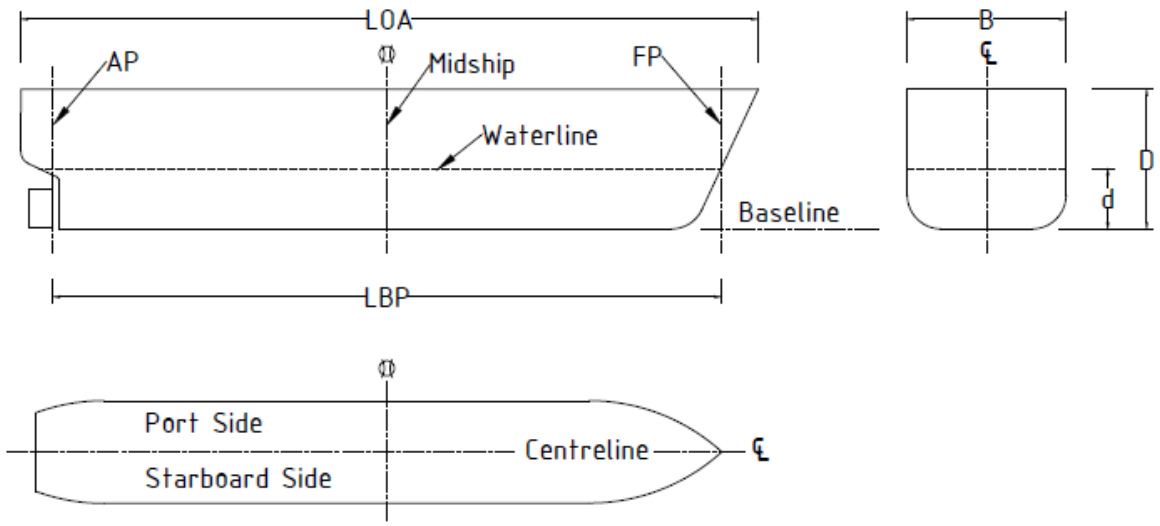


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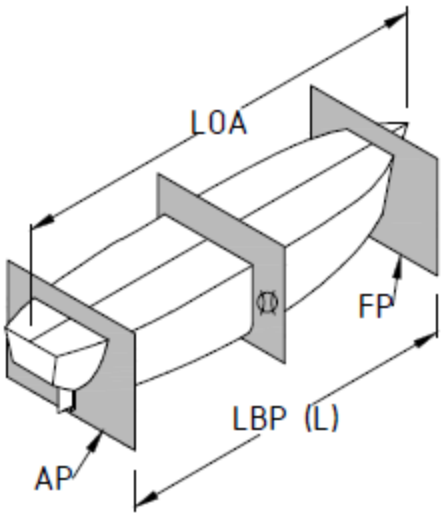


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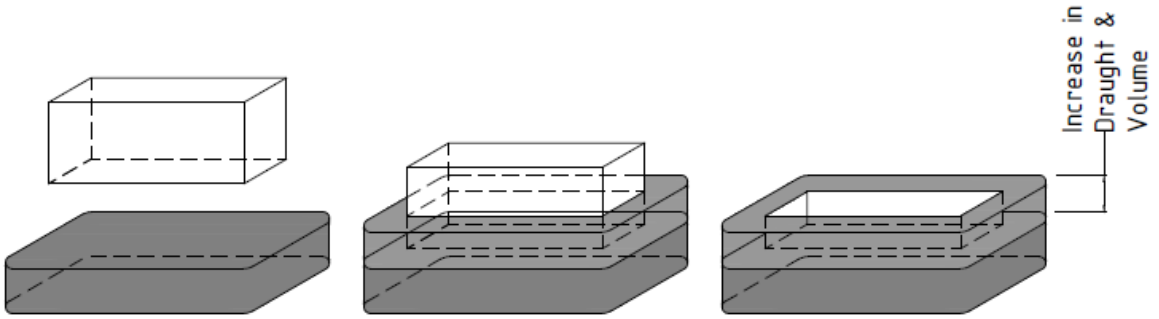


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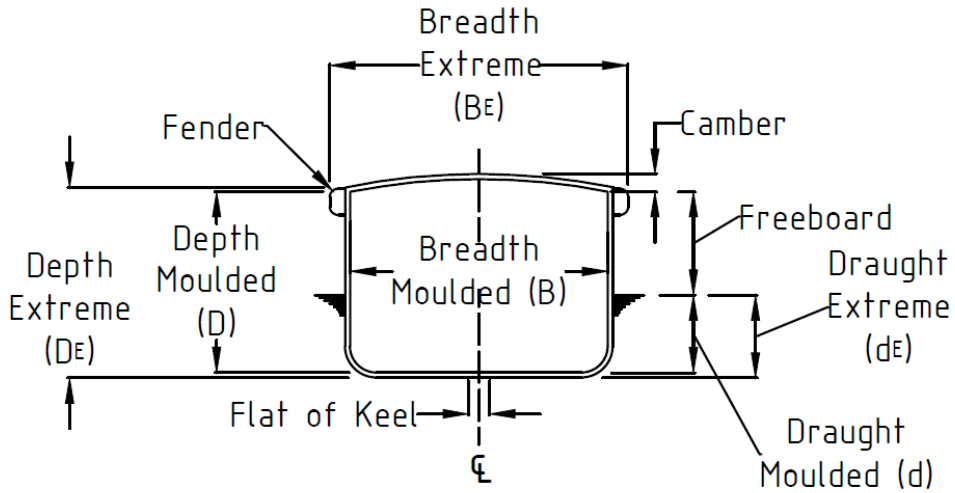


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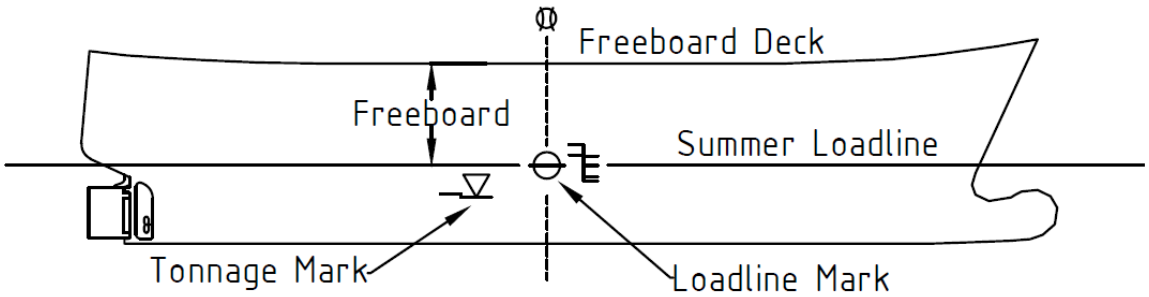


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    The letters beside the circular marks indicate the assigning authority, e.g. AB refers to American Bureau of Shipping, etc.

41. DRAUGHT MARKS: Figures welded on the bow, midship and stern of each side of a ship's shell plating to indicate draught. The distance is read from the lower edge of each number. Draught measured at the bow is called *forward draught* and at the stern is called *aft draught*.

42. TONNAGE MARK: A mark placed on each side of a vessel amidships to indicate any draught limitations imposed as a result of exempting the spaces between the second and upper decks from tonnage.

43. BAR DRAUGHT: Maximum draught which a ship can take to pass over a bar or sand bank. Bar is sand or silt forming a bank across the mouth of a river or harbor entrance. This build-up reduces the water depth and limits the draught of a ship which can cross it.

44. AIR DRAUGHT: The vertical distance measured from the ship's waterline to the highest point on the ship; usually comes into consideration when the ship has to sail under overhead bridges in a river.

45. BILGE: The section of the ship's hull, usually curved, which joins the vertical sides to the horizontal bottom.

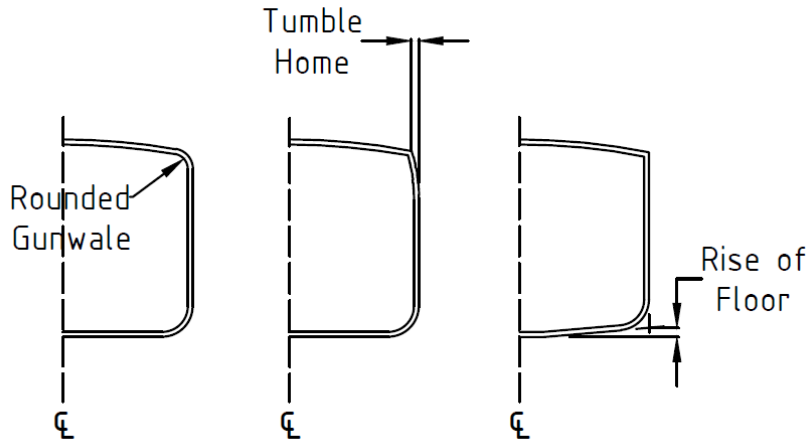


Figure 1.8

46. BILGE RADIUS: The radius of the circular arc forming the bilge.

47. RISE OF FLOOR: Where the transverse midship section at the bottom of a ship is not horizontal, the rise of floor is the height above the baseline, measured at the shipside.

48. CAMBER or ROUND OF BEAM: Curvature of decks in the transverse direction measured as the height of the deck at the centreline above the height of the side as reference.

49. TUMBLE HOME: The amount by which the side of the ship falls in from the breadth moulded line. Rarely employed in modern ships.

50. SHEER: Curvature of decks in the longitudinal direction measured as height of deck at the ship side at any point above the height of deck amidships as reference.

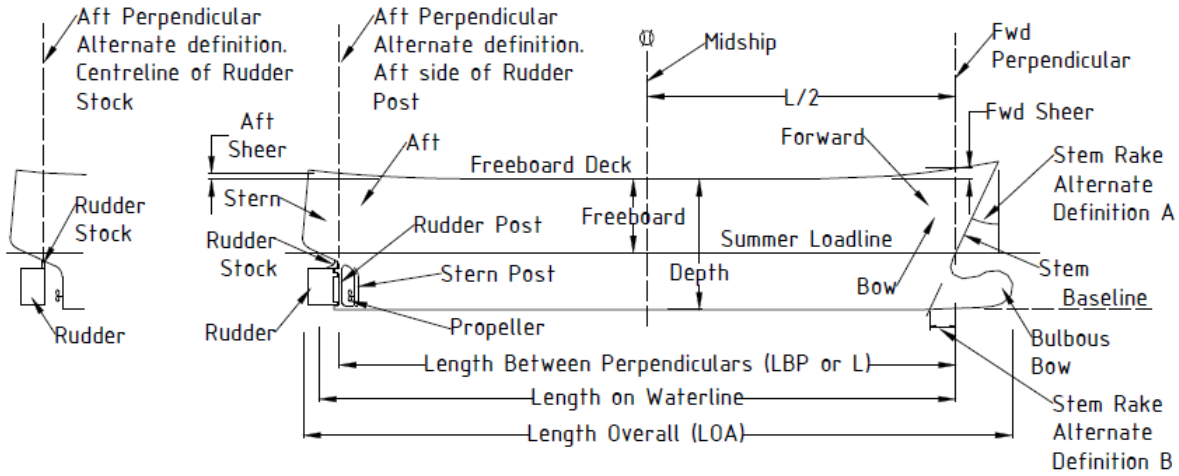


Figure 1.9

51. RAKE OF STEM: The inclination of the stem to the vertical defined by either the angle to the vertical or the distance between the intersection of the stem produced with the baseline and the forward perpendicular.

52. ENTRANCE: The gradual reduction in section towards the stem from the parallel middle body to encourage smooth flow of water around the hull.

53. RUN: The gradual reduction in section towards the stern from the parallel middle body to guide and encourage smooth flow of water receding away from the hull and into the propeller.

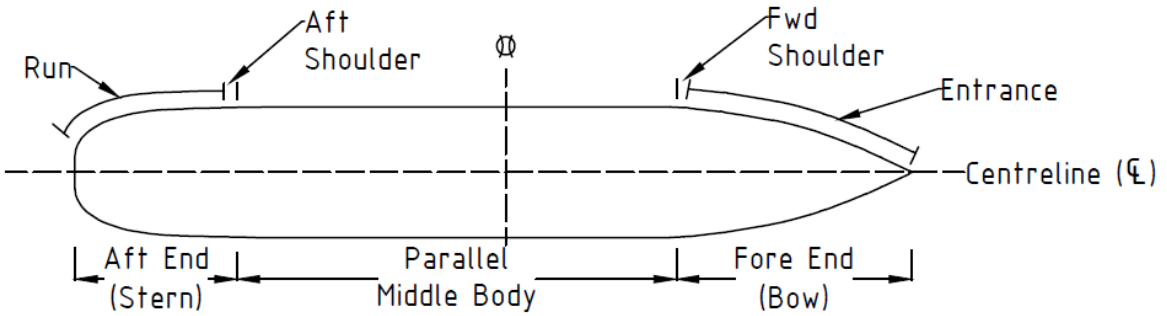


Figure 1.10

54. FORWARD SHOULDER: The point where the entrance joins the parallel middle body.

55. AFT SHOULDER: The point where the run joins the parallel middle body.

56. BALLAST: Any weight in solid or liquid form taken on a ship to increase draught, to change trim, or to improve the stability.

57. DESIGN DRAUGHT: Draught as used for design calculation before actual construction, which has a tendency to distort the final reading.

58. DRAUGHT SURVEY: Survey carried out to determine the cargo weight on board by measuring the ship's draught.

59. SUBDIVISION LOAD LINE: Waterline used to determine the subdivision of vessel for compliance with SOLAS ; the deepest subdivision load line is the waterline which corresponds to the greatest draught permitted by subdivision requirements

60. SSW-Summer Salt Water: Refers to the draught of the ship in salt water when it is loaded to its summer loadline in the open sea.

2–GENERAL SHIP KNOWLEDGE

GENERAL ARRANGEMENTS

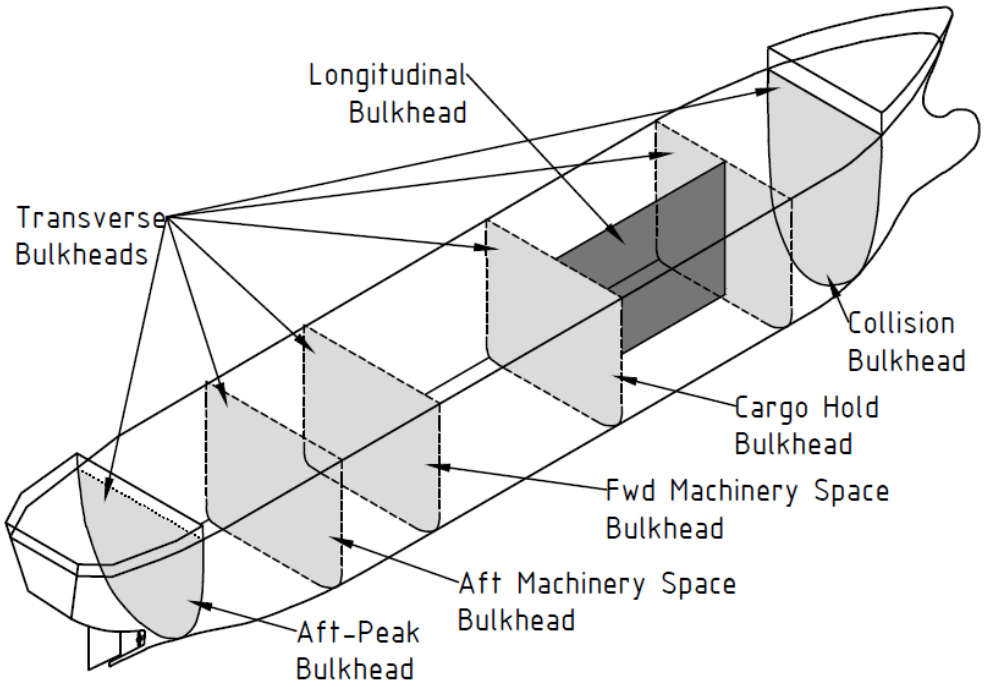


Figure 2.1 – Subdivisions

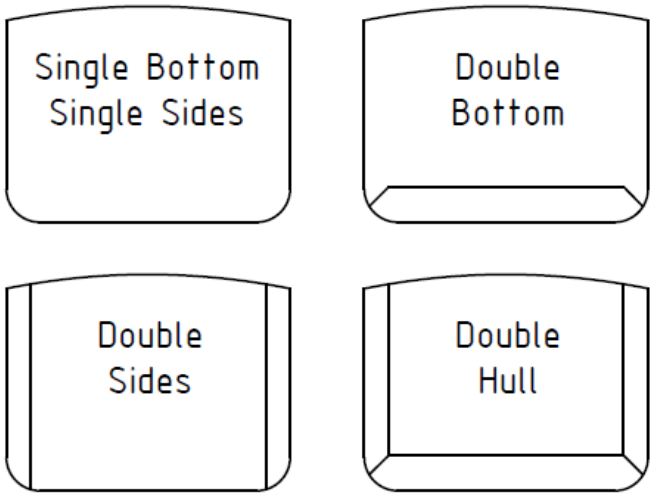


Figure 2.2 – Hull Forms

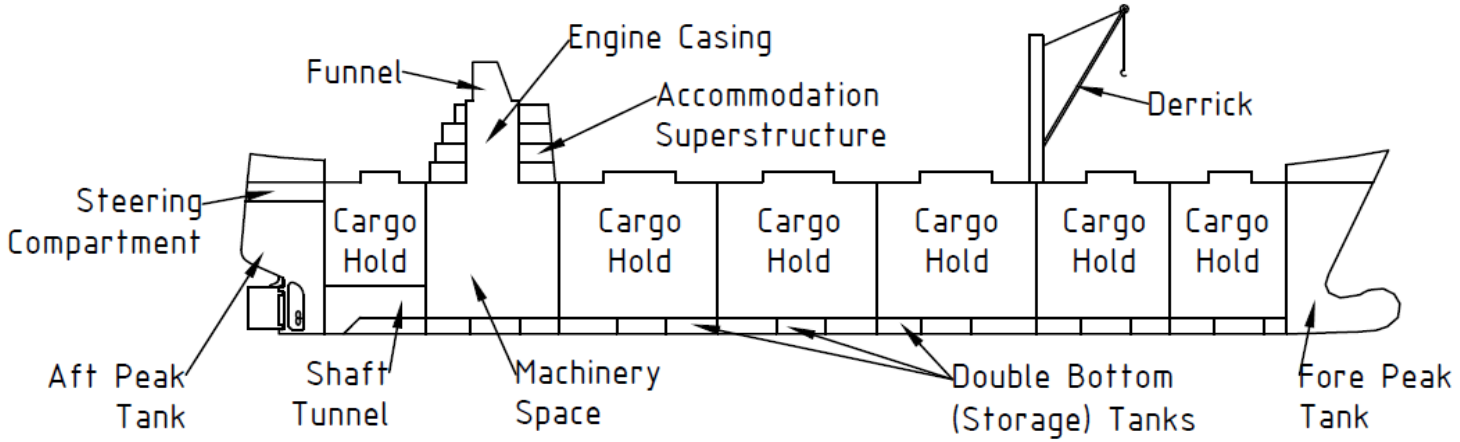


Figure 2.3 – Dry Cargo Ship

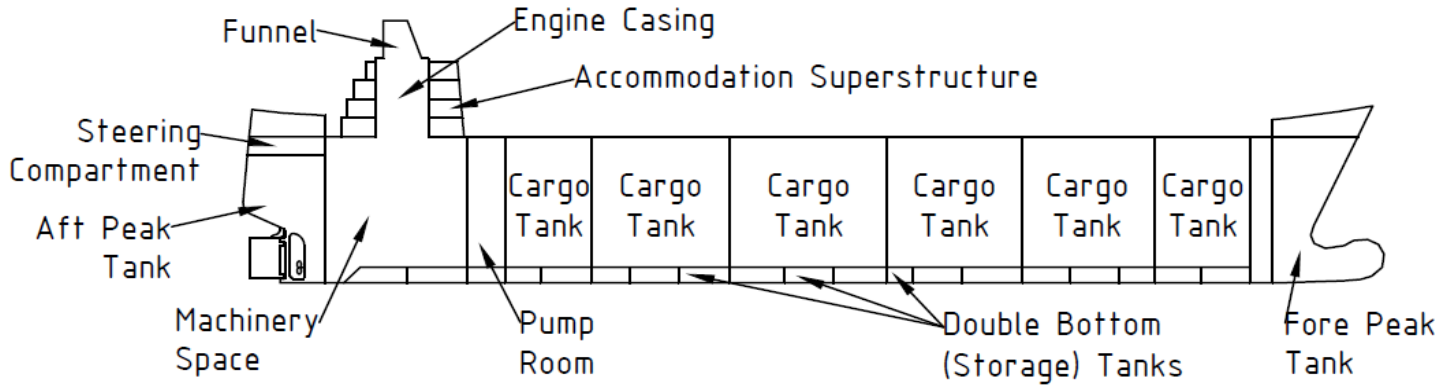


Figure 2.4 – Liquid Cargo Ship

SHIP SYSTEMS

Main System

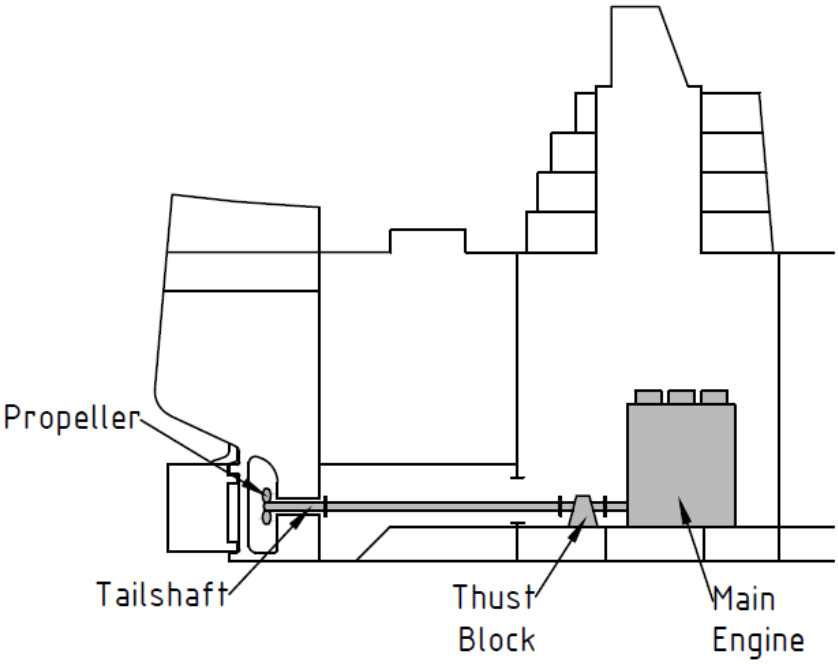


Figure 2.5 – Propulsion Plant (Mechanical)

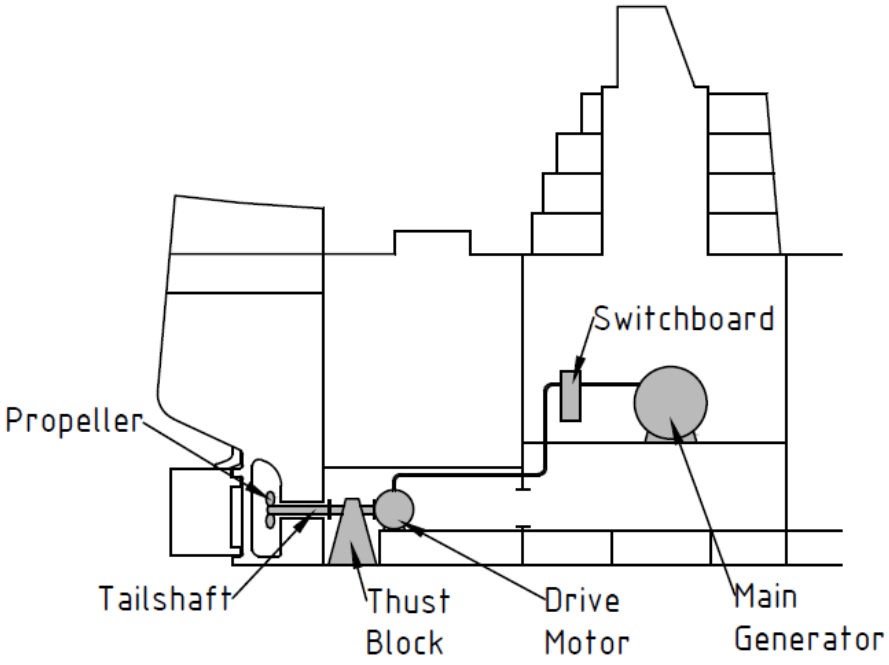


Figure 2.6 – Propulsion Plant (Electrical)

Auxiliaries Systems

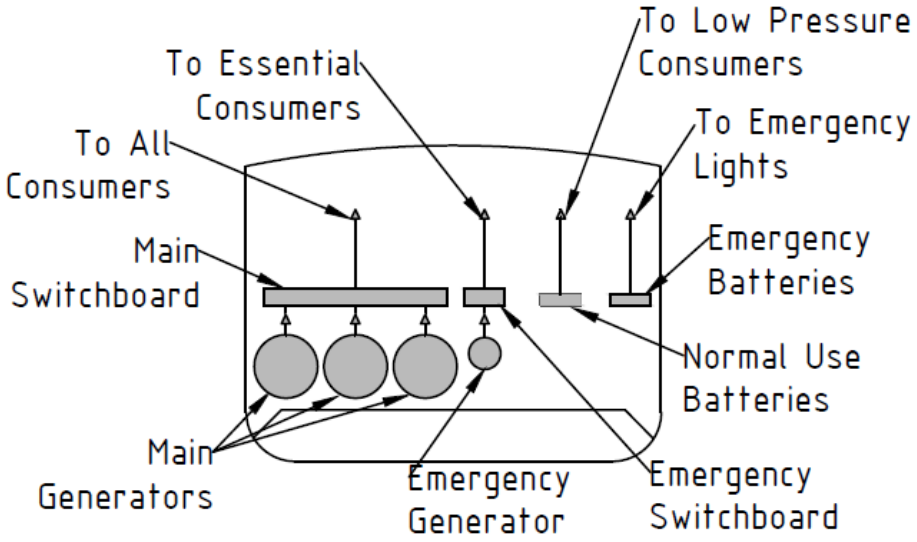


Figure 2.7 – Electrical System

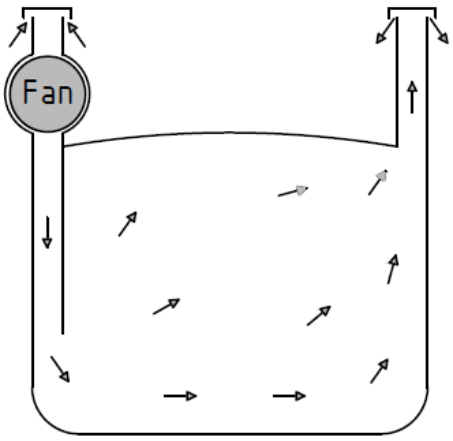


Figure 2.8 – Ventilation System

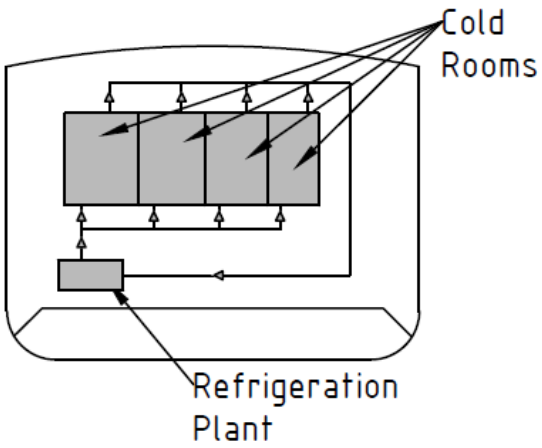


Figure 2.9 – Domestic Fridge System

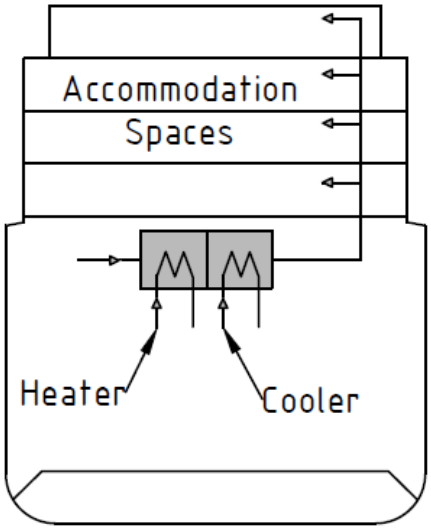


Figure 2.10 – Heating & Air Conditioning Systems

Pumping Systems

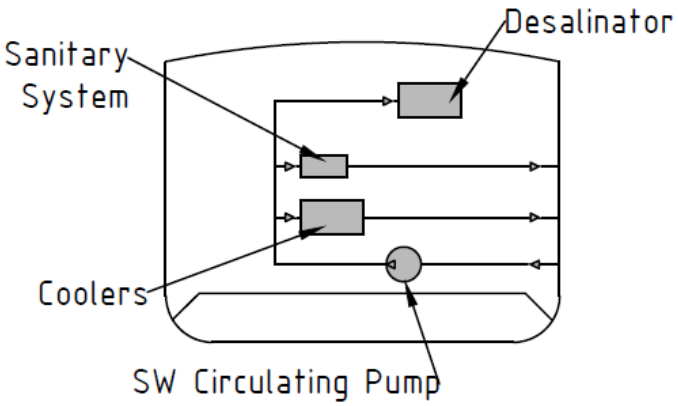


Figure 2.11 – Sea Water Systems

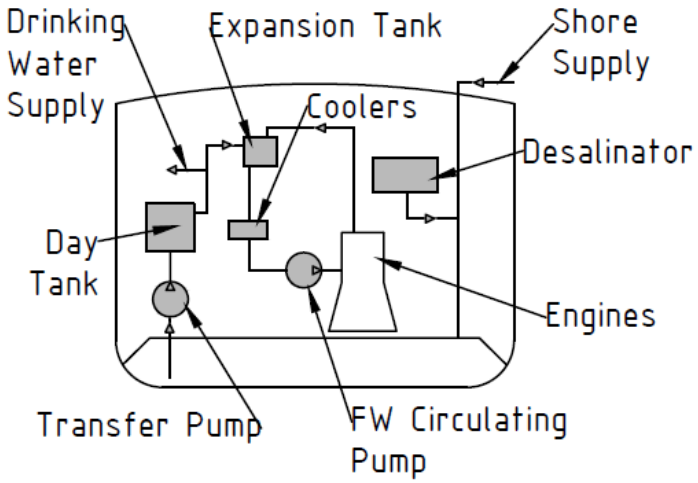


Figure 2.12 – Fresh Water Systems

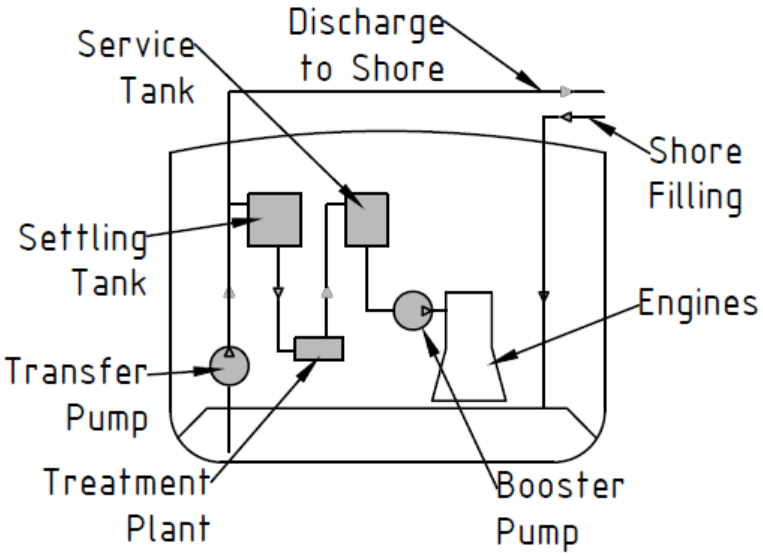


Figure 2.13 – Fuel System

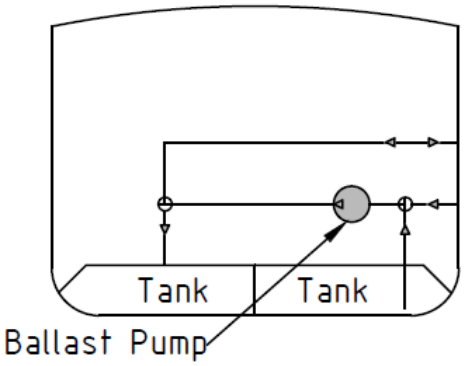


Figure 2.14 – Ballast System

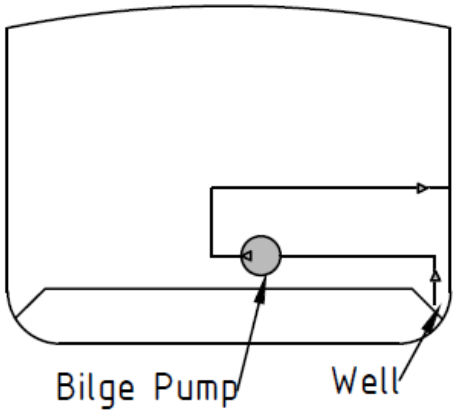


Figure 2.15 – Bilge System

Navigation & Ship Handling Systems

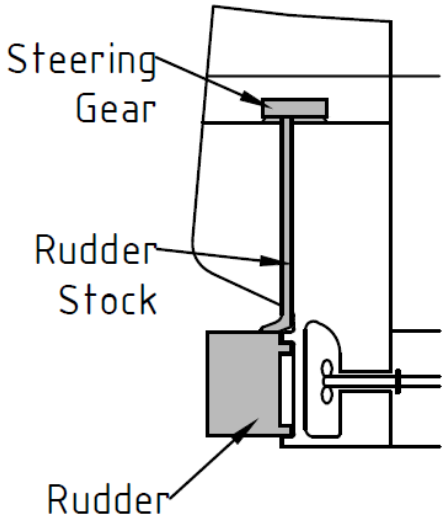


Figure 2.16 – Steering System

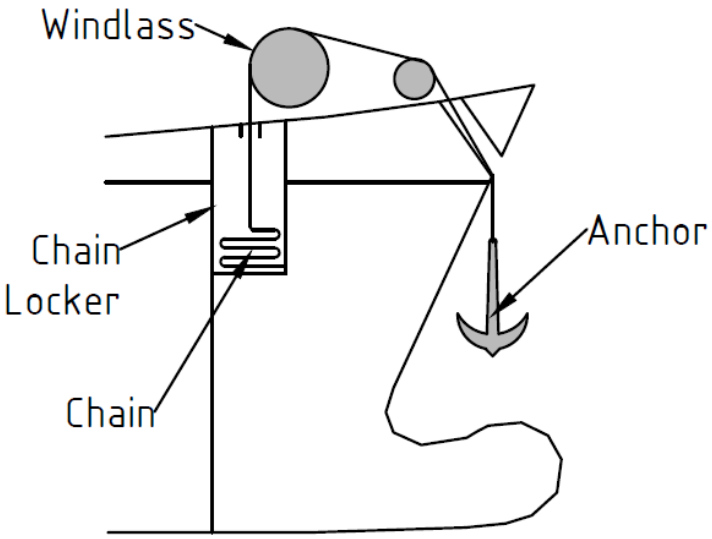


Figure 2.17 – Anchor Handling System

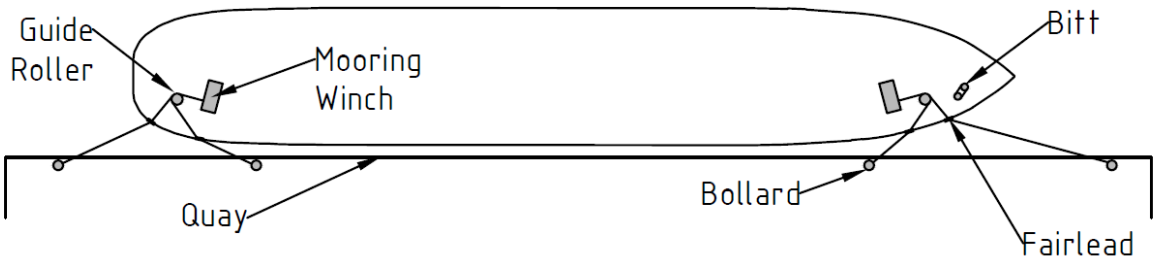


Figure 2.18 – Mooring System

Fire & Safety Systems

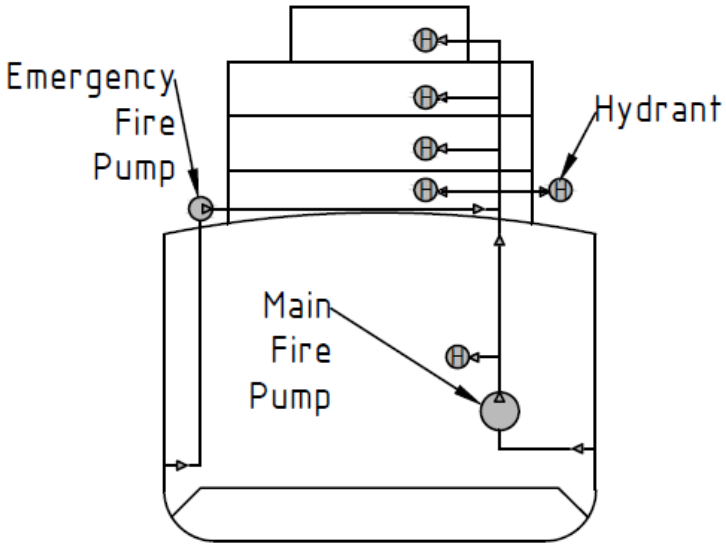


Figure 2.19 – Main Fire System

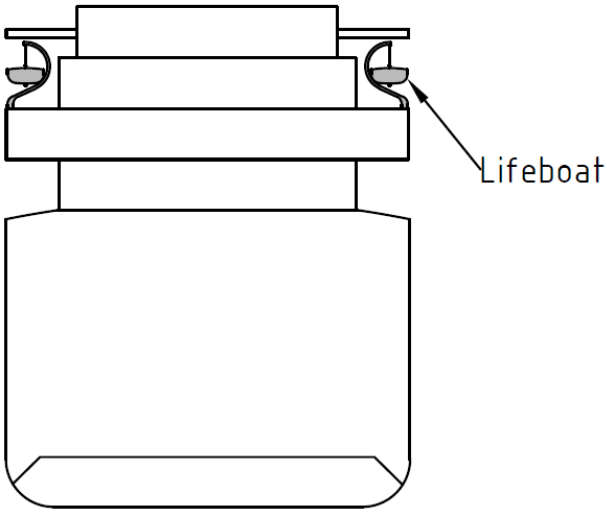


Figure 2.20 – Lifeboat System

3–COEFFICIENTS OF FORM

3.1 REQUIREMENTS OF A SHIP'S HULL

A ship's hull must:

1. Provide sufficient buoyancy to support expected loads. This requires that it is voluminous or bulky.
2. Have its centroid positioned within a certain range in the fore-aft direction to ensure adequate stability and motion control. The best practice is to have it around the midsection region and this is achieved by having a length to breadth ratio which is greater than four. That is one reason why ships hulls are usually slender in form.
3. Avoid generating undue resistance to motion so as to enable travel at good speeds without using fuel excessively and hence making them uneconomical. This is again achieved by making the hull form slender and in addition providing entrances and runs.

The above requirements are described as fineness, fatness, bulkiness, slimness etc, and are quantified mathematically by coefficients.

3.2 BLOCK COEFFICIENT (C_B)

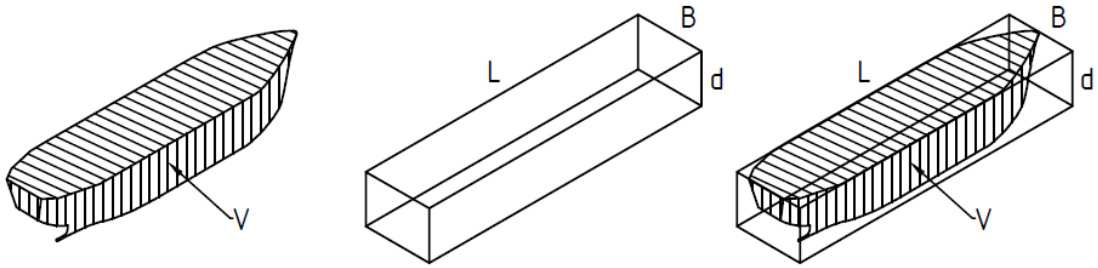


Figure 3.1 – CONCEPT OF BLOCK COEFFICIENT

$$C_B = \frac{\nabla}{LBd}$$

The ratio of the displaced volume to the volume of a circumscribing rectangular box of the same principal dimensions of length, breadth and draught as that of the ship.

3.3 MIDSHIP SECTION AREA COEFFICIENT (C_M)

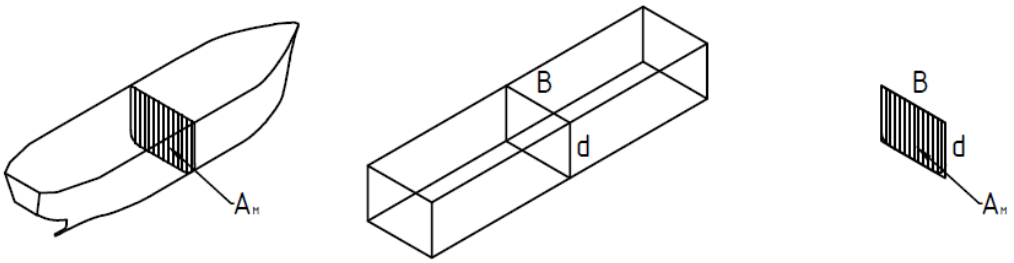


Figure 3.2 – CONCEPT OF MIDSHIP SECTION AREA COEFFICIENT

$$C_M = \frac{A_M}{Bd}$$

The ratio of the midship section area to the area of a circumscribing rectangle of the same principal dimensions of breadth and draught as that of the ship.

3.4 WATERPLANE AREA COEFFICIENT (C_W)

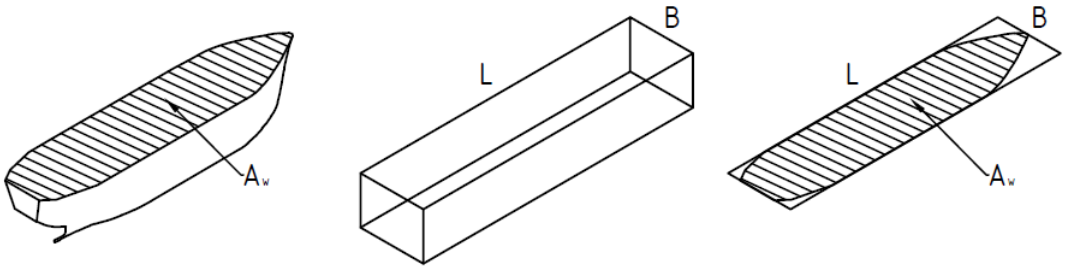


Figure 3.3 – CONCEPT OF WATERPLANE AREA COEFFICIENT

$$C_W = \frac{A_W}{LB}$$

The ratio of the waterplane area to the area of a circumscribing rectangle of the same principal dimensions of length and breadth as that of the ship.

3.5 "LONGITUDINAL" PRISMATIC COEFFICIENT (C_P)

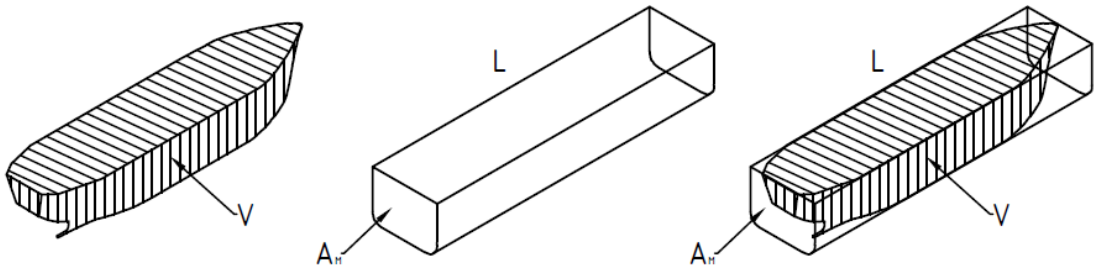


Figure 3.4 – CONCEPT OF "LONGITUDINAL" PRISMATIC COEFFICIENT

$$C_P = \frac{\nabla}{A_M L}$$

The ratio of the displaced volume to a circumscribing volume generated by multiplying the midship section area by the length of the ship.

3.6 VERTICAL PRISMATIC COEFFICIENT (C_{PV})

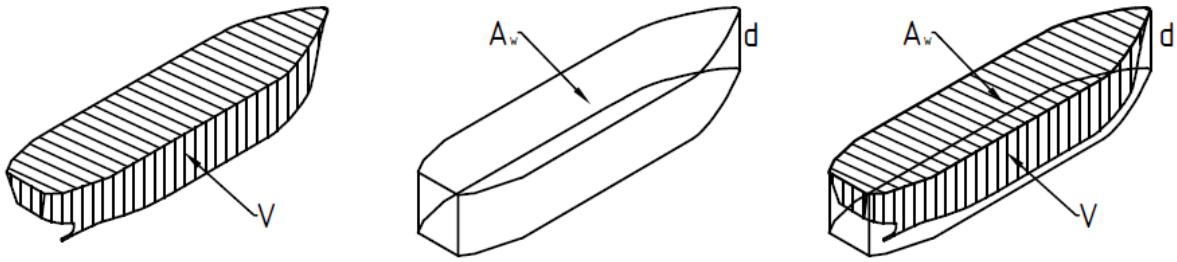


Figure 3.5 – CONCEPT OF VERTICAL PRISMATIC COEFFICIENT

$$C_{PV} = \frac{\nabla}{A_W d}$$

The ratio of the displaced volume to a circumscribing volume generated by multiplying the waterplne area by the draught of the ship.

IN SUMMARY: $C_B = C_M \times C_P = C_W \times C_{PV}$

3.7 NOTES ON BLOCK COEFFICIENT

- The block coefficient (C_B)
  - Links LBP, B and draught d and hence C_W, C_M, C_P & C_{PV}.
  - Is used to estimate ship size at the earlier stages of design.
  - Is used to determine the number of persons (N) a lifeboat can safely carry.

    Where: $N = \frac{(L \times B \times d \times C_B)}{Volume\ of\ a\ person}$

    And for lifeboat lengths:
    - 7.3 m or more the "volume of a person" is taken as 0.283
    - 4.9 m the "volume of a person" is taken as 0.396
    - For intermediate boat lengths (i.e. between 4.9 m and 7.3 m) the values are interpolated.
- In shallow waters (i.e. when the Depth of Influence (F_D)) is below a certain threshold ships with higher C_B will:
  - Experience:
    - Loss of speed.
    - Decrease in propeller revolutions.
    - Higher increase in squat.
  - Become sluggish to manoeuvre.

  Where: $(F_D = \frac{4.44d}{{C_B}^{1.3}})$ As per C.B. Barrass
- Thrust generated by a ship is proportional to the speed of advance (V_a) which in turn is proportional to the wake speed (V_W) and hence the wake fraction w_t. Since ships with higher C_B have higher w_t, they hence suffer higher thrust losses.

  Where: $w_t = \frac{C_B}{2} - 0.05$ As per D.W. Taylor
- The frequency of vibration in ships is partially affected by entrained water (i.e. water dragged along). Since ships with higher C_B have bigger wetted surface areas they are hence more affected in that respect.

4–SHIP GEOMETRY AND HYDROSTATIC CALCULATIONS

4.1 SHIP GEOMETRY AND TABLE OF OFFSETS

To enable stability, trim and powering requirements of a ship to be accurately evaluated, it is necessary to calculate various areas, volumes, centroids, etc. This information put together is referred to as the hydrostatic data.

For most parts a ship's hull form does not follow a perfect geometrical figure, hence the curves which represent its outside surface cannot be expressed by a function. Therefore to enable its hydrostatic data to be calculated, measurements are taken from a set of drawings produced by cutting the hull in three imaginary and mutually orthogonal planes to produce a table of offsets as follows:

4.1.1 Body Plan

This is the set of drawings obtained by cutting longitudinally at eleven equally spaced sections (or stations) from the AP designated as zero (0) to the FP designated as ten (10) to produce ordinates. Midship is section (5). For neatness of presentation only half of sections 6 to 10 are shown and to the left of drawings, while to the right side only half sections of 1 to 4 are shown. Section 5 is represented in full and serves as reference.

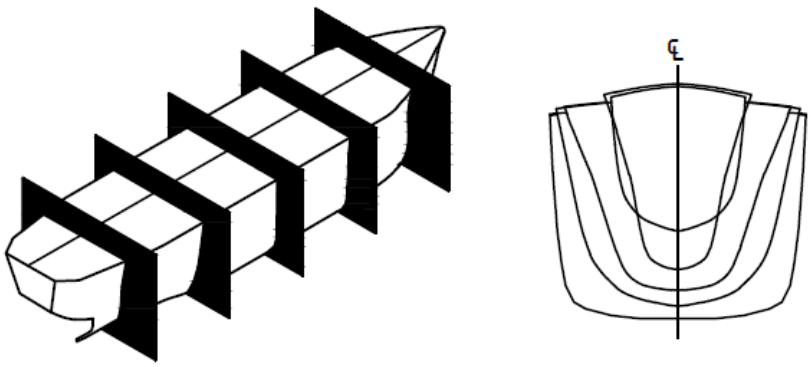


Figure 4.1.1 - Development of Body Plan

4.1.2 Breadth Plan

This is the set of drawings obtained by cutting vertically usually at 1 m intervals from the baseline designated as zero (0) to the designed draught. Drawings produced are usually symmetrical about the centreline which serves as reference.

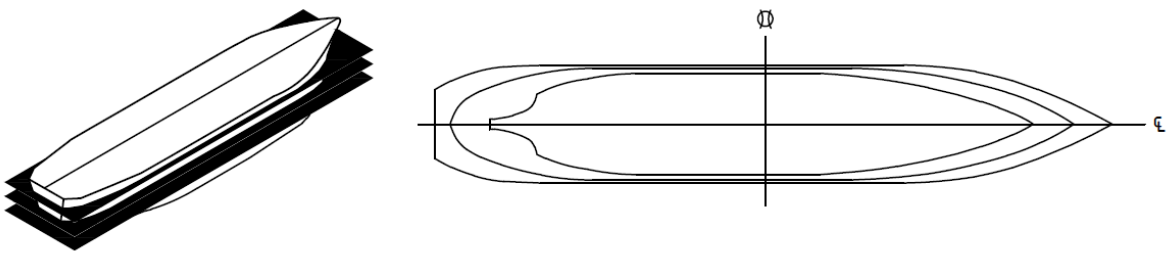


Figure 4.1.2 – Development of Breadth Plan

4.1.3 Sheer Profile

This is the third set of drawings obtained by cutting transversely to port or to starboard from the centreline as reference.

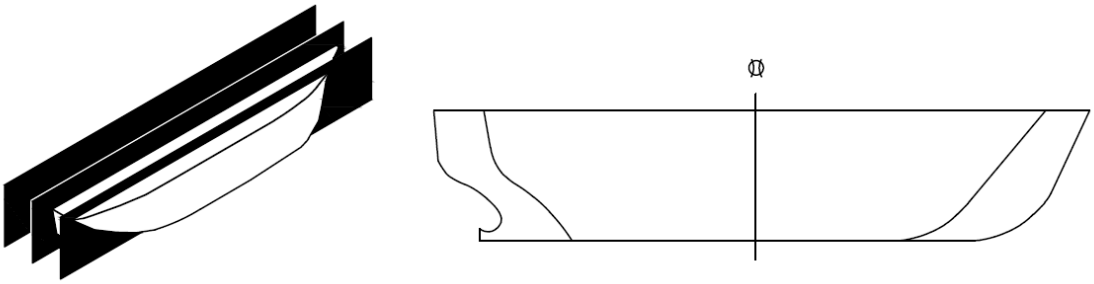


Figure 4.1.3 – Development of Sheer Profile

4.1.4 Lines Plan

The three sets of drawings so produced when put together in one representation is called the Lines Plan. In this the Body Plan is superimposed on the Sheer Profile, while the Breadth Plan is presented below them. The Breadth Plan is presented in half only because of symmetry and is then referred to as the Half-breadth Plan. The ordinates measured from the centreline are referred to as offsets.

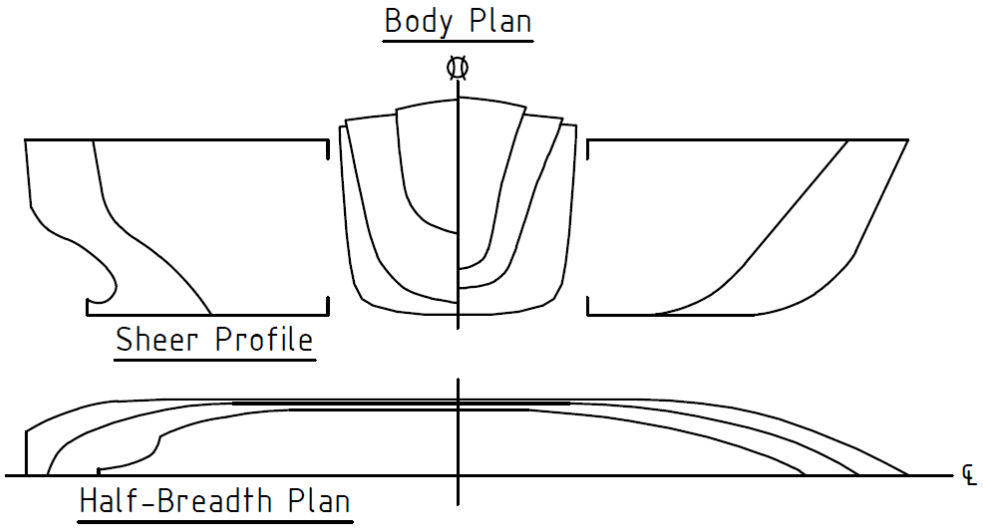


Figure 4.1.4 – LINES PLAN

4.1.5 Table of Offsets

With the Lines Plan completed, offsets or ordinates are then measured to scale at various draughts and eleven sections from the aft perpendicular to the forward perpendicular to produce a Body Plan as in Figure 4.1.6 using the procedure illustrated in Figure 4.1.5. From the Body Plan a Table of Offsets is compiled as in Table 4.1.1. Ordinates may then be taken from this Table to calculate areas, volumes, centroids and moments. When put together the total calculations produced is referred to as the ship's hydrostatic data.

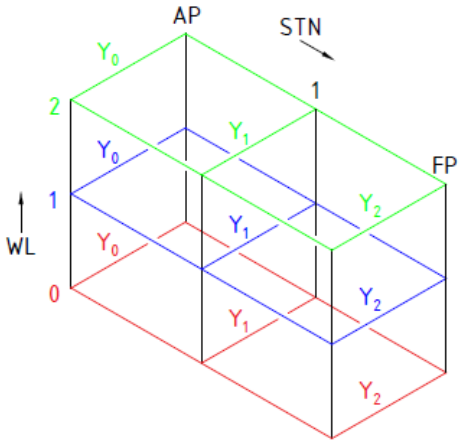


TABLE OF OFFSETS			
WL	SECTION		
	AP	1	FP
2	Y_0	Y_1	Y_2
1	Y_0	Y_1	Y_2
0	Y_0	Y_1	Y_2

Figure 4.1.5 – Procedure for Picking Ordinates

SHIP PARTICULARS
Name: M/V NAUTICO
Displacement: 12 620 Tonne

LOA: 162.00 m
LBP x B x D: 150.00 x 19.00 x 11.00 m
d: 7.00 m

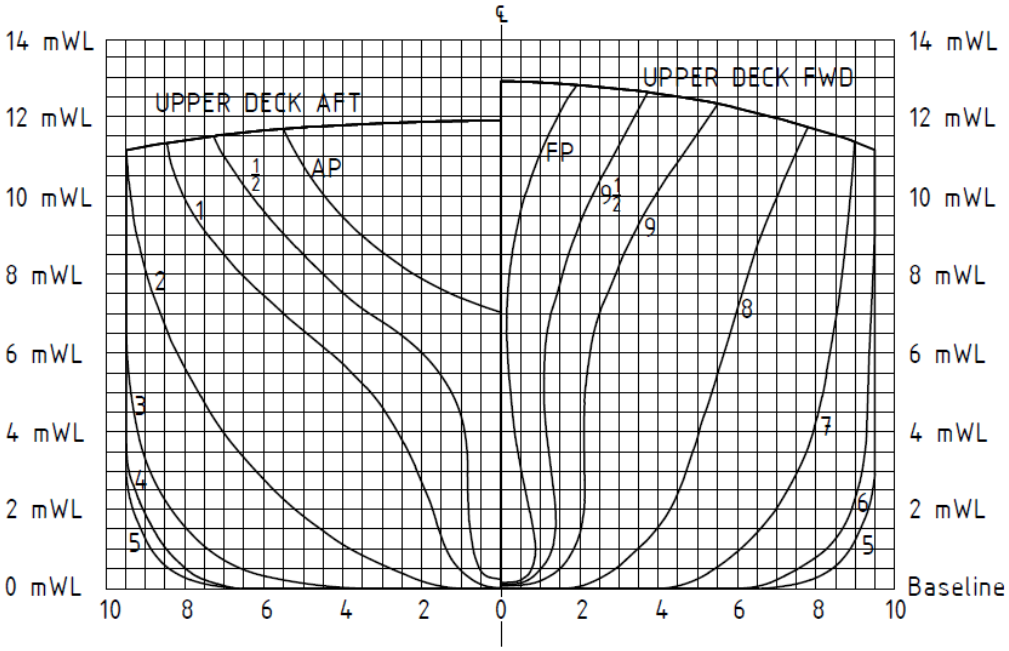


Figure 4.1.6 – View of Ordinates in Body Plan

Name: M/V NAUTICO Displacement:12 620 Tonne LOA: 162 m LBP x B x D: 150 x 19 x 11 m d: 7 m

Up. Dk. Sheer Ab. BL	11.90	11.70	11.50	11.30	11.20	11.20	11.20	11.20	11.20	11.35	11.75	12.30	12.60	12.80	12.90
Stn. / WL	Aft End	AP	½	1	2	3	4	5	6	7	8	9	9½	FP	Fore End
	Half-Breadths														
11.20 m (Up. Dk.)		5.25	7.20	8.40	11.20	9.50	9.50	9.50	9.50	8.95	7.55	4.75	2.95	1.05	
10 m		4.50	6.30	8.00	9.35	9.50	9.50	9.50	9.50	8.80	7.00	3.90	2.25	0.60	
9 m		3.50	5.45	7.40	9.25	9.50	9.50	9.50	9.49	8.75	6.60	3.30	1.90	0.30	
8 m		2.24	4.50	6.55	9.00	9.50	9.50	9.50	9.45	8.65	6.25	2.90	1.60	0.20	
7 m			3.35	5.50	8.60	9.50	9.50	9.50	9.40	8.50	5.90	2.50	1.25	0.15	
6 m			2.00	4.30	8.20	9.45	9.50	9.50	9.35	8.30	5.70	2.25	1.15	0.15	
5 m			1.25	3.30	7.65	9.35	9.50	9.50	9.30	8.20	5.30	2.15	1.10	0.25	
4 m			0.90	2.60	7.05	9.20	9.50	9.50	9.25	7.90	5.05	2.15	1.20	0.40	
3 m			0.85	2.15	6.30	8.85	9.40	9.45	9.20	7.50	4.70	2.15	1.25	0.50	
2 m			0.80	1.75	5.30	8.30	9.10	9.35	8.85	6.95	4.25	2.15	1.40	0.75	
1 m			0.70	1.40	3.80	7.40	8.50	8.80	8.30	6.10	3.50	1.80	1.30	0.85	
0 m (BL)					1.20	3.35	6.50	7.00	6.00	4.00	1.50				

Table 4.1.1 – TABLE OF OFFSETS

4.2 HYDROSTATIC DATA CALCULATIONS

4.2.1 Use of Definite Integration

Using the rules of integration, calculation of areas, volumes, centroids and moments under a curve that is defined by a definite function is carried out as follows:

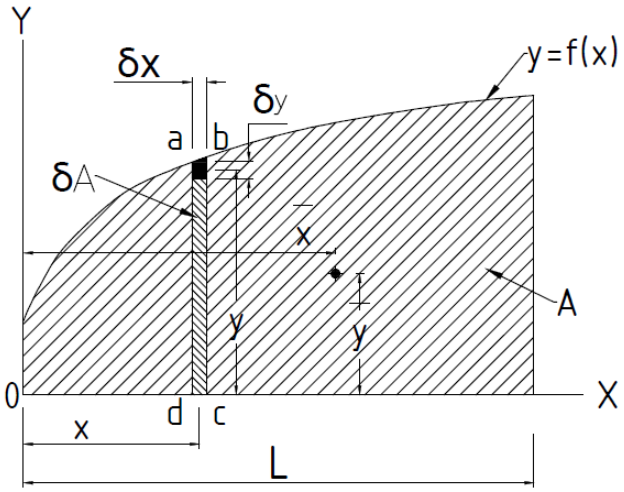


Figure 4.2.1

AREA

Let (δA) be the area of the elementary rectangle (abcd) which is made up a number of elementary squares each of area (δy.δx) as shown in the Figure 4.2.1. Then:

$$\delta A = \delta x \sum \delta y = y \delta x$$

And:

$$A = \sum \delta A = \sum y \delta x$$

Putting it in terms of a definite integral:

$$A = \int_0^L y dx$$

FIRST MOMENT OF AREA ABOUT OY

Let δM_{OY} be the first moment of area of the elementary rectangle (δA) about axis OY. Then:

$$\delta M_{OY} = \delta A . x = xy \delta x$$

And:

$$M_{OY} = \sum xy \delta x = \int_0^L xy dx$$

FIRST MOMENT OF AREA ABOUT OX

Let δM_{OX} be the first moment of area of the elementary rectangle (δA) about axis OX. Then:

$$\delta M_{OX} = \delta A . \frac{y}{2} = \frac{y^2}{2} \delta x$$

And:

$$M_{OX} = \sum \frac{y^2}{2} \delta x = \int_0^L \frac{y^2}{2} dx = \frac{1}{2} \int_0^L y^2 dx$$

CENTROID OF AREA ABOUT OY	CENTROID OF AREA ABOUT OX
Centroid of area is determined by dividing the first moment of the area by the area. Hence: $$\bar{x} = \frac{M_{OY}}{A} = \frac{\int_0^L xydx}{\int_0^L ydx}$$	Again centroid equals the first moment of the area divided by the area. Hence: $$\bar{y} = \frac{M_{OX}}{A} = \frac{\frac{1}{2}\int_0^L y^2dx}{\int_0^L ydx}$$
SECOND MOMENT OF AREA ABOUT OY Let δI_{OY} be the second moment of area of the elementary rectangle (δA) about axis OY. Then: $$\delta I_{OY} = y\delta x.x^2 = x^2y\delta x$$ And: $$I_{OY} = \sum x^2y\delta x = \int_O^L x^2ydx$$	SECOND MOMENT OF AREA ABOUT OX Starting from the elementary square ($\delta y.\delta y$), let δI_{OX} be the second moment of the elementary rectangle (δA) about axis OX. Then: $$\delta I_{OX} = (\delta y\delta x)y^2 = y^2\delta y.\delta x$$ And: $$I_{OX} = \sum y^2\delta y \sum \delta x = \sum\sum (y^2\,\delta y).\delta x$$ $$I_{OX} = \int_0^L \int_0^y (y^2\,dy).dx$$ $$I_{OX} = \int_0^L \frac{y^3}{3}dx = \frac{1}{3}\int_0^L y^3dx$$

Table 4.2.1 – INTEGRATING RULES

4.2.2 Use of Approximate Integration

Since curves that describe the hull form cannot be expressed by definite functions, approximate integration methods are employed to calculate the hydrostatic data of a ship. Examples of such methods are SIMPSON'S RULES, TRAPEZOIDAL RULE, TCHEBYCHEFF'S RULES, GAUSS RULES, etc. For the calcularions in this exercise the Simpson's First Rule shall be used.

4.2.2.1 Simpson's First Rule

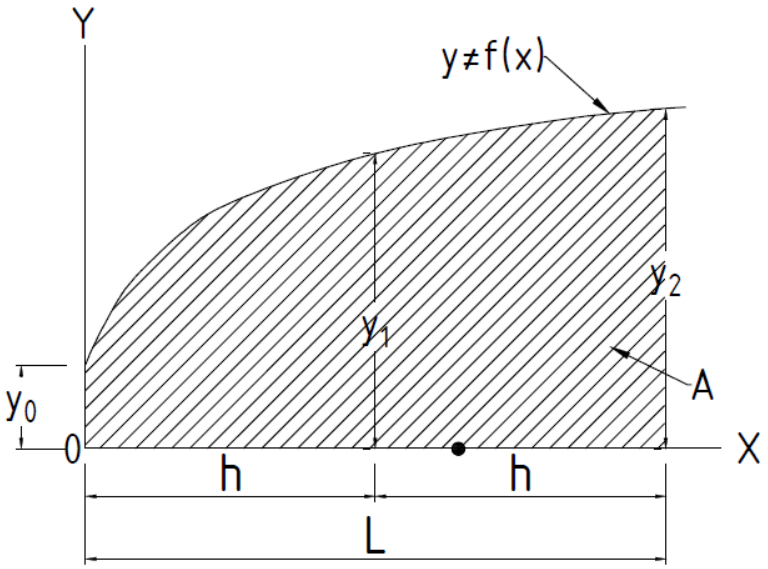


Figure 4.2.2

To obtain the area (A) under a curve, erect THREE ORDINATES (y_0, y_1 and y_2) at equal or 'common' spacing or intervals (h) along the X-axis as shown in Figure 4.2.2. Multiply the ordinates by the quotients 1, 4 and 1, referred to as 'Simpson's Multipliers' (SM), respectivley, and then sum up the three products obtained. Finally multiply the sum so obtained by the common interval and divide the results by three to obtain the area:

$$A = \frac{h}{3}(1y_0 + 4y_1 + 1y_2)$$

Comparison of Calculations by Definite Integration and by Simpson's First Rule:

Property	By definite integration	By Simpson's First Rule
Area	$A = \int_0^L y dx$	$A = \frac{h}{3}(1y_0 + 4y_1 + 1y_2)$
Longitudinal Centre of Flotation	$\bar{x} = \frac{\int_0^L yx dx}{\int_0^L y dx}$	$\bar{x} = \frac{\frac{h}{3}(1y_0x_0 + 4y_1x_1 + 1y_2x_2)}{A}$
Transverse Centre of Flotation	$\bar{y} = \frac{\int_0^L \frac{y^2}{2} dx}{\int_0^L y dx}$	$\bar{y} = \frac{\frac{h}{3}\left(1\frac{{y_0}^2}{2} + 4\frac{{y_1}^2}{2} + 1\frac{{y_2}^2}{2}]\right)}{A}$
Second Moment of Area About Transverse Axis	$I_{OY} = \int_0^L yx^2 dx$	$I_{OY} = \frac{h}{3}(1y_0{x_0}^2 + 4y_1{x_1}^2 + 1y_2{x_2}^2])$
Second Moment of Area About Longitudinal Axis	$I_{OX} = \int_0^L \frac{y^3}{3} \delta x$	$I_{0X} = \frac{h}{3}\left(1\frac{{y_0}^3}{3} + 4\frac{{y_1}^3}{3} + 1\frac{{y_2}^3}{3}\right)$

Table 4.2.2 – COMPARISON TABLE

4.2.2.2 Application of Simpson's Rule to the calculation of a ship's Waterplane Area (A_W)

With respect to the zero draught, the half-breadth plan will appear as per the detail in Figure 4.2.4.

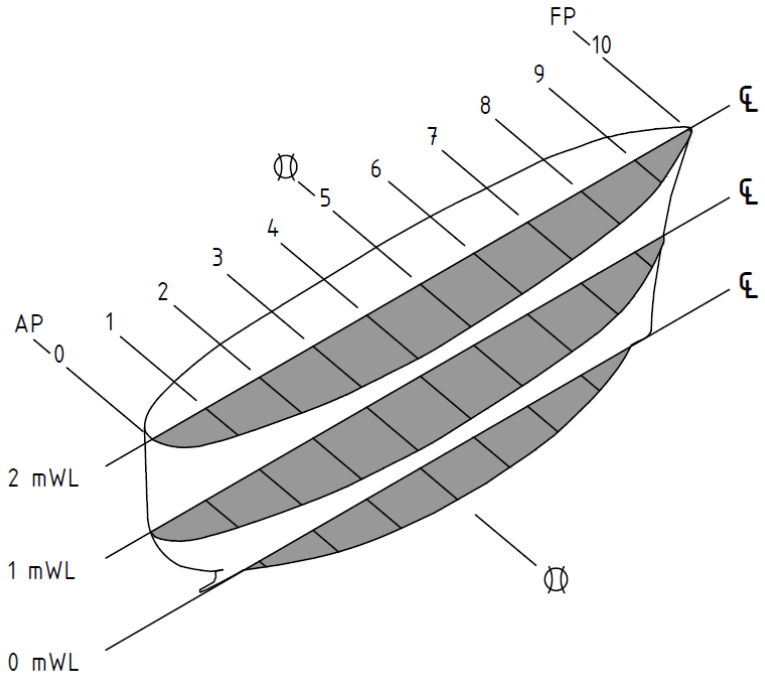


Figure 4.2.3 – Waterplane Areas (Isometric View)

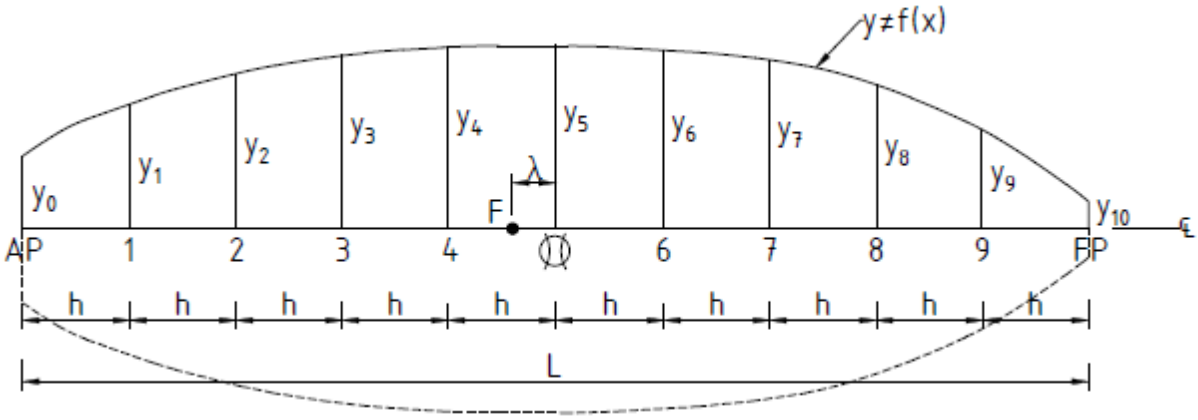


Figure 4.2.4 – Details of Waterplane Area at the Baseline

Starting from the AP let the half-breadths or half-ordinates, taken at common intervals ($h_{LONGITUDINAL}$) or (h_L) or (h) for short be represented by y_0 to y_{10}. Let the centroid of area (F) be at a distance (λ) from midship..

By taking three sets of half-ordinates at a time so as to satisfy the Rule, calculate the total area as follows:

$$A_{W0} = 2\left[\frac{h_L}{3}(1y_0 + 4y_1 + 1y_2) + \frac{h_L}{3}(1y_2 + 4y_3 + 1y_4) + \frac{h_L}{3}(1y_4 + 4y_5 + 1y_6) + \frac{h_L}{3}(1y_6 + 4y_7 + 1y_8) + \frac{h_L}{3}(1y_8 + 4y_9 + 1y_{10})\right]$$

$$A_{W0} = 2\frac{h_L}{3}(1y_0 + 4y_1 + 2y_2 + 4y_3 + 2y_4 + 4y_5 + 2y_6 + 4y_7 + 2y_8 + 4y_9 + 1y_{10})$$

To simplify and ease computation, calculations may be presented vertically in columns as shown in Table 4.2.3.

a	b	c			d = b.c
SECTION	½-ORDINATE	SM			AREA PRODUCT
		Bits		Combined	
AP	y_0	1		1	$1y_0$
1	y_1	4		4	$4y_1$
2	y_2	1	1	2	$2y_2$
3	y_3		4	4	$4y_3$
4	y_4	1	1	2	$2y_4$
5	y_5	4		4	$4y_5$
6	y_6	1	1	2	$2y_6$
7	y_7		4	4	$4y_7$
8	y_8	1	1	2	$2y_8$
9	y_9	4		4	$4y_9$
FP	y_{10}	1		1	$1y_{10}$
					$\sum AREA\ PRODUCT$

Table 4.2.3 – Table for Calculation of Waterplane Area using Half-Ordinates

Final Calculation:

$$A_W = 2\frac{h_L}{3}\sum AREA\ PRODUCT$$

4.2.2.3 Calculation of (A_W) with ***intermediate*** half-ordinates introduced

To improve upon accuracy, it is sometimes the practice to introduce additional ordinates at the ends where the curve changes gradient rapidly. These are referred to as *INTERMEDIATE ORDINATES.* They are placed halfway between the normal ordinates wherever they are introduced, thus making the spacing between them equal to half the common interval.

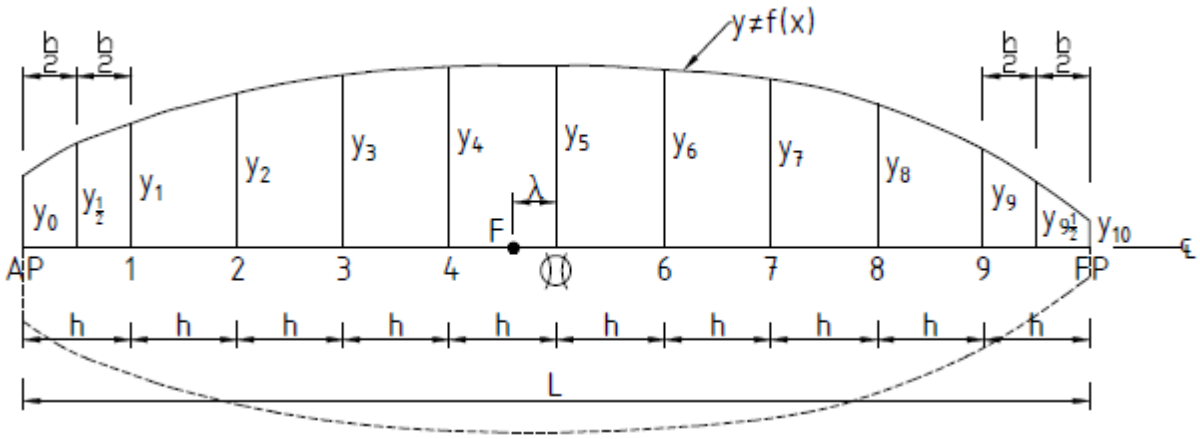


Figure 4.2.5 – View of Waterplane Area with Intermediate Half-Ordinates

For such an arrangement the computation is modified as follows:

$$A_{W0} = 2\left[\frac{h_L/2}{3}(1y_0 + 4y_{½} + 1y_1) + \frac{h_L}{3}(1y_1 + 4y_2 + 1y_3) + \frac{h_L}{3}(1y_3 + 4y_4 + 1y_5)\right.$$
$$+ \frac{h_L}{3}(1y_5 + 4y_6 + 1y_7) + \frac{h_L}{3}(1y_7 + 4y_8 + 1y_9)$$
$$\left. + \frac{h_L/2}{3}(1y_9 + 4y_{9½} + 1y_{10})\right]$$

$$A_{W0} = 2\frac{h_L}{3}(½y_0 + 2y_{½} + 1½y_1 + 4y_2 + 2y_3 + 4y_4 + 2y_5 + 4y_6 + 2y_7 + 4y_8 + 1½y_9$$
$$+ 2y_{9½} + ½y_{10})$$

a	b	c			d = b.c
SECTION	½-ORDINATE	SM			AREA PRODUCT
		Sets		Final	
AP	y_0	½		½	$½y_0$
½	$y_{½}$	2		2	$2y_{½}$
1	y_1	½	1	1½	$1½y_1$
2	y_2		4	4	$4y_2$
3	y_3	1	1	2	$2y_3$
4	y_4	4		4	$4y_4$
5	y_5	1	1	2	$2y_5$
6	y_6		4	4	$4y_6$
7	y_7	1	1	2	$2y_7$
8	y_8	4		4	$4y_8$
9	y_9	1	½	1½	$1½y_9$
9½	$y_{9½}$		2	2	$2y_{9½}$
FP	y_{10}		½	½	$½y_{10}$
					$\sum AREA\ PRODUCT$

Table 4.2.4 – Table for Calculation of Waterplane Area with Intermediate Half-Ordinates Included

Final Calculation:

$$A_W = 2\frac{h_L}{3}\sum AREA\ PRODUCT$$

4.2.2.4 Calculation of LCF_{MS}; TCF_{CL}; I_{MS}; I_{CL}; I_{CF} – With reference to Midship and Centreline

By convention the reference axis for measuring transverse and longitudinal centroids and moments are the centerline and midship respectively. On that basis the calculation of LCF_{MS} and I_{MS} are therefore computed by considering (x) levers to be positive when going aft, and negative when going forward as illustrated in Figure 4.2.6. Now since the (x) values are multiples of the common interval, they are usually expressed in terms of (h) as: $x_0 = 0h$; $x_1 = 1h$; $x_2 = 2h$; $x_3 = 3h$; $x_4 = 4h$ and $x_5 = 5h$. With respect to TCF_{CL} and I_{CL} the (y) levers are considered positive when going to port side and negative when going starboard.

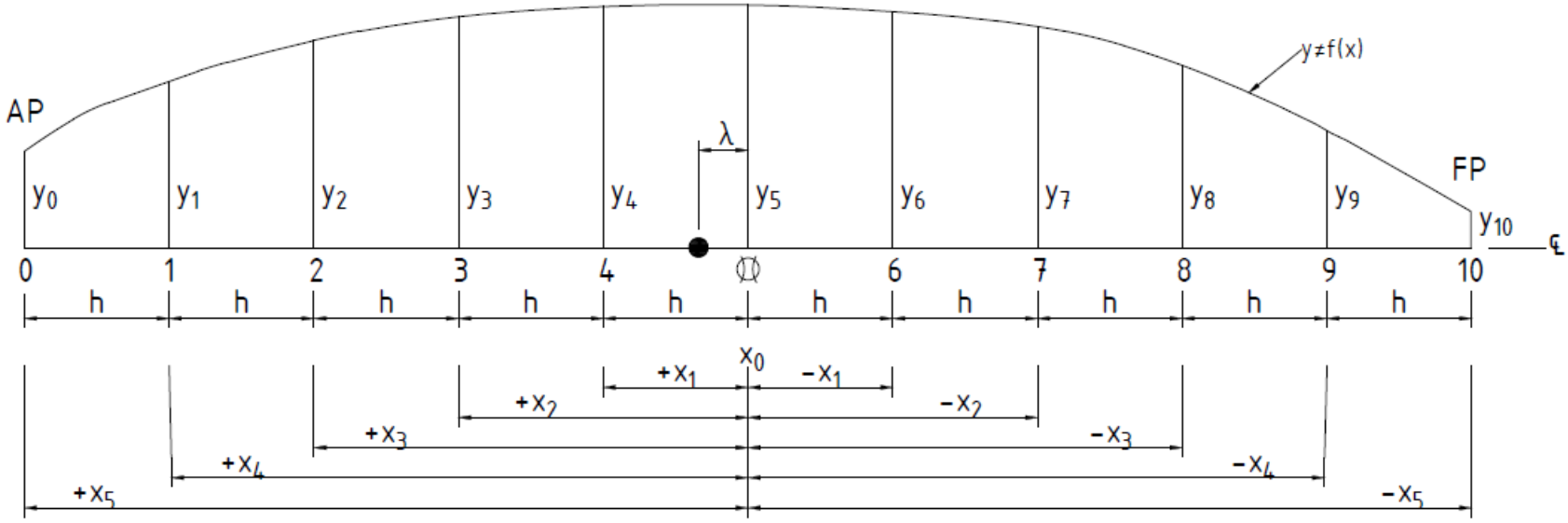


Figure 4.2.6 – Moments with Midship and Centreline as References

a	b	c			d = bc	e	f = de	g = ef	j= +b/2	k= -b/2	l = j+k	m = c x l	h = b^3	i = ch
SECTION	½-ORD	SM			AREA PRODUCT	$LEVER_{MS}$	FM_{MS} PRODUCT	I_{MS} PRODUCT	$LEVER_P$	$LEVER_S$	$LEVER_{CL}$	FM_{CL} PRODUCT	$(½\text{-ORD})^3$	I_{CL} PRODUCT
		Sets		Comb										
AP	y_0	1		1	$1y_0$	$5h_L$	$5h_Ly_0$	$25h_L^2y_0$	$+½y_0$	$-½y_0$	0	0	y_0^3	$1y_0^3$
1	y_1	4		4	$4y_1$	$4h_L$	$16h_Ly_1$	$64h_L^2y_1$	$+½y_1$	$-½y_1$	0	0	y_1^3	$4y_1^3$
2	y_2	1	1	2	$2y_2$	$3h_L$	$6h_Ly_2$	$18h_L^2y_2$	$+½y_2$	$-½y_2$	0	0	y_2^3	$2y_2^3$
3	y_3		4	4	$4y_3$	$2h_L$	$8h_Ly_3$	$16h_L^2y_3$	$+½y_3$	$-½y_3$	0	0	y_3^3	$4y_3^3$
4	y_4	1	1	2	$2y_4$	$1h_L$	$2h_Ly_4$	$2h_L^2y_4$	$+½y_4$	$-½y_4$	0	0	y_4^3	$2y_4^3$
5	y_5	4		4	$4y_5$	$0h_L$	$0h_Ly_5$	$0h_L^2y_5$	$+½y_5$	$-½y_5$	0	0	y_5^3	$4y_5^3$
6	y_6	1	1	2	$2y_6$	$-1h_L$	$-2h_Ly_6$	$-2h_L^2y_6$	$+½y_6$	$-½y_6$	0	0	y_6^3	$2y_6^3$
7	y_7		4	4	$4y_7$	$-2h_L$	$-8h_Ly_7$	$-16h_L^2y_7$	$+½y_7$	$-½y_7$	0	0	y_7^3	$4y_7^3$
8	y_8	1	1	2	$2y_8$	$-3h_L$	$-6h_Ly_8$	$-18h_L^2y_8$	$+½y_8$	$-½y_8$	0	0	y_8^3	$2y_8^3$
9	y_9	4		4	$4y_9$	$-4h_L$	$-16h_Ly_9$	$-64h_L^2y_9$	$+½y_9$	$-½y_9$	0	0	y_9^3	$4y_9^3$
FP	y_{10}	1		1	$1y_{10}$	$-5h_L$	$-5h_Ly_{10}$	$-25h_L^2y_{10}$	$+½y_{10}$	$-½y_{10}$	0	0	y_{10}^3	$1y_{10}^3$
					$\sum AREA\ PROD$		$\sum FM_{MS}\ PROD$	$\sum I_{MS}\ PROD$				0		$\sum I_{CL}\ PROD$

Table 4.2.5 – Table for Calculation of Moments with Midship and Centreline as References

Final Calculations:

$$A_W = 2\frac{h_L}{3}\sum AREA\ PRODUCT$$

$$LCF_{MS} = \frac{2\frac{h_L}{3}\sum FM_{MS}\ PROD}{A_W}$$

$$I_{MS} = 2\frac{h_L}{3}\sum I_{MS}\ PROD$$

$$I_F = I_{MS} - A_W\lambda^2$$

$$I_{CL} = 2\frac{h_L}{3}\sum I_{CL}\ PROD$$

$$TCF_{CL} = \frac{0}{A_W}$$

4.2.2.5 Calculation of LCF_{MS}; I_{MS}; I_{CL}; I_{CF} – With reference to Aft Perpendicular and Centreline

Instead of midship the AP may be used as reference for taking moments as illustrated in Figure 4.2.7. Corrections are thereafter made to transfer the results to midship.

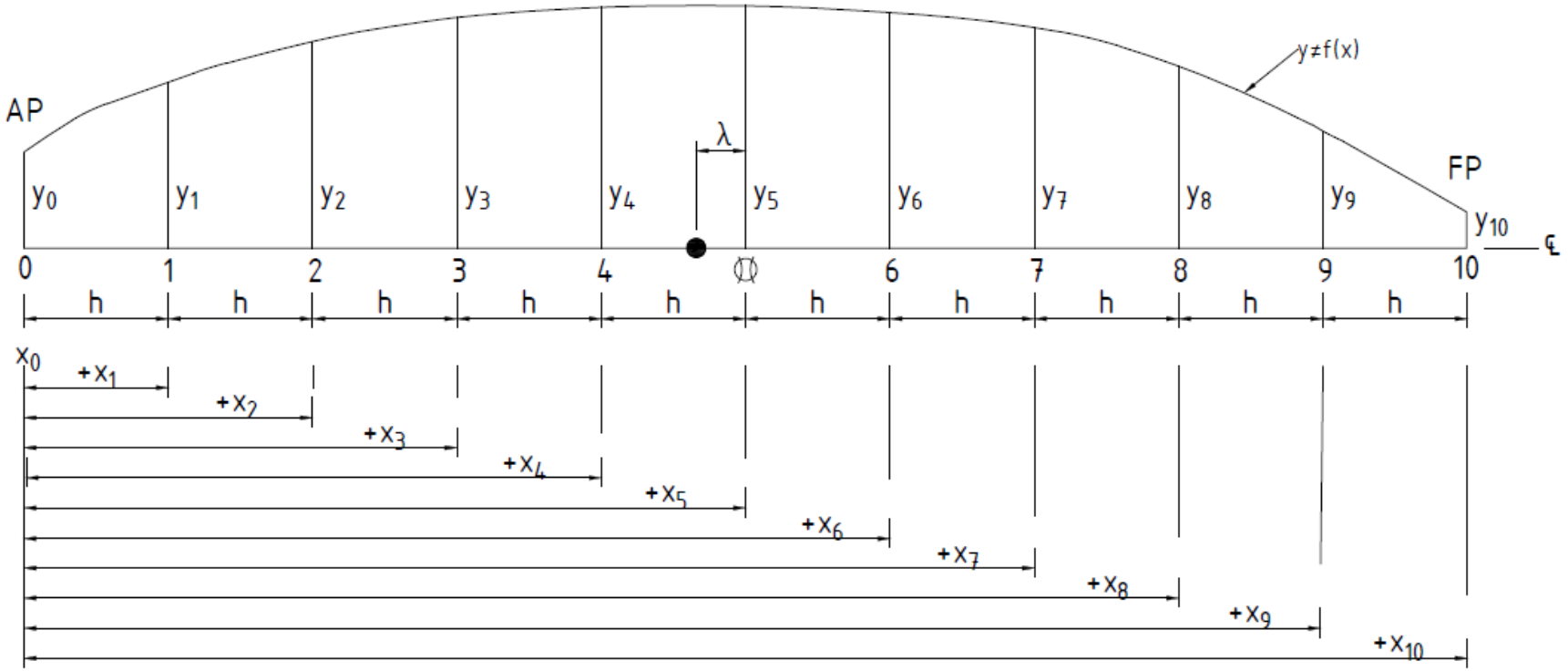


Figure 4.2.7 – Moments with AP and Centreline as References

a	b	c			d = bc	e	f = de	g = ef	j= +b/2	k= -b/2	l = j+k	m = c x l	h = b^3	i = ch
SECTION	½-ORD	SM			AREA PRODUCT	$LEVER_{AP}$	FM_{AP} PRODUCT	I_{AP} PRODUCT	$LEVER_P$	$LEVER_S$	$LEVER_{CL}$	FM_{CL} PRODUCT	$(½\text{-ORD})^3$	I_{CL} PRODUCT
		Sets		Comb										
AP	y_0	1		1	$1y_0$	$0h_Ly_0$	$0h_Ly_0$	$0h_L^2y_0$	$+½y_0$	$-½y_0$	0	0	y_0^3	$1y_0^3$
1	y_1	4		4	$4y_1$	$1h_Ly_1$	$4h_Ly_1$	$4h_L^2y_1$	$+½y_1$	$-½y_1$	0	0	y_1^3	$4y_1^3$
2	y_2	1	1	2	$2y_2$	$2h_Ly_2$	$4h_Ly_2$	$8h_L^2y_2$	$+½y_2$	$-½y_2$	0	0	y_2^3	$2y_2^3$
3	y_3		4	4	$4y_3$	$3h_Ly_3$	$12h_Ly_3$	$36h_L^2y_3$	$+½y_3$	$-½y_3$	0	0	y_3^3	$4y_3^3$
4	y_4	1	1	2	$2y_4$	$4h_Ly_4$	$8h_Ly_4$	$32h_L^2y_4$	$+½y_4$	$-½y_4$	0	0	y_4^3	$2y_4^3$
5	y_5	4		4	$4y_5$	$5h_Ly_5$	$20h_Ly_5$	$100h_L^2y_5$	$+½y_5$	$-½y_5$	0	0	y_5^3	$4y_5^3$
6	y_6	1	1	2	$2y_6$	$6h_Ly_6$	$12h_Ly_6$	$72h_L^2y_6$	$+½y_6$	$-½y_6$	0	0	y_6^3	$2y_6^3$
7	y_7		4	4	$4y_7$	$7h_Ly_7$	$28h_Ly_7$	$196h_L^2y_7$	$+½y_7$	$-½y_7$	0	0	y_7^3	$4y_7^3$
8	y_8	1	1	2	$2y_8$	$8h_Ly_8$	$16h_Ly_8$	$128h_L^2y_8$	$+½y_8$	$-½y_8$	0	0	y_8^3	$2y_8^3$
9	y_9	4		4	$4y_9$	$9h_Ly_9$	$36h_Ly_9$	$324h_L^2y_9$	$+½y_9$	$-½y_9$	0	0	y_9^3	$4y_9^3$
FP	y_{10}	1		1	$1y_{10}$	$10h_Ly_{10}$	$10h_Ly_{10}$	$100h_L^2y_{10}$	$+½y_{10}$	$-½y_{10}$	0	0	y_{10}^3	$1y_{10}^3$
					$\sum AREA\ PROD$		$\sum FM_{AP}\ PROD$	$\sum I_{AP}\ PROD$				0		$\sum I_{CL}\ PROD$

Table 4.2.6 – Table for Calculation of Area and Moments with AP as Reference

Final Calculations:

$$A_W = 2\frac{h_L}{3}\sum AREA\ PRODUCT$$

$$LCF_{AP} = \frac{2\frac{h_L}{3}\sum FM_{AP}\ PROD}{A_W}$$

$$LCF_{MS} = \frac{L}{2} - LCF_{AP}$$

$$I_{AP} = 2\frac{h_L}{3}\sum I_{AP}\ PROD$$

$$I_{MS} = I_{AP} - A_W\left(\frac{L}{2}\right)^2$$

$$I_F = I_{MS} - A_W\lambda^2$$

$$I_{CL} = 2\frac{h_L}{3}\sum I_{CL}\ PROD$$

$$TCF_{CL} = \frac{0}{A_W}$$

4.2.2.6 Calculation of Volume of Dis[placement (∇) by Computatlly Verticallly

By repeating the calculation of waterplane areas for all waterlines up to the load line, the volume of displacement is determined as follows. In the following calculation use three waterplane areas (A_{W0}, A_{W1} and A_{W2}), and let the common interval be ($h_{VERTICAL}$) or (h_V) or (h) for short.

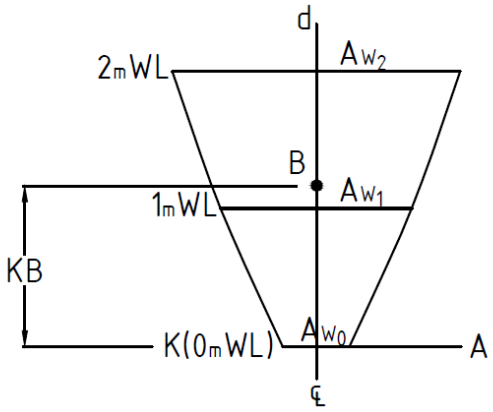


Figure 4.2.8 – Waterplane Area Curve - Symmetrical View

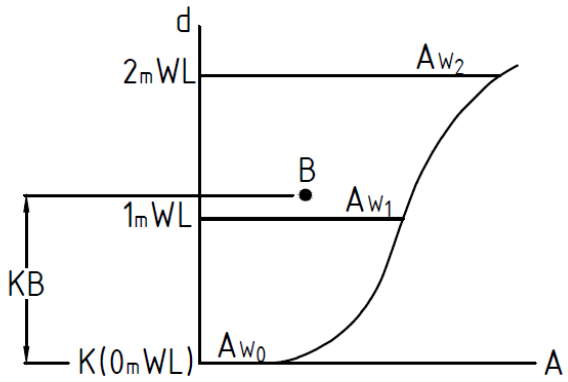


Figure 4.2.9 – Waterplane Area Curve - Conventional View

a	b	c	d = bc
WL	A	SM	VOLUME PRODUCT
0	A_{W0}	1	$1A_{W0}$
1	A_{W1}	4	$4A_{W1}$
2	A_{W2}	1	$1A_{W2}$
			$\sum VOL_{PROD}$

Table 4.2.7 – Table for Calculation Volume (V)

Final Calculation:

$$\nabla = \frac{h_V}{3}(1A_{W0} + 4A_{W1} + 1A_{W2}) = \frac{h_V}{3}\sum VOL_{PROD}$$

4.2.2.7 Calculation of Centre of Volume (LCB_K) or (KB)

By taking first moment of volumes and dividing by the volume so obtained, the vertical centre of buoyancy is then determined as follows.

a	b	c	d = bc	e	f = de
WL	A	SM	VOLUME PRODUCT	$LEVER_K$	FM_K PRODUCT
0	A_{W0}	1	$1A_{W0}$	$0h_V$	0
1	A_{W1}	4	$4A_{W1}$	$1h_V$	$4h_VA_{W1}$
2	A_{W2}	1	$1A_{W2}$	$2h_V$	$2h_VA_{W2}$
			$\sum_{PROD}^{VOL}$		$\sum_{PROD}^{FM_K}$

Table 4.2.8 – Table for Calculation Volume (V)

Final Calculation:

$$KB = \frac{\frac{h_V}{3}\sum_{PROD}^{FM_K}}{\frac{h_V}{3}\sum_{PROD}^{VOL}}$$

4.4.2.7 Calculation of Volume of Displacement (∇) by Computing Longitudinally

At each station and to a particular waterline, calculate cross-sectional areas and integrate the results so obtained over the length of ship to determine volumes, as follows.

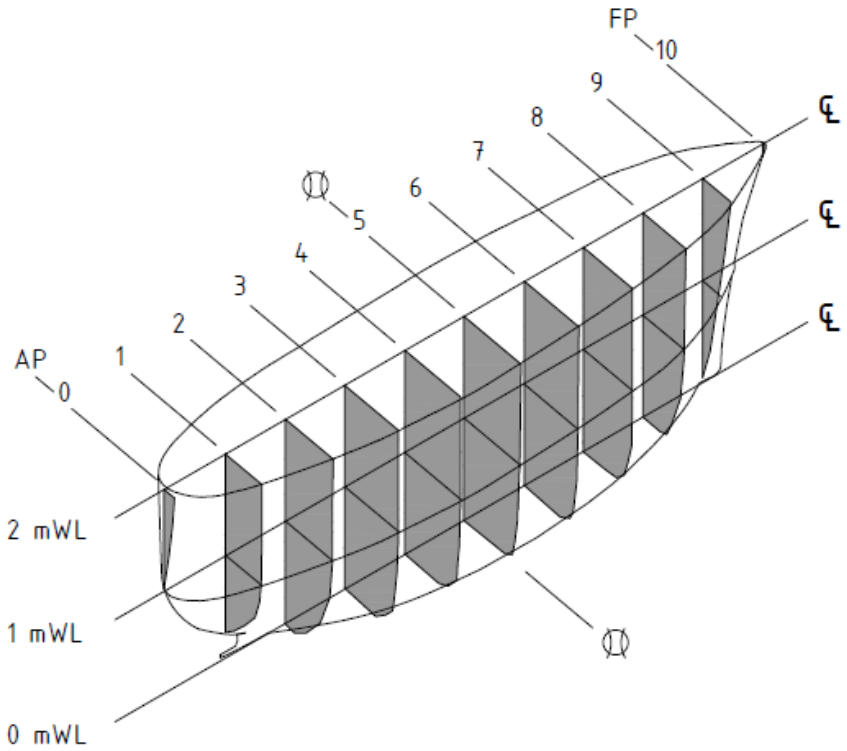


Figure 4.2.10 – Cross-Sectional Areas (Isometric View)

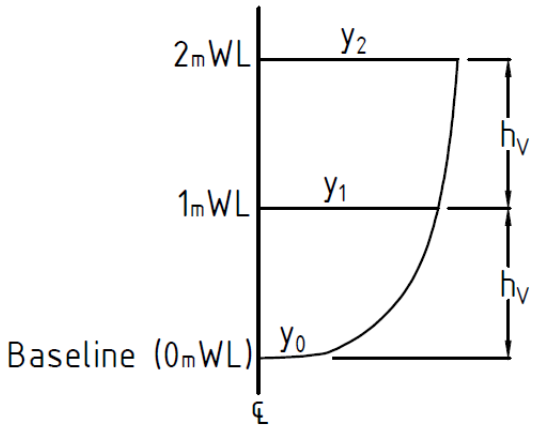


Figure 4.2.11 – Cross-Sectional Area (Each Section)

a	b	c	d = bc
WL	A	SM	AREA PRODUCT
0	y_0	1	$1y_0$
1	y_1	4	$4y_1$
2	y_2	1	$1y_2$
			$\sum AREA\ PROD$

Table 4.2.9 – Table for Calculation of Cross-Sectional Area

Final Calculation:

$$\text{CSA} = \frac{h_V}{3}\sum AREA\ PROD$$

Repeated above calculation for each section and the present the results so obtained as a Cross-Sectional Area curve up to that draught as shown in Figures 4.2.12 and 4.2.13.

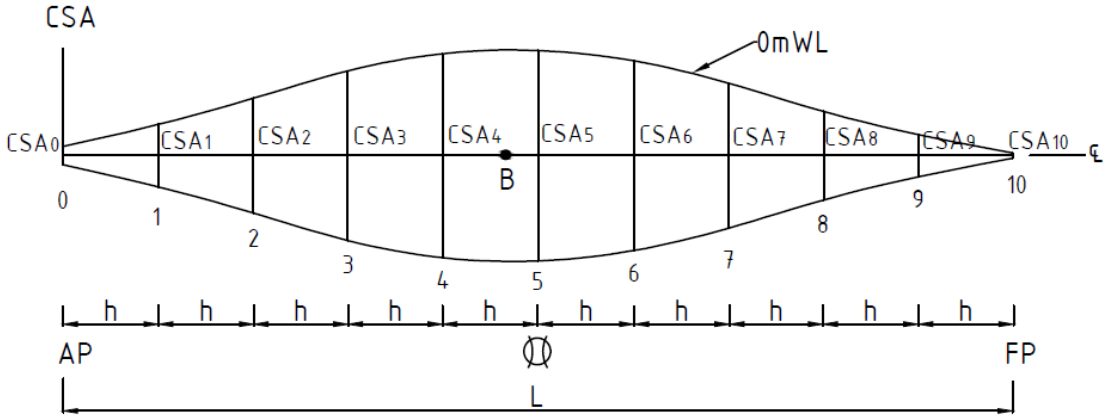


Figure 4.2.12 – Cross-Sectional Area Curve - Symmetrical View

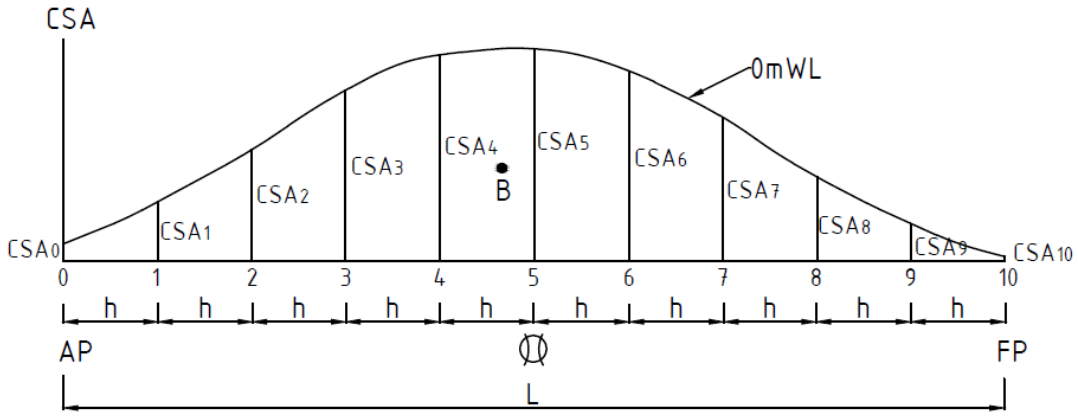


Figure 4.2.13 – Cross-Sectional Area Curve - Conventional View

4.4.2.8 Calculation of Longitudinal Centre of Buoyancy with Reference to Midship (LCB_{MS})

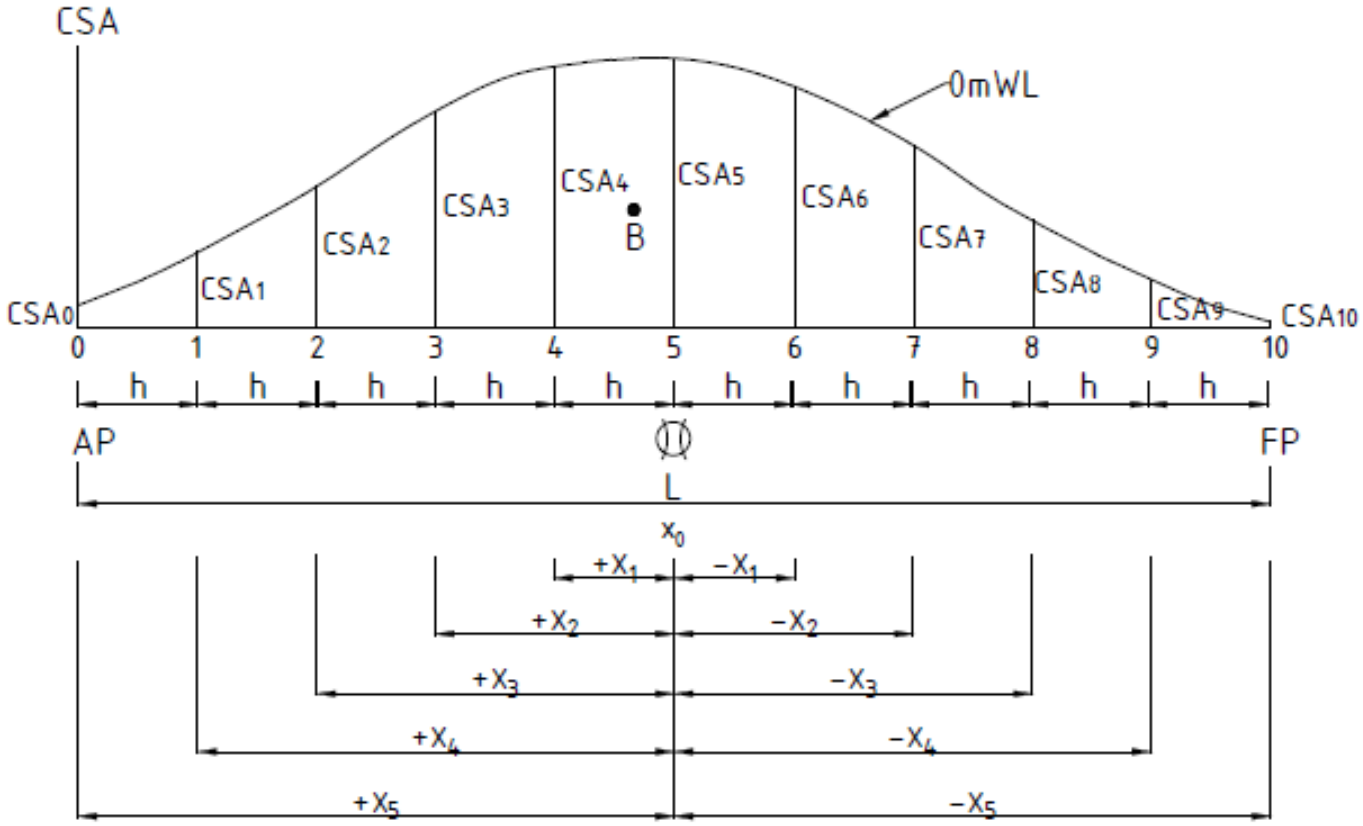


Figure 4.2.14 – Cross-Sectional Area Curve – Midship Reference

a	b	c			d = bc	Reference - Midship	
						e	f = de
SECTION	CSA	SM			VOLUME PRODUCT	LEVER	FM_{MS} PRODUCT
		Sets		Final			
AP	CSA_0	1		1	$1CSA_0$	$5h_L$	$5h_LCSA_0$
1	CSA_1	4		4	$4CSA_1$	$4h_L$	$16h_LCSA_1$
2	CSA_2	1	1	2	$2CSA_2$	$3h_L$	$6h_LCSA_2$
3	CSA_3		4	4	$4CSA_3$	$2h_L$	$8h_LCSA_3$
4	CSA_4	1	1	2	$2CSA_4$	$1h_L$	$2h_LCSA_4$
5	CSA_5	4		4	$4CSA_5$	$0h_L$	$0h_LCSA_5$
6	CSA_6	1	1	2	$2CSA_6$	$-1h_L$	$-2h_LCSA_6$
7	CSA_7		4	4	$4CSA_7$	$-2h_L$	$-8h_LCSA_7$
8	CSA_8	1	1	2	$2CSA_8$	$-3h_L$	$-6h_LCSA_8$
9	CSA_9	4		4	$4CSA_9$	$-4h_L$	$-16h_LCSA_9$
FP	CSA_{10}	1		1	$1CSA_{10}$	$-5h_L$	$-5h_LCSA_{10}$
					$\sum_{PROD}^{VOL}$		$\sum_{PROD}^{FM_{MS}}$

Table 4.2.10 – Table for Calculation of LCB with Reference to Midship

Final Calcuations:

$$\nabla = \frac{h_L}{3} \sum_{PROD}^{VOL}$$

$$LCB_{MS} = \frac{\frac{h_L}{3} \sum_{PROD}^{FM_{MS}}}{\frac{h_L}{3} \sum_{PROD}^{VOL}}$$

4.4.2.9 Calculation of Longitudinal Centre of Buoyancy (LCB_{MS}) by first Referencing to AP

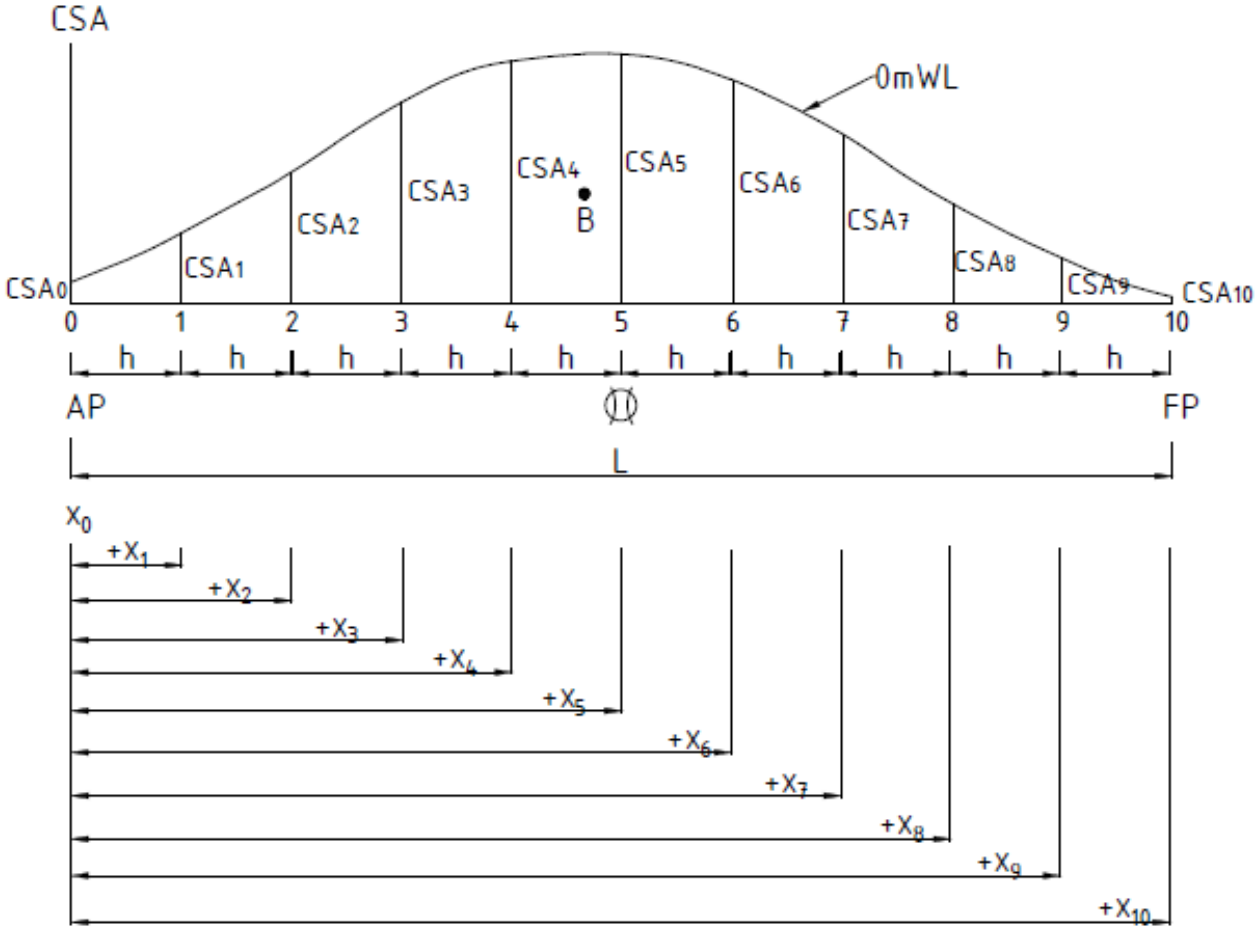


Figure 4.2.15 – Cross-Sectional Area Curve – AP Reference

a	b	c			d = bc	Reference - AP	
						e	f = de
SECTION	CSA	SM			VOLUME PRODUCT	LEVER	FM_{AP} PRODUCT
		Sets		Final			
AP	CSA_0	1		1	$1CSA_0$	$0h_L$	$0h_LCSA_0$
1	CSA_1	4		4	$4CSA_1$	$1h_L$	$4h_LCSA_1$
2	CSA_2	1	1	2	$2CSA_2$	$2h_L$	$4h_LCSA_2$
3	CSA_3		4	4	$4CSA_3$	$3h_L$	$12h_LCSA_3$
4	CSA_4	1	1	2	$2CSA_4$	$4h_L$	$8h_LCSA_4$
5	CSA_5	4		4	$4CSA_5$	$5h_L$	$20h_LCSA_5$
6	CSA_6	1	1	2	$2CSA_6$	$6h_L$	$12h_LCSA_6$
7	CSA_7		4	4	$4CSA_7$	$7h_L$	$28h_LCSA_7$
8	CSA_8	1	1	2	$2CSA_8$	$8h_L$	$16h_LCSA_8$
9	CSA_9	4		4	$4CSA_9$	$9h_L$	$36h_LCSA_9$
FP	CSA_{10}	1		1	$1CSA_{10}$	$10h_L$	$10h_LCSA_{100}$
					$\sum_{PROD} VOL$		$\sum_{PROD} FM_{AP}$

Table 4.2.12 – Table for Calculation of LCB by First Referencing to AP

$$\nabla = \frac{h_L}{3} \sum_{PROD} VOL$$

$$LCB_{AP} = \frac{\frac{h_L}{3} \sum_{PROD} FM_{AP}}{\frac{h_L}{3} \sum_{PROD} VOL}$$

$$LCB_{MS} = \frac{L}{2} \pm LCB_{AP}$$

4.2.3 By Approximate Formulae

Vertical centre of buoyancy:		
1	$KB = 0.535d$	
2	$KB = \frac{d}{1 + {C_B}/{C_W}}$	
3	$KB = \frac{d}{2}$	For box-shaped vessels
4	$KB = \frac{2}{3}d$	For triangular-shaped vessels
5	$KB = 0.7d$	For yachts
6	$KB = \frac{a}{\Delta}$	Where (a) is area above displacement curve up to the respective draught
Block coefficient @ any draught ($d_?$)		
	$C_{B?} = C_{BSLWL} \times \left(\frac{d_?}{d_{SLWL}}\right)^x$	Where: $? = Any\ draught$ $x = 4.5 \times e^{-5 \times C_{BSLWL}}$ $e = 2.718$
Waterplane area coefficient @ SLWL		
1	$C_{WSLWL} = C_{BSLWL} + K$	Where: $K = \frac{1}{3}(1 - C_B)$ Or: 0.17 for Tugs 0.14 for Container ships 0.13 for Passenger liners 0.10 for General cargo ships 0.70 for Oil tankers 0.60 for VLCCs 0.50 for ULCCs
2	$C_{WSLWL} = \left(\frac{2}{3} \times C_{BSLWL}\right) + \frac{1}{3}$	
BM_T		
	$BM_T = \frac{\eta \times B^2}{d \times C_B}$	Where (η) is referred to as "Inertia Coefficient" and for Merchant ships with C_W of 0.692-0.893: $\eta = 0.084 \times (C_W)^2$

Moment to change trim a centimetre		
	$MCT1_{cm} = \frac{k \times TPC^2}{B}$	Where: k = 7.31 for General Cargo ships k = 7.80 for Oil Tankers
BM_L		
1	$BM_L = \frac{\eta \times L^2}{d \times C_B}$	Where for Merchant ships with C_W of 0.692-0.893: $\eta = \frac{3}{40} \times (C_W)^2$
2	$BM_L = \frac{1377 \times (C_W)^2}{d \times C_B}$	
Comparison between BM_T & BM_L		
	$\frac{BM_T}{BM_L} = 1.12 \times \left(\frac{B}{L}\right)^2$	
Comparison between d, Δ, C_B & C_W		
	$\frac{d_L}{d_{SLWL}} = \left(\frac{\Delta_L}{\Delta_{SLWL}}\right)^{C_B/C_W}$	

Table 4.5.1 – Approximate Methods of Calculation

4.3 Summary

4.3.1 Hydrostatic Data

With exception of cross-sectional areas, on completion of the calculation of areas, volumes, centroids and other properties of a ship's hull form, the results so obtained for various draughts up to the LWL are then presented in a table called the *Hydrostatic Data*. Table 4.3.1 below is the Hydrostatic Data for the vessel M/V NAUTICO.

Draught (m)	A_W	LCF_M	TCF_{CL}	I_M	I_{LCF}	I_{CL}	V	KB	BM_T	KM_T	BM_L	KM_L	LCB_M	C_B	C_M	C_W	C_P	C_{PV}	S	$MCT1_{CM}$
	m^2	m (+=A)	m (+=P)	m^4	m^4	m^4	m^3	m	m	m	m	m	m (+=A)						m^2	T.m
10	2345	6.51	0.00	3389906	3290472	59062	19110	5.44	3.09	8.53	172.18	177.62	7.33	0.670	0.979	0.822	0.684	0.910	2234	224.85
9	2275	6.55	0.00	3137704	3040010	56513														
8	2197	6.12	0.00	2867288	2784884	53586	14563	4.32	3.68	8.00	191.23	195.55	5.57	0.638	0.974	0.770	0.655	0.867	2079	190.30
7 (LWL)	2099	5.24	0.00	2523994	2466353	50425														
6	2017	4.07	0.00	2260013	2226544	47823	10360	3.23	4.62	7.85	214.92	218.15	3.60	0.605	0.966	0.707	0.627	0.822	1920	152.15
5	1941	3.19	0.00	2049525	2029824	45203														
4	1881	2.33	0.00	1915763	1905565	42722	6473	2.16	6.60	8.77	294.41	296.57	1.92	0.567	0.949	0.660	0.598	0.771	1747	130.21
3	1809	1.70	0.00	1790325	1785072	39352														
2	1698	0.82	0.00	1641769	1640617	33896	2867	1.09	11.82	12.91	572.16	573.25	0.13	0.503	0.904	0.595	0.556	0.683	1498	112.11
1	1502	-0.03	0.00	1364456	1364455	25934														
0 m (BL)	895	-0.70	0.00	463500	463057	9574	0	0.00	∞	∞	∞	∞	0.00			0.314			0	0.00

Table 4.3.1 – HYDROSTATIC DATA

The data may also be presented diagrammatically in the form of curves. It is then referred to as *Hydrostatic Curves*. They are plotted against a common draught placed vertically. Each curve have a specific scale placed horizontally because of the wide variation in the units in which the data is measured. Taking readings from the curves therefore requires to be done with care to avoid mistakes. Figure 4.3.1 is the set of Hydrostatic Curves for M/V NAUTICO.

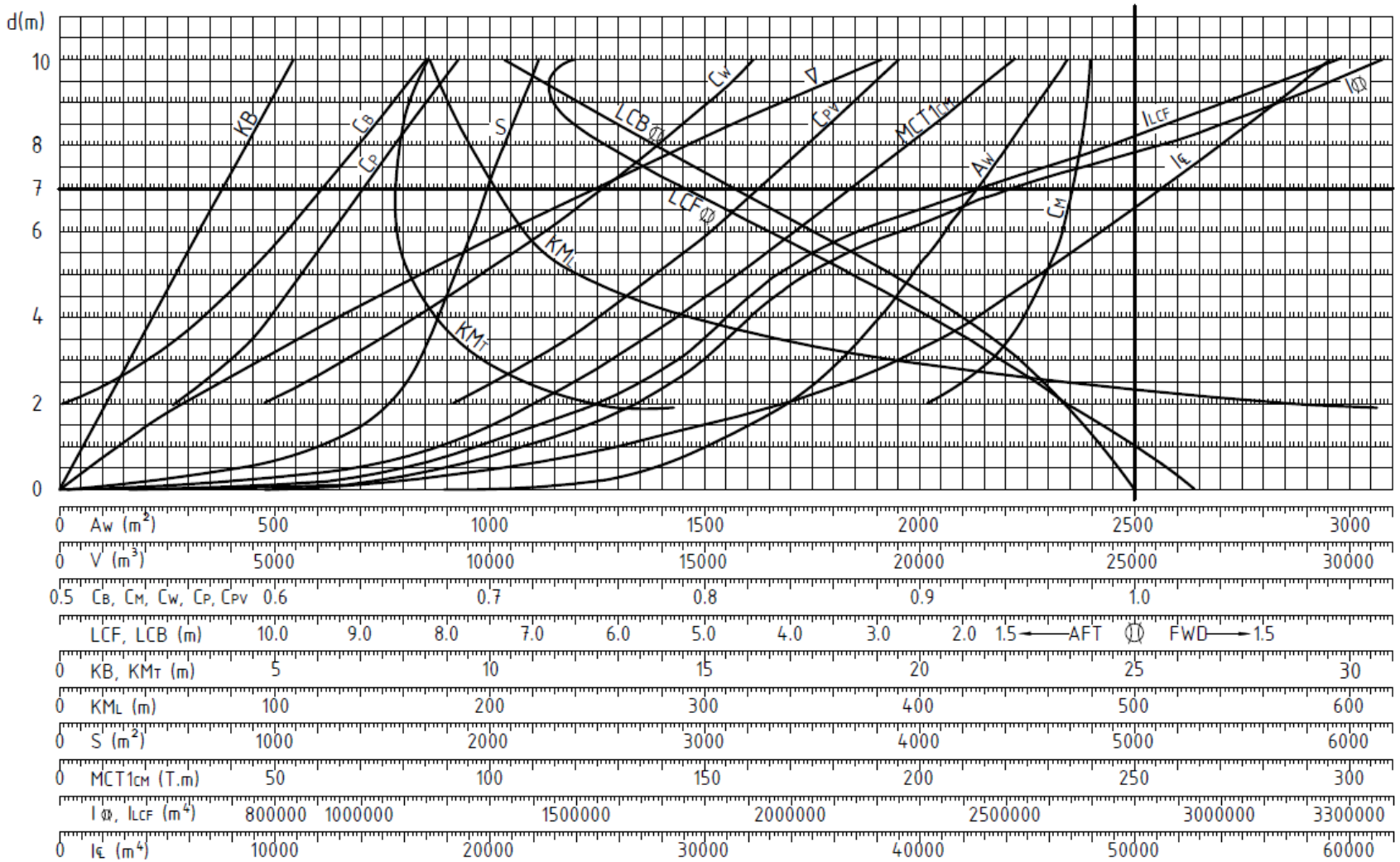


Figure 4.3.1 – Hydrostatic Curves

4.3.3 Cross-Sectional Area Curves

For every draught, cross-sectional areas are calculated for each section from aft to fore as shown in Figure 4.3.2. After that a set of curves for various waterlines up to the LWL are then plotted to produce the Cross-Sectional Area Curves shown diagrammatically in Figure 4.3.3.

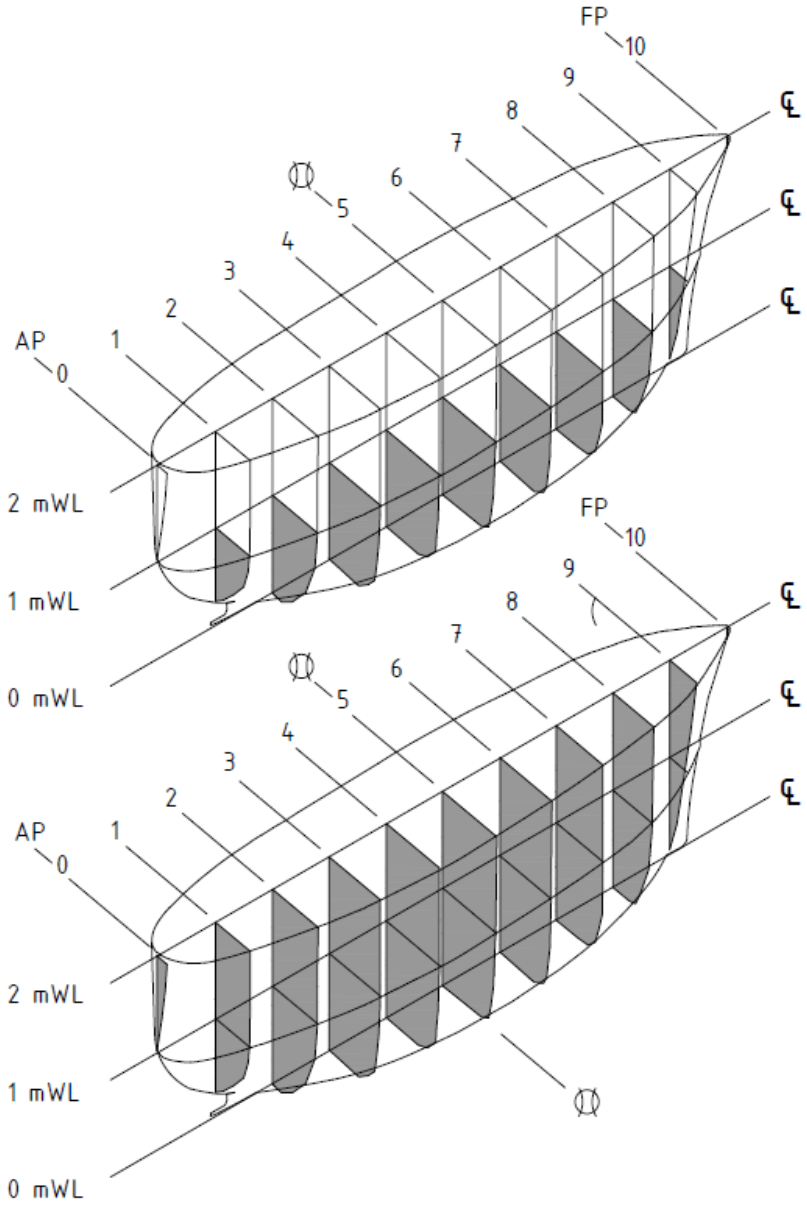


Figure 4.3.2 – Cross-Sectional Areas

CROSS SECTIONAL AREAS (CSA) (m^2)													
Draught (m)	AP	½	1	2	3	4	5	6	7	8	9	9½	FP
10	15	46	78	141	176	185	186	181	155	103	47	27	8
9													
8	1	24	49	104	138	147	148	143	120	77	34	19	7
7													
6	0	11	27	70	100	109	110	106	86	53	24	14	6
5													
4	0	6	14	40	63	71	72	69	53	32	15	9	5
3													
2	0	2	5	14	28	33	34	32	24	13	6	4	3
1													
0 (BL)	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 4.3.3 – Table of Cross-Sectional Areas

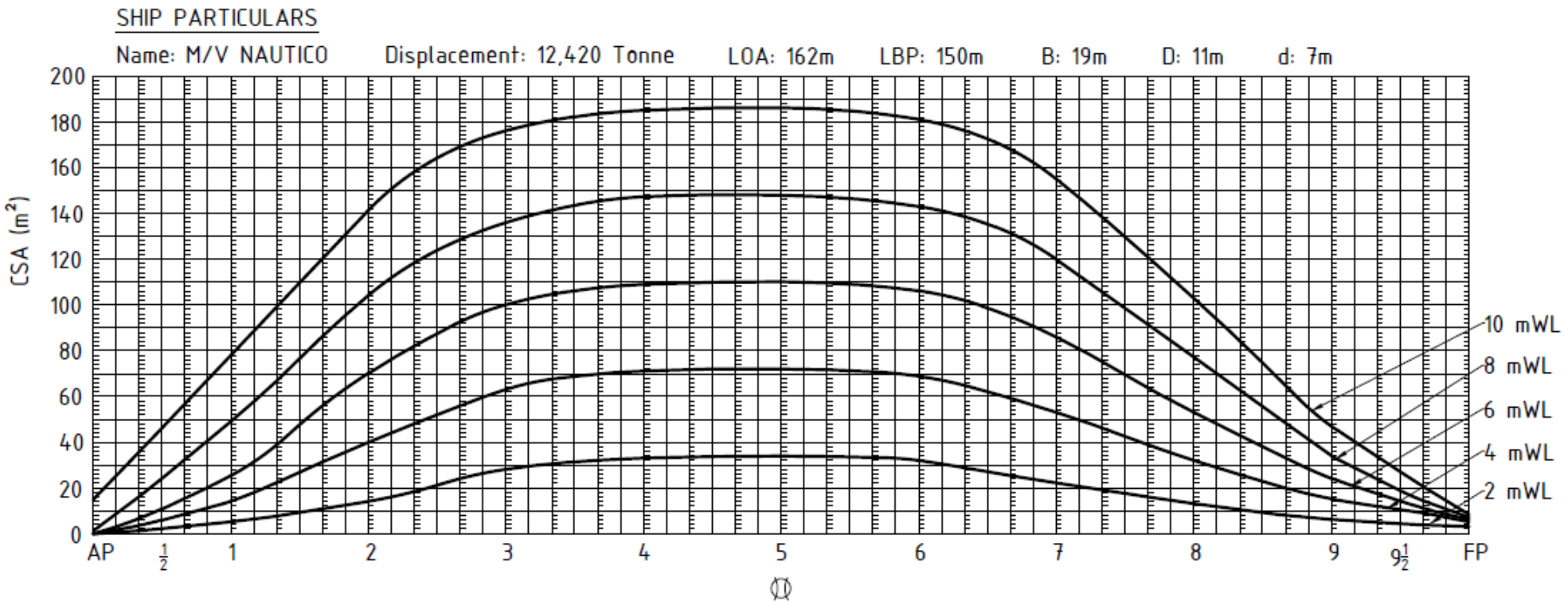


Figure 4.3.3 – Cross-Sectional Areas Curves

4.4 BONJEAN CURVES

These are curves of cros-sectional areas plotted against draught and drawn on the sheer profile of a vessel. It enables volumes of displacement to be calculated for waterlines that are not parallel to the baseline.

To develop the Bonjean Curves, draw the sheer profile of the vessel to scale as in Figure 4.4.1(a).

From the CSA curves in Figure 4.3.2, tabulate the values of CSAs at all waterlines from the baseline to the deck for every section, starting from the AP, and use the data collected to plot curves as in Figure 4.4.1(b).

Next superimpose the curves so obtained in b) on the sheer profile a) to obtain the final set of curves in c) called the Bonjean Curves.

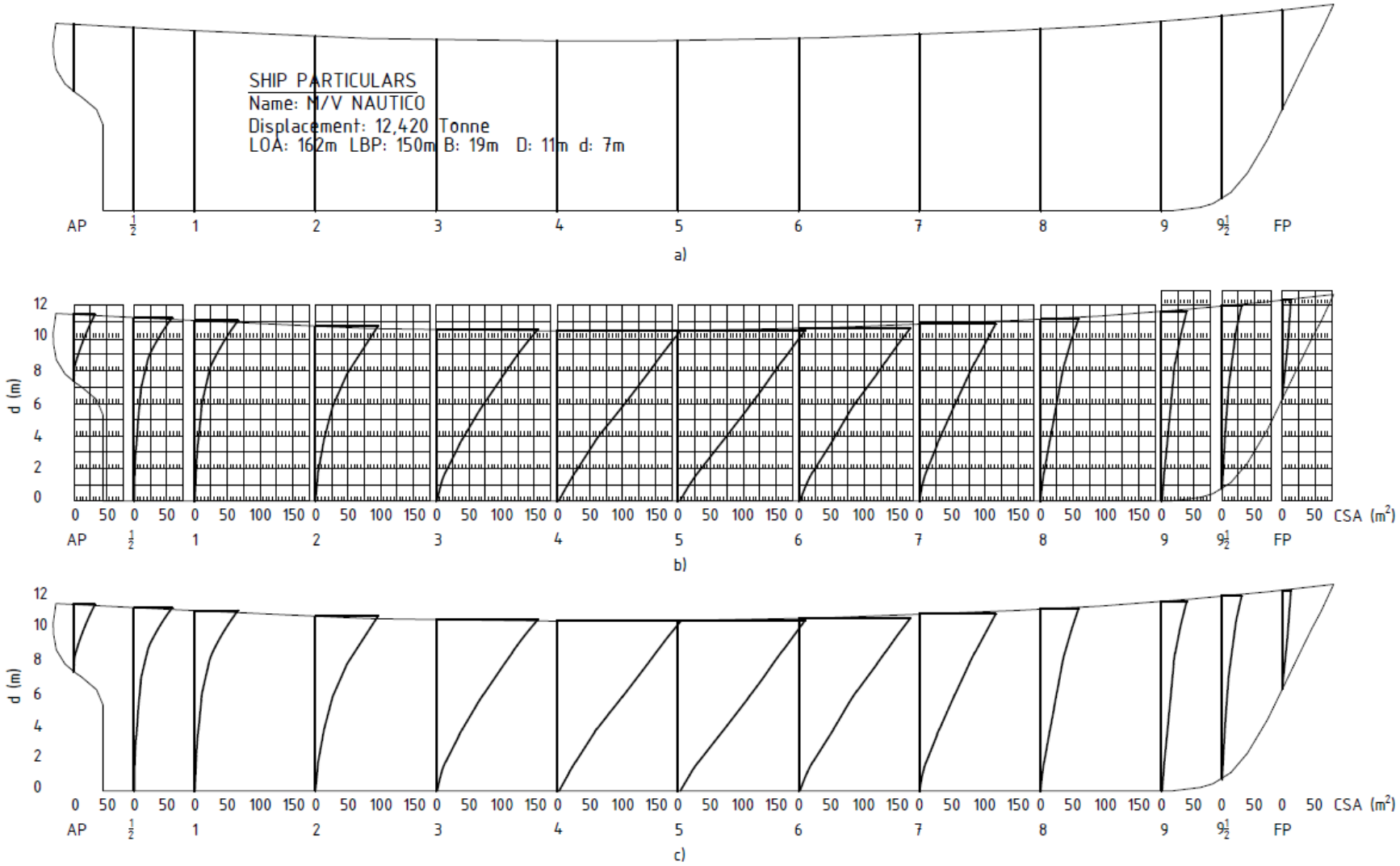


Figure 4.4.1 – BONJEAN Curves of Areas

Example

A ship of length 150 m is floating at waterline WL as in Figure 4.7.3.

From its Bonjean Curves the cross-sectional areas at that waterline are:

Station	AP	½	1	2	3	4	5	6	7	8	9	9½	FP
CSA (m^2)	23	52	80	121	139	130	112	90	58	28	8	4	0

Determine the volume displaced and the position of its longitudinal centre of buoyancy from midship.

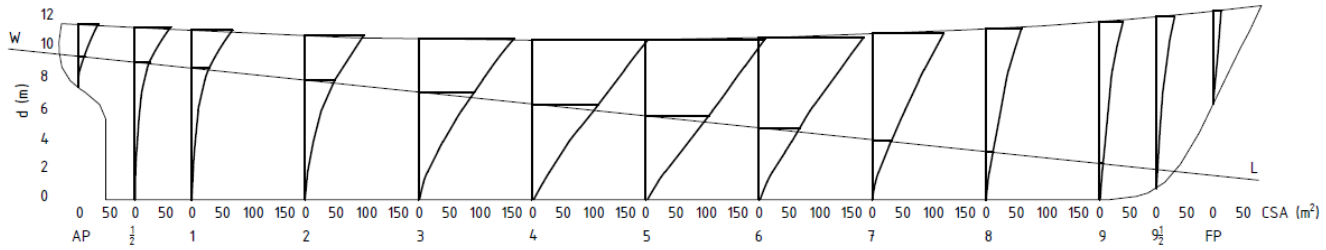


Figure 4.4.2

Solution

Using Simpson's First Rule compute the products for volume of displacement and longitudinal centre of buoyancy as in Table 4.7.1.

SECTION	CSA	SM			VOLUME PRODUCT	LEVER	FM_{MS} PRODUCT
		Sets		Final			
AP	23	½		½	11.5	$5h_L$	$57.5h_L$
½	52	2		2	104.0	$4½h_L$	$468.0h_L$
1	80	½	1	1½	120.0	$4h_L$	$480.0h_L$
2	121		4	4	484.0	$3h_L$	$1452.0h_L$
3	139	1	1	2	278.0	$2h_L$	$556.0h_L$
4	130	4		4	520.0	$1h_L$	$520.0h_L$
5	112	1	1	2	224.0	$0h_L$	**$3533.5h_L$**
6	90		4	4	360.0	$-1h_L$	$-360.0h_L$
7	58	1	1	2	116.0	$-2h_L$	$-232.0h_L$
8	28	4		4	112.0	$-3h_L$	$336.0h_L$
9	8	1	½	1½	12.0	$-4h_L$	$48.0h_L$
9½	4		2	2	8.0	$-9½h_L$	$76.0h_L$
FP	0		½	1	0.0	$-5h_L$	0
					2349.5		**$-1052.0h_L$**

Table 4.4.1

$$h_L = \frac{150}{10} = 15 \text{ m}$$

$$V = \frac{15}{3} \times 2349.5 = \underline{11747.5\ m^2}$$

$$LCB_{MS} = \frac{\left[\left(\frac{15}{3}\right)^2 \times (3533.5 - 1052)\right]}{11747.5} = \underline{5.28\ m\ A}$$

5–FLOTATION AND RESERVE BUOYANCY

5.1 FLOTATION AND SURVIVABILITY

A ship at sea is subjected to external loads due to vagaries of weather caused by wind and waves. This results in heavy rolling, pitching and heaving motions. In addition to these, loads may also be generated internally to affect a ship's situation. To survive and remain floating a ship must therefore have adequate buoyancy at all times to combat the constantly changing weather. To achieve this let us first examine the nature of the forces that tend to sink a ship at sea.

5.1.1 Effect of External Downward Forces

Consider a vessel floating initially in equilibrium vertically under the influence of its displacement Δ acting downwards and buoyancy B acting upwards. Under this condition the Principle of Flotation is satisfied and:

$$\Delta = B$$

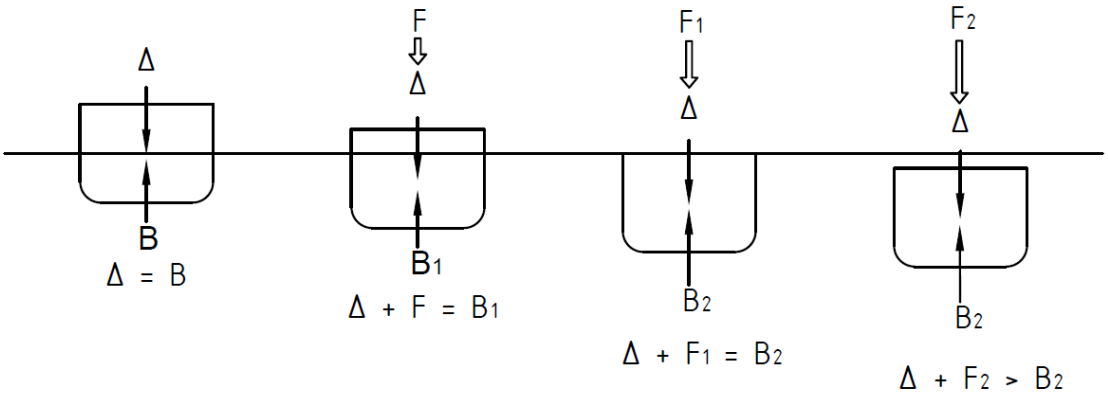


Figure 5.1 - Effect of External Downward Forces

When now the vessel is pushed down by an external force F, for it to continue floating, buoyancy must have to be increased to B_1 to give:

$$\Delta + F = B_1$$

Increase the force to F_1 and for the vessel to remain floating buoyancy must again increase to the maximum B_2 to give:

$$\Delta + F_1 = B_2$$

The vessel is now at the limit of floating, hence if the force should be increased further to F_2 it has a situation where:

$$\Delta + F_2 > B_2$$

At this stage the vessel will sink. It is said to have gone beyond the limit of survivability because the deck has submerged. If therefore a vessel is initially loaded as in the third case to a displacement Δ_1 instead of Δ, the situation is as follows:

$$\Delta_1 = \Delta + F_2 = B_2$$

It will then have no reserve to fall on when external forces act and she will sink on the first impact.

5.1.2 Effect of Symmetrical Flooding

Flooding may be caused through burst piping or breaching of the hull in one way or the other. This apparently results in increase in displacement which if not controlled will result in a behavior similar to that caused by external downward forces illustrated earlier and sinking.

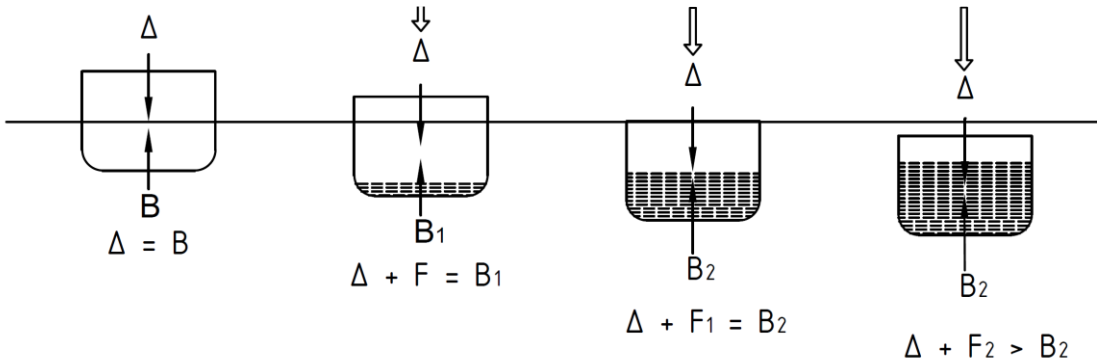


Figure 5.2 - Effect of Symmetrical Flooding

5.1.3 Effect of External Transverse Moments

The actions of external forces may act in the form of moments which tend to roll a ship over to port or to starboard. As soon as the deck edge becomes submerged it will have gone beyond the limit of survivability and will sink. The exact moment that occurs will therefore depend upon the height of deck above water initially, or the reserve buoyancy.

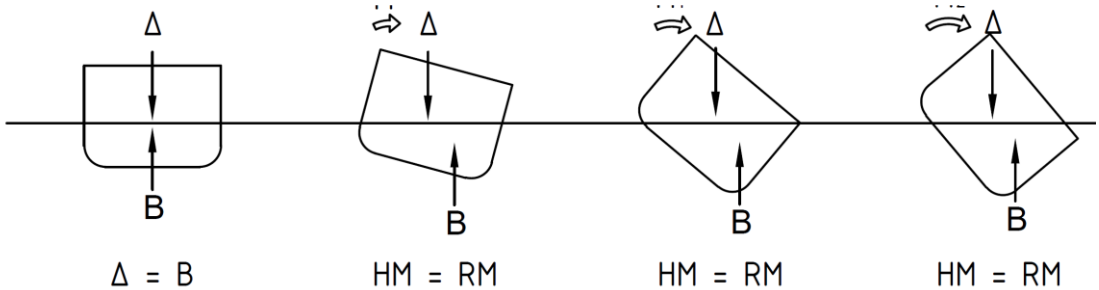


Figure 5.3 - Effect of External Transverse Moments

5.1.4 Effect of Unsymmetrical Flooding

When flooding is not central moments are generated to roll the ship over to port or to starboard. The result in a situation similar to moments caused by external forces.

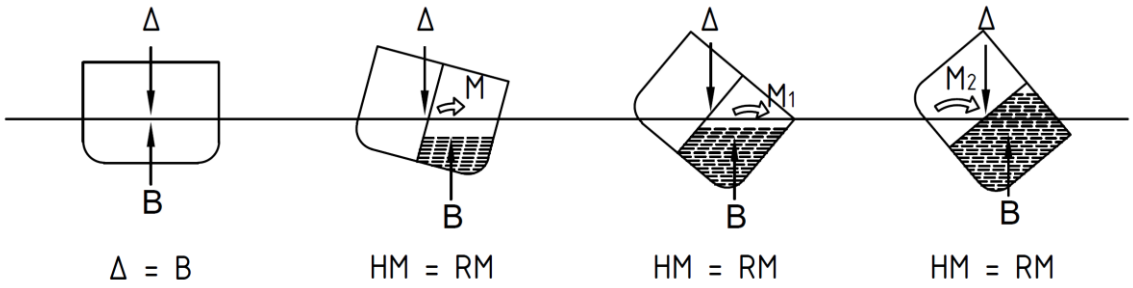


Figure 5.4 - Effect of Unsymmetrical Flooding

5.1.5 Effect of External Longitudinal Moments

This is the situation when moments due to the external forces act longitudinally. The vessel trims seriously to fore or to aft and may go beyond the limit of survivability and sink.

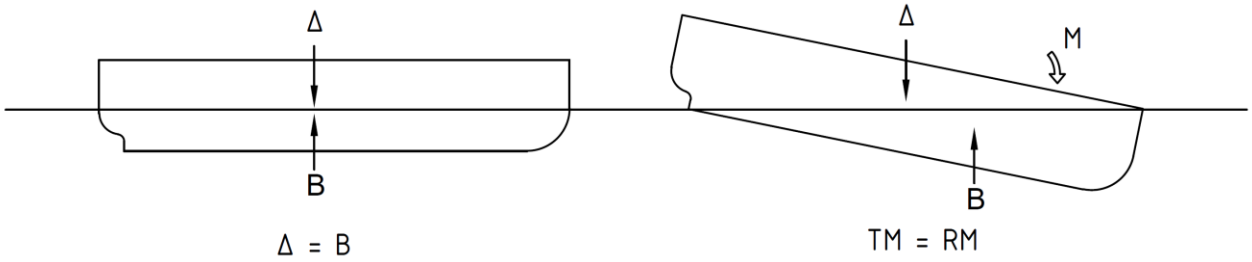


Figure 5.5 - Effect of External Longitudinal Moments

5.1.6 Effect of Fore/Aft Flooding

Similar to longitudinal moments caused by external forces, unsymmetrical flooding in the fore-aft direction may also cause a vessel to be lost if enough buoyancy is not maintained initially.

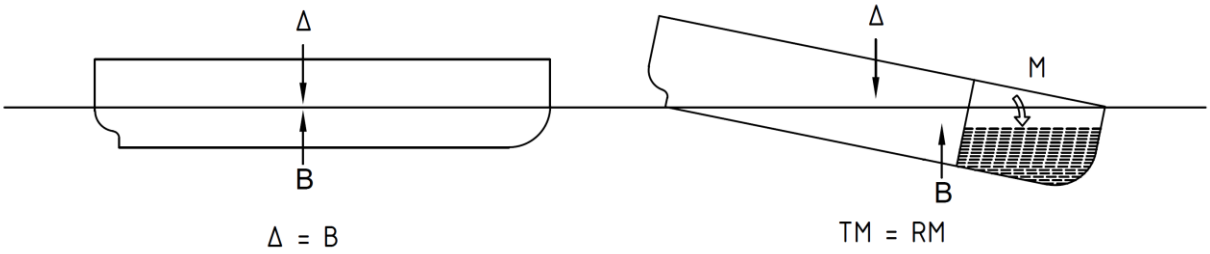


Figure 5.6 - Effect of Fore/Aft Flooding

5.2 FREEBOARD

To prevent the occurrence of any of the situations described in paragraph 6.1 from occurring. It is good practice for ships to maintain reserve buoyancy when fully loaded. It should have been expected therefore that the survival instincts of human beings should be enough to ensure self-regulation when it comes to the maintenance of reserve buoyancy. Unfortunately due to greed that has most of the time not being the case. Especially since ship owning has largely been separated from ship operation by captains and crew. Reserve buoyancy is therefore enforced by law worldwide through the provisions of The International Convention on Load Lines, 1966. It is measured in

terms of Freeboard, i.e. the distance from the deck line at the side to the draught line when a ship is loaded. The markings indicate the minimum freeboard which occurs under fully loaded conditions.

5.2.1 Load Line Marks

The International Convention on Load Lines requires that no ship to which it applies shall proceed to sea on an international voyage unless it has been surveyed, marked and provided with an International Load Line Certificate (1966), or where appropriate, an International Load Line Exemption Certificate.

The load lines and letters shall be painted in white or yellow on a dark background or in black on a light background. They shall also be permanently marked, usually embossed, on both sides of the ship amidships.

Because of wide variations in weather conditions and densities of water worldwide different freeboard markings are used for different waters, appropriately referred to as "zones", in accordance with the following criteria:

Zone	Criteria	How Assigned
S - Summer	Areas with not more than 10 percent winds of force 8 Beaufort (34 knots) or more.	Obtained from Freeboard Table, Reg. 28.
T – Tropical	Areas with not more than 1 percent winds of force 8 Beaufort (34 knots) or more. Not more than one tropical storm in 10 years in an area of 5º square in any one separate calendar month.	$S - \frac{1}{48} \times d_{SM}$
W – Winter	Areas within the Summer zone which seasonally experience weather conditions that are more severe than what usually pertains in the Summer zone.	$S + \frac{1}{48} \times d_{SM}$
WNA – Winter North Atlantic	Special case of Winter zone limited to the North Atlantic area only, for which the seasonal winter weather is longer than the general Atlantic area.	
	For ships of 100 m length or more.	$W + 50\ mm$
	For ships of less than 100 m length	$W + y$ (Where y ≥ 50 mm)
F – Fresh Water	For loading in fresh water in the Summer zone	$S - \frac{\Delta_S}{400TPC_S} mm$
TF – Tropical Fresh Water	For loading in fresh water in the Tropical zone	$T - \frac{\Delta_S}{400TPC_S} mm$

d_{SM} = Summer Moulded Draught
Δ_S = Displacement in Salt Water

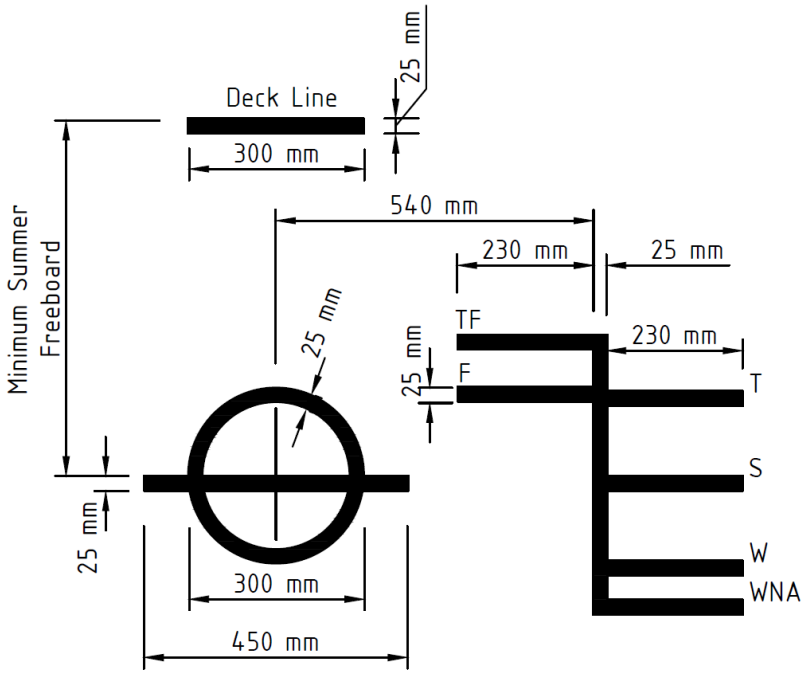


Figure 5.7 – Load Line Mark and lines to be used with this mark

5.3 Tonne Per Centimetre Immersion

A useful means of approximating the amounts of load added to or removed from a ship is the use of Tonne Per Centimetre (TPC). TPC is the increase in displacement per unit increase in mean draught.

Consider a vessel initially displacing a mass Δ and volume ∇. By the adding mass (m), its displacement and volume will increase to Δ_1 and volume ∇_1 respectively. This is equivalent to an increase in mean draught (s), referred to as sinkage.

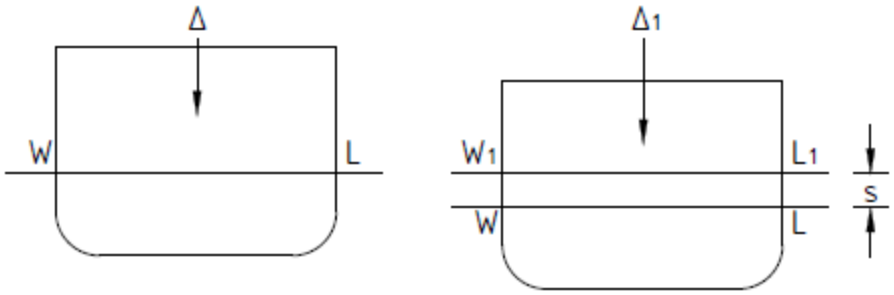


Figure 5.8 – Sinkage due to Added Mass

On assumption that the change in waterplane area from waterline WL to W_1L_1 is negligible, then:

$$\Delta_1 - \Delta = A_W \times s \times \rho$$

Or it can be said that $(\Delta_1 - \Delta)$ tonnes is required to increase or decrease draught by (s) metres.

Therefore for a change in mean draught by one metre, mass in tonnes required will be:

$$(A_W \times 1 \times \rho)$$

Or:

$$TPM = A_W \rho$$

With reference to a centimetre change this will then be referred as Tonne Per Centimetre immersion:

$$TPC = \frac{A_W \rho}{100}$$

TPC is quoted usually without dimensions.

A curve of TPC plotted against draught will have the same form as the waterplane area curve. TPC may therefore be obtained by extracting from the A_W curve and multiplying the results by the density.

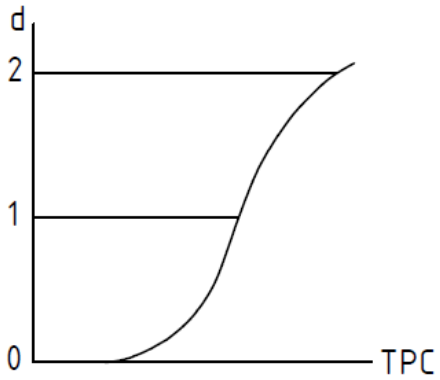


Figure 5.9 – TPC Curve

5.4 Fresh Water Allowance

When a ship sails from sea water of density 1.025 tonne/m^3 into fresh water of lower density without change in displacement, then as a result of the difference in densities there will be change in volume of displacement.

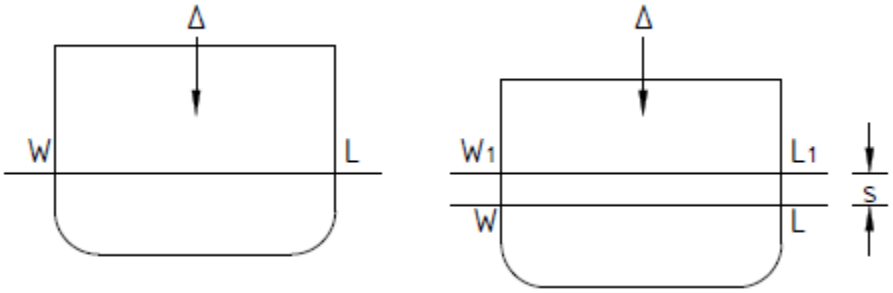


Figure 5.10 – Sinkage due to Change in Density

Assuming the A_W remains constant and representing densities of sea water and fresh water by (ρ_S) and (ρ_F) respectively, then:

$$s = \frac{\nabla_F - \nabla_S}{A_W} = \frac{\frac{\Delta}{\rho_F} - \frac{\Delta}{\rho_S}}{A_W}$$

Therefore

$$s = \frac{\Delta}{A_W}\left(\frac{1}{\rho_F} - \frac{1}{\rho_S}\right) = \frac{\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \text{ m}$$

Or

$$s = \frac{100\Delta}{A_W}\left(\frac{\rho_S - \rho_F}{\rho_F \rho_S}\right) \text{ cm}$$

But

$$\frac{100}{A_W \rho_S} = \frac{1}{TPC_S}$$

Therefore

$$s = \frac{\Delta}{TPC_S}\left(\frac{\rho_S - \rho_F}{\rho_F}\right) \text{ cm}$$

For the special case of sailing into pure fresh water of ρ_F = 1.000 tonne/m^3

$$s = \frac{\Delta}{TPC_S}\left(\frac{1.025 - 1.000}{1.000}\right) = \frac{\Delta}{TPC_S}\left(\frac{0.025}{1.000}\right) = \frac{\Delta}{40TPC_S} \text{ cm}$$

This change in density is referred to as the Fresh Water Allowance (FWA). i.e.:

$$FWA = \frac{\Delta}{40TPC_S} \text{ cm}$$

5.5 Deadweight Scale

The deadweight scale is as drawing that provides a method for estimating the additional draft or for determining the extra load that can be taken onboard when a vessel is being loaded in water of density less than that of salt water. The following scale displays readings for freeboard, deadweight in salt water and in fresh water, draft of the ship (mean), displacement in salt water and in fresh water, tonnes per cm (TPC) in salt water and in fresh water, and moment to change trim 1 cm (MCTC).

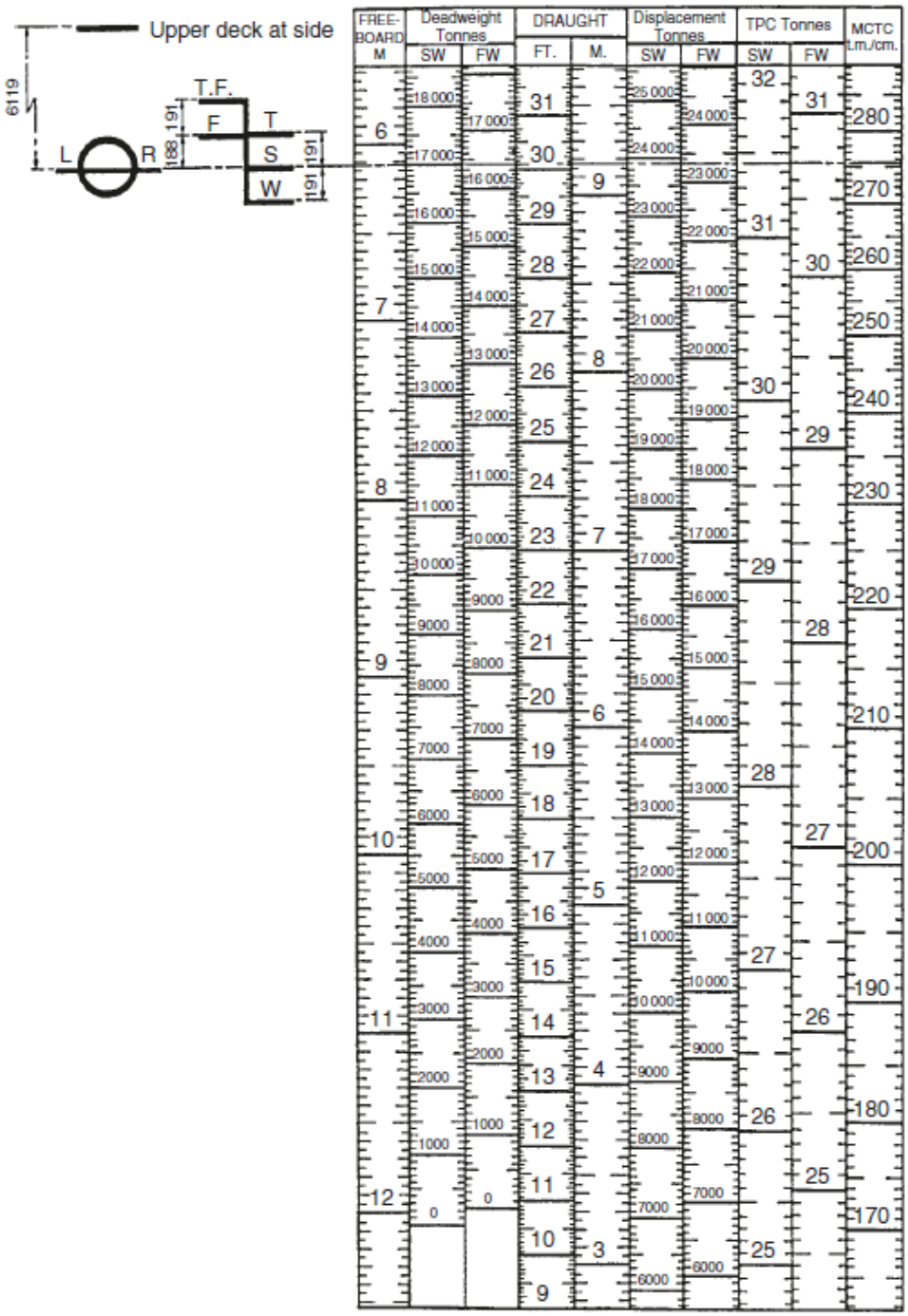


Figure 5.11 - Deadweight Scale

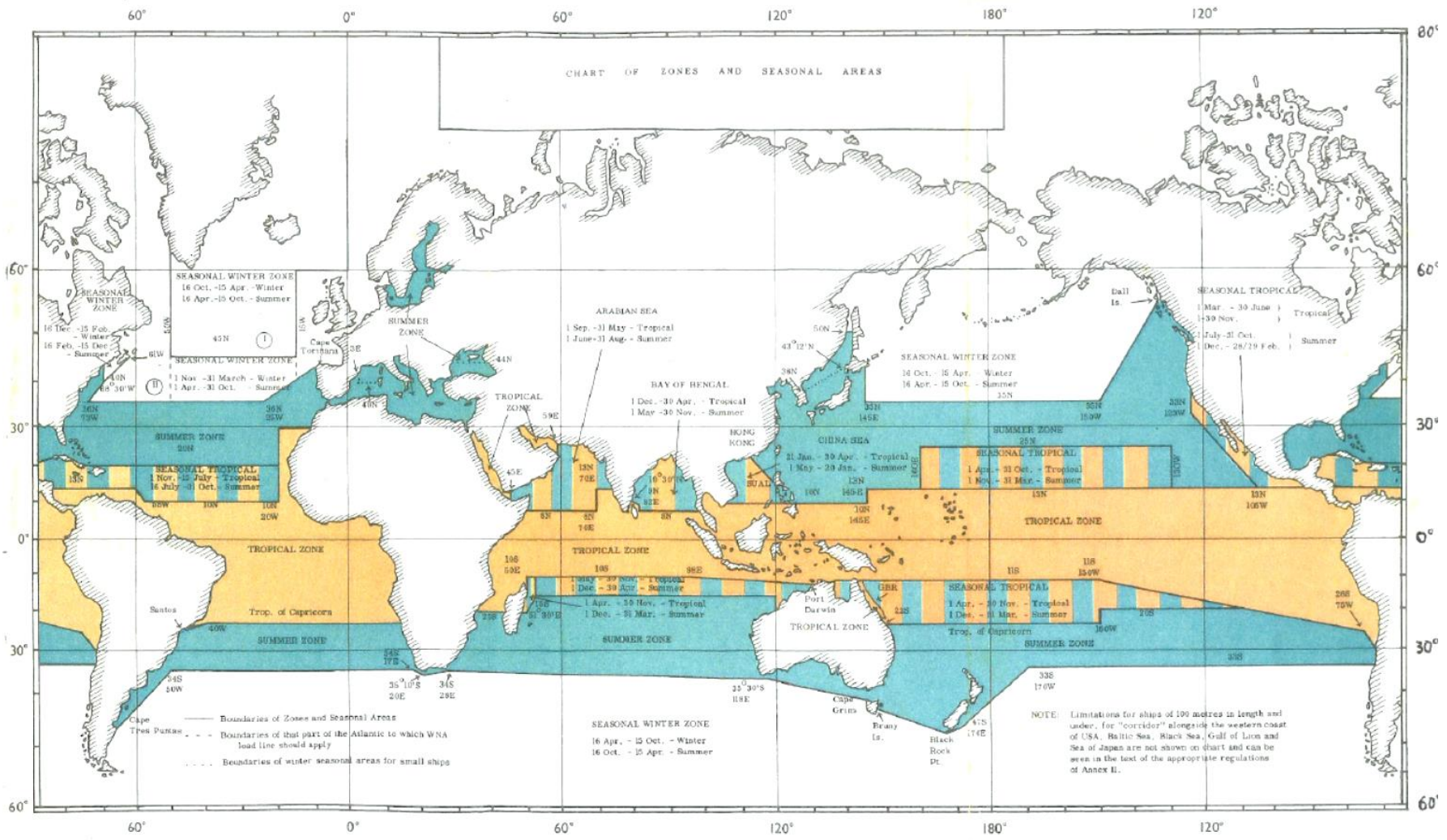


Figure 5.12 – Chart of Zones and Seasonal Areas

6–LOADED CENTRE OF GRAVITY

6.1 GENERAL PRINCIPLES

6.1.1 Effect of Adding a Mass to a Body

Consider a homogenous bar of mass M to be initially supported at its centre of gravity (C of G), (G), in a horizontal posture as shown in Figure 6.1.

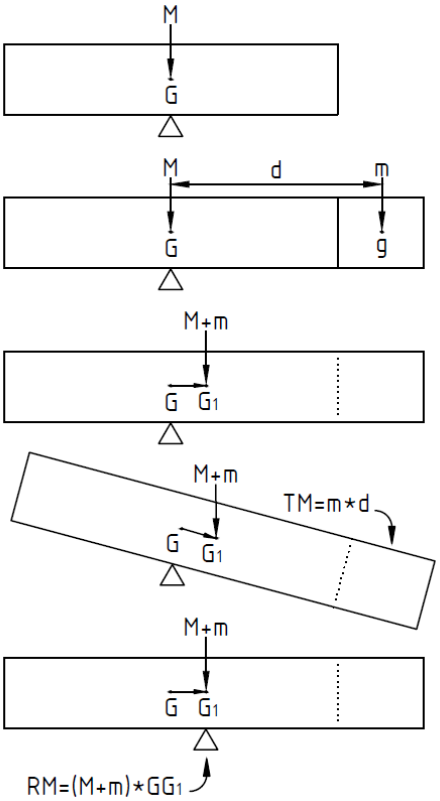


Figure 6.1 – Effect of Adding a Mass

If now a similar homogenous bar with mass (m) and C of G, (g) is attached at a distance (d) as shown, the result will be a new bar of mass (M+m) with C of G at (G_1).

With the new bar still supported at G, a Tilting Moment (TM) will come into action to rotate the bar clockwise.

To restore the new bar to the original horizontal condition, an equal but opposing Righting Moment (RM) will have to be applied at G_1.

Whereby: $$\mathrm{RM} = \mathrm{TM}$$

$$(\mathrm{M} + \mathrm{m}) * \mathrm{GG}_1 = \mathrm{m} * \mathrm{d}$$

Therefore the shift in C of G:

$$GG_1 = \frac{m*d}{M+m}$$

Summary

When a mass is *added* to a body the C of G of the body *shifts towards* the C of G of the *added mass* by an amount equal to the product of the mass added, and the distance between the C of Gs of the mass and the body, divided by the new total mass.

6.1.2 Effect of Removing a Mass from a Body

Consider this time that the mass is rather removed as shown in Figure 6.2. This will result in a new bar of mass (M-m) with C of G at (G_1).

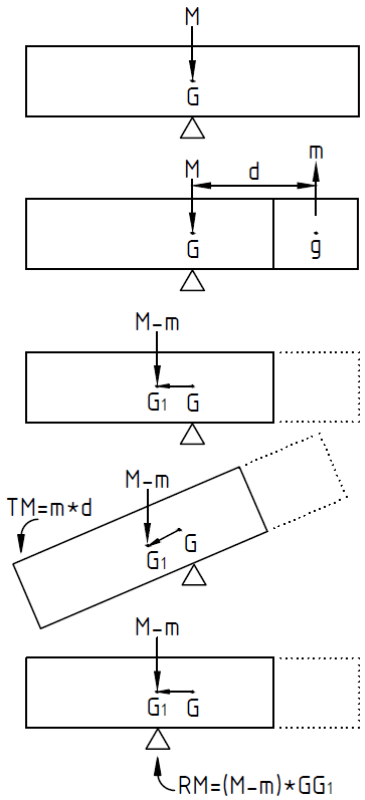


Figure 6.2 – Effect of Removing a Mass

With the new bar still supported at G, a TM will come into action to rotate the bar anticlockwise.

Again to restore the new bar to the original horizontal condition, an equal but opposing RM will have to be applied at G_1.

Now: $$RM = TM$$

$$(M - m) * GG_1 = m * d$$

Therefore the shift in C of G:

$$GG_1 = \frac{m * d}{M - m}$$

Summary

When a mass is *removed* from a body the C of G of the body *shifts away* from the C of G of the *removed mass* by an amount equal to the product of the mass removed and the distance between the C of Gs of the mass and the body, divided by the new total mass.

6.1.3 Effect of Shifting a Mass within a Body

Consider a body of total mass M to be composed of six small masses of sizes (m) each as shown in Figure 6.3.

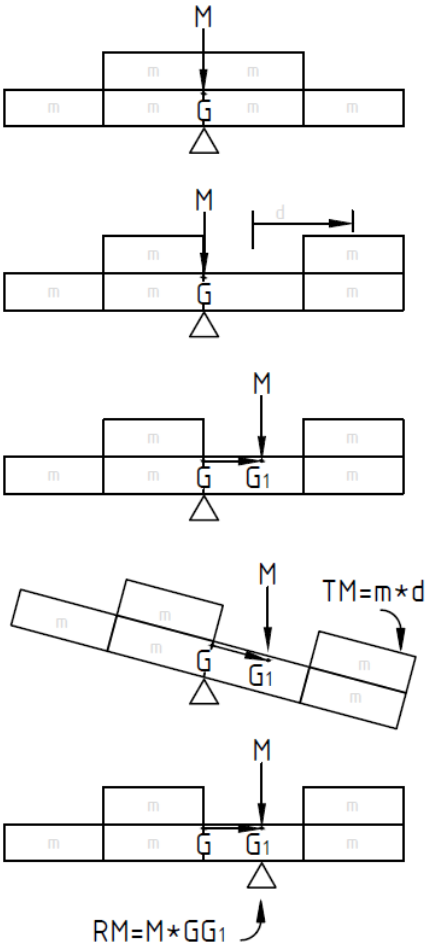


Figure 6.3 – Effect of Shifting a Mass

Now shift one mass to the right as shown. The C of G of the body will immediately shift to G_1.

With the bar still supported at G, a TM will come into action to rotate the bar clockwise.

To restore the body back to the original horizontal condition, an equal but opposing RM will have to be applied at G_1.

Now: $$RM = TM$$

$$M * GG_1 = m * d$$

Therefore the shift in C of G:

$$GG_1 = \frac{m * d}{M}$$

Summary

When a mass is *shifted within* a body the C of G of the body *shifts parallel* to the C of G of the *shifted mass* by an amount equal to the product of the mass shifted and the distance shifted, divided by the total mass.

6.1.4 Effect of Suspending a Mass

In the left diagram of Figure 6.4 a mass (m) is attached by rope to the lifting gear and ready to be hoisted. In this slack condition there is no tension in the rope and the mass may be considered as not forming part of the gear.

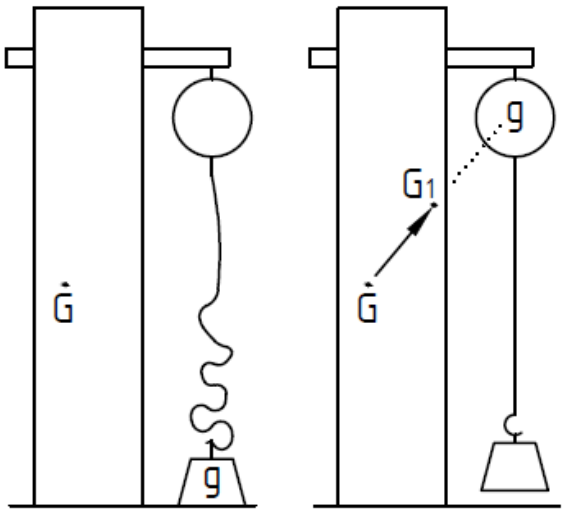


Figure 6.4 – Effect of Suspending a Mass

Immediately the mass is lift off the ground as in the right diagram the rope becomes tensioned and pressure is put on the hook at the top of the gear.

Summary

When a mass is *suspended* the C of G of the mass *shifts* to the point of suspension, and no matter how far away the mass is from the point of suspension, the effect remains the same.

6.2 POSITION OF CENTRE OF GRAVITY OF A SHIP

The centre of gravity of a ship, being the centroid of a three-dimensional body is described by three coordinates. These are:

6.2.1 The **Vertical Centre of Gravity** (VCG_K)

This position is measured with respect to the keel. It is usually written as KG.

6.2.2 The **Longitudinal Centre of Gravity** (LCG_{MS})

It is measured with respect to midship. Its fore or aft direction hence has to be specified in every particular case.

6.2.3 The **Transverse Centre of Gravity** (TCG_{CL})

Measured with respect to the centreline, the TCG_{CL} will usually have a zero value since ships are normally constructed with symmetrical hull forms and balanced distribution of machinery and equipment port–starboard.

6.3 LIGHTSHIP CENTRE OF GRAVITY

The Longitudinal position in the lightship condition (LCG_{MSL}) is at the same distance from midship as LCB_L as shown in Figure 6.5(a) and need not further be calculated.

In respect of the vertical position KG_L, Figure 6.5(b), this is determined by conducting an experiment on a ship when construction is completed, and unless a ship is modified, the position remains constant throughout its service life. The method of conducting the experiment will be discussed later in Chapter 10.

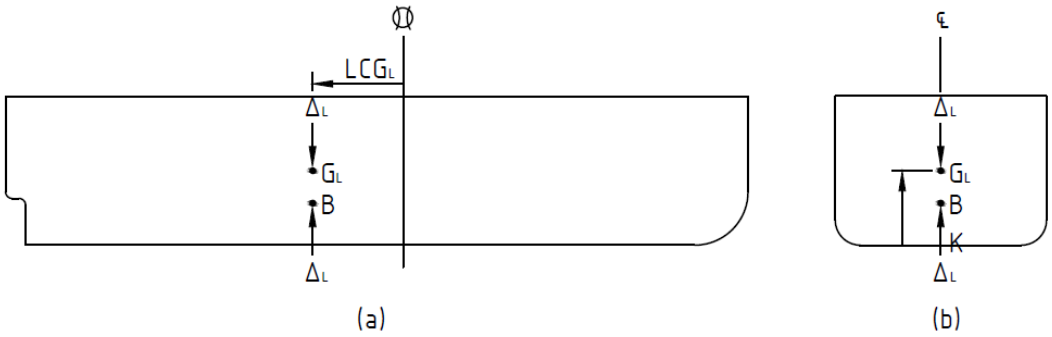


Figure 6.5 – Lightship Centre of Gravity

The transverse position usually coincides with the centreline and will therefore have a value of zero.

6.4 LOADED CENTRE OF GRAVITY

In any condition of loading over and above the lightship, the three positions of the centre of gravity are determined by taking moments as follows:

6.4.1 Determination considering added and removed masses together first followed by computation of shifted masses

6.4.1.1 Determination of KG

Consider a vessel initially empty with lightship displacement Δ_L to have its C of G at G_L and hence a lightship C of G (KG_L) as in Figure 6.5(b).

Now add masses m_1, m_2, m_4 and m_6 at heights z_1, z_2, z_4 and z_6 respectively, and discharge masses m_3 and m_5 from heights z_3 and z_5 respectively.

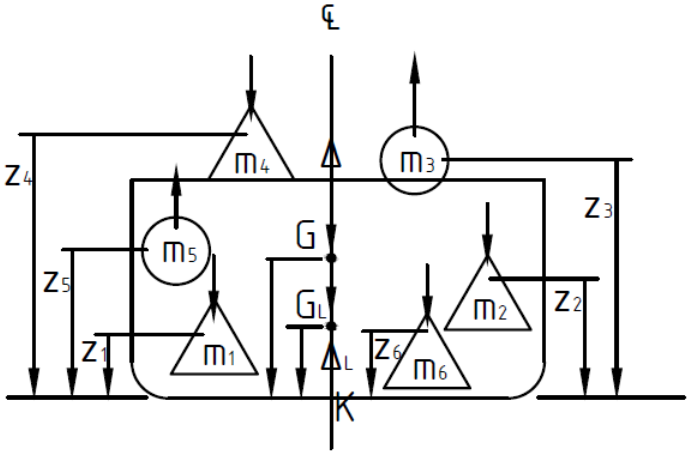


Figure 6.6 – Loaded KG

Its new total mass will be: $\Delta = \Delta_L + m_1 + m_2 + m_4 + m_6 - m_3 - m_5$

With a new C of G (KG). By moments of masses about the keel:

$$\text{Final moment} = \text{Sum of moments of individual masses}$$

$$\Delta KG = \Delta_L KG_L + m_1 z_1 + m_2 z_2 + m_4 z_4 + m_6 z_6 - m_3 z_3 - m_6 z_5$$

$$KG = \frac{\Delta_L KG_L + m_1 z_1 + m_2 z_2 + m_4 z_4 + m_6 z_6 - m_3 z_3 - m_6 z_5}{\Delta}$$

For ease of computation and to reduce making mistakes the above calculations may be presented vertically as follows:

Masses	Kg	Moments
Δ_L	KG_L	$\Delta_L KG_L$
m_1	z_1	$m_1 z_1$
m_2	z_2	$m_2 z_2$
$-m_3$	z_3	$-m_3 z_3$
m_4	z_4	$m_4 z_4$
$-m_5$	z_5	$-m_5 z_5$
m_6	z_6	$m_6 z_6$
$\Delta=\sum$(Masses)		$\sum$(Moments)

Table 6.1 – Calculation for KG

Therefore $$KG = \frac{\sum(\text{Moments})}{\sum(\text{Masses})}$$

6.4.1.2 Determination of LCG_{MS}

If now moments are taken with reference to midship (MS), then by the vertical computation method, and considering aft moments to be positive and forward moments to be negative:

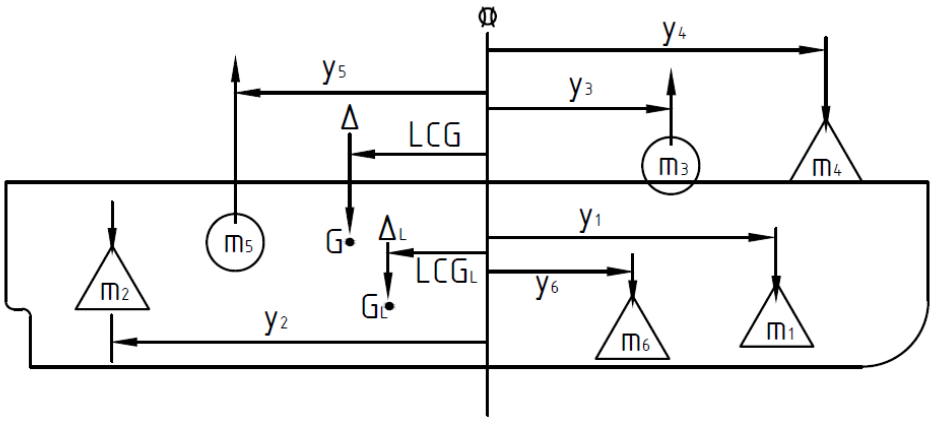


Figure 6.7 – Loaded LCG

Masses	Lcg_{MS}	Moments	
		Aft (+)	Fwd (-)
Δ_L	$LCG_{LA}A$	$\Delta_L LCG_L$	
m_1	y_1F		m_1y_1
m_2	y_2A	m_2y_2	
$-m_3$	y_3F		$-m_3y_3$
m_4	y_4F		m_4y_4
$-m_5$	y_5A	$-m_5y_5$	
m_6	y_6A		m_6y_6
$\Delta=\sum$(Masses)		$\sum$(Moments A)	$\sum$(Moments F)

Table 6.2 – Calculation for LCG (Method A_

$$\mathrm{LCG_{MS}} = \frac{\sum(\text{Moments A}) - \sum(\text{Moments F})}{\sum(\text{masses})}$$

Masses	Lcg_{MS}	Moments
		Aft (+); Fwd (-)
Δ_L	$LCG_{LA}A$	$\Delta_L LCG_L$
m_1	y_1F	$-m_1y_1$
m_2	y_2A	m_2y_2
$-m_3$	y_3F	m_3y_3
m_4	y_4F	$-m_4y_4$
$-m_5$	y_5A	$-m_5y_5$
m_6	y_6A	$-m_6y_6$
$\Delta=\sum$(Masses)		$\sum$(Moments)

Table 6.3 – Calculation for LCG (Method B)_

$$\mathrm{LCG_{MS}} = \frac{\sum(\text{Moments})}{\sum(\text{masses})}$$

6.4.1.3 Determination of TCG_{CL}

If now moments are taken with reference to midship (MS), then by the vertical computation method, and considering aft moments to be positive and forward moments to be negative:

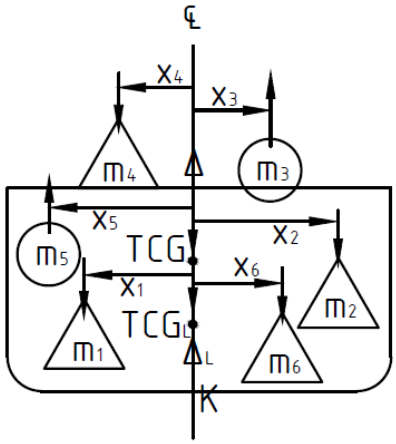


Figure 6.8 – Loaded TCG

Method A

Masses	Vcg_{CL}	Moments	
		Port (+)	Stbd (-)
Δ_L	VCG_{CL}	0	0
m_1	y_1P	m_1y_1	
m_2	y_2S		m_2y_2
$-m_3$	y_3P	$-m_3y_3$	
m_4	y_4S		m_4y_4
$-m_5$	y_5P	$-m_5y_5$	
m_6	y_6S		m_6y_6
$\Delta=\sum$(Masses)		$\sum(\text{Moments P})$	$\sum(\text{Moments S})$

Table 6.4 – Calculation for TCG (Method A)

$$\text{VCG}_{\text{CL}} = \frac{\sum(\text{Moments P}) - \sum(\text{Moments S})}{\sum(\text{masses})}$$

Method B

Masses	Vcg_{MS}	Moments
		Port (+); Stbd (-)
Δ_L	0	0
m_1	y_1P	m_1y_1
m_2	y_2S	$-m_2y_2$
$-m_3$	y_3P	$-m_3y_3$
m_4	y_4S	$-m_4y_4$
$-m_5$	y_5P	$-m_5y_5$
m_6	y_6S	$-m_6y_6$
$\Delta=\sum$(Masses)		$\sum(\text{Moments})$

Table 6.5 – Calculation for TCG (Method B)

$$\text{LCG}_{\text{MS}} = \frac{\sum(\text{Moments})}{\sum(\text{masses})}$$

6.4.1.4 Determination of KG, LCG_{MS} and TCG_{CL} when masses are shifted
When an intermediate position of the C of G for added and removed masses has been obtained as above, the results so obtained are then used as basis in a second exercise to calculate the effect of any shifted mass (m_S) to complete computation.

7–TRANSVERSE STATICAL STABILITY (INITIAL)

7.1 EQUILIBRIUM

When a vessel floats it is said to be in equilibrium vertically under the influence of its displacement (Δ) acting downwards through the Centre of Gravity (G), and force of buoyancy of the same value (Δ) acting upwards in opposition through the Centre of Buoyancy (B).

A vessel in equilibrium with the forces acting along the centerline is said to be *UPRIGHT* as shown in Figure 7.1(a). If however a vessel is in equilibrium inclined then it is said to be *LISTED* as shown in Figure 7.1(b).

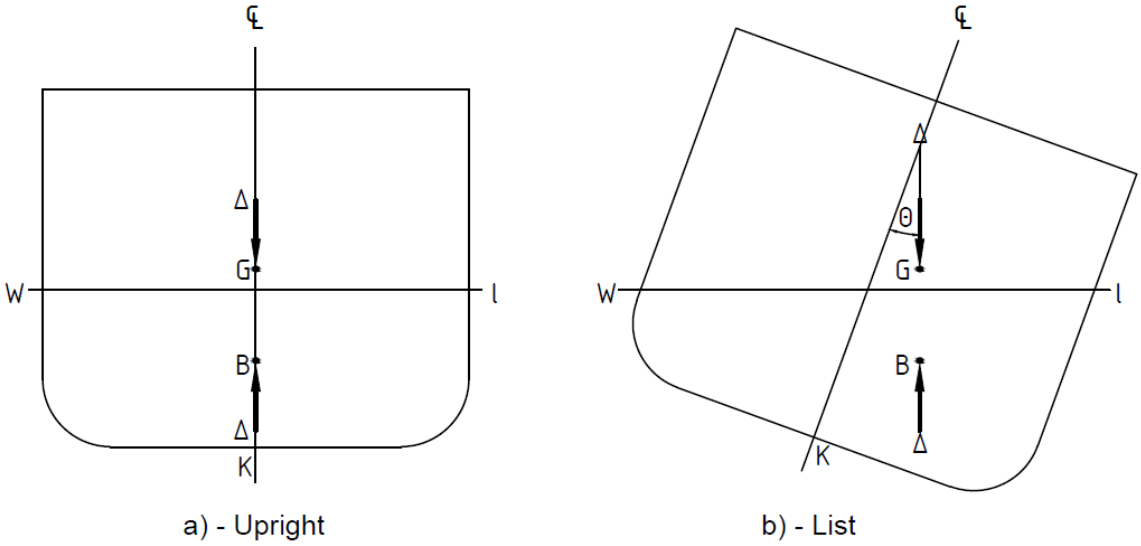


Figure 7.1 – Equilibrium Condition

7.2 HEEL AND HEELING MOMENT

When a wind force acts transversely on the above water surface of a hull the tendency is to push it sideways off its track. This is however resisted by the underwater lateral resistance (R), thereby generating a moment instead which causes the vessel to incline to an angle (Θ).

Again due to wave action a vessel may be forced to incline in a similar manner as that caused by wind forces.

Such transversely induced inclination is referred to as *HEELING* as shown in Figure 7.2(b). The moment of force causing heel is called the *HEELING MOMENT* (HM), while the angle of inclination Θ is referred to as the *ANGLE OF HEEL*.

By definition, the ***point of intersection*** of the lines of action of the force of buoyancy acting through B when vessel is upright and through B_1 when vessel is heeled is called the *TRANSVERSE METACENTRE*. In general the position M_T is the point of intersection of the lines of action of the force of buoyancy for any two angles of heel, displacement remaining constant.

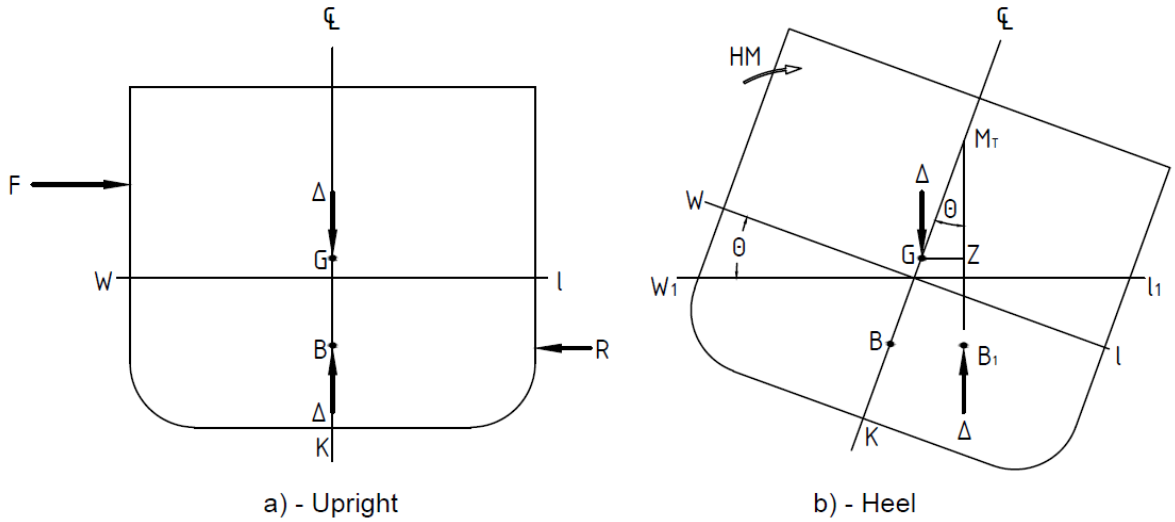


Figure 7.2 – Dynamic Condition

7.3 RIGHTING MOMENT AND TRANSVERSE STABILITY

7.3.1 Stable Equilibrium

As a vessel begins to heel the actions of the forces of buoyancy and displacement will form a couple to oppose the HM. This couple is referred to as *RIGHTING MOMENT* (RM). It increases with heel until it eventually neutralizes the effect of the HM, at which stage the vessel remains in dynamic equilibrium at the angle reached. If at this stage the HM should be removed the RM takes over to bring the vessel back to equilibrium upright or listed as the case may be.

This ability of a ship to right itself when heeled defines the concept of *TRANSVERSE STATICAL STABILITY*, and the particular situation as illustrated, in which the couple generated is able to oppose the RM is referred to as *STABLE EQUILIBRIUM*. The couple arm GZ is called the *RIGHTING LEVER*, and the distance GM_T the *METACENTRIC HEIGHT*. GZ and GM_T are considered to be positive under this situation.

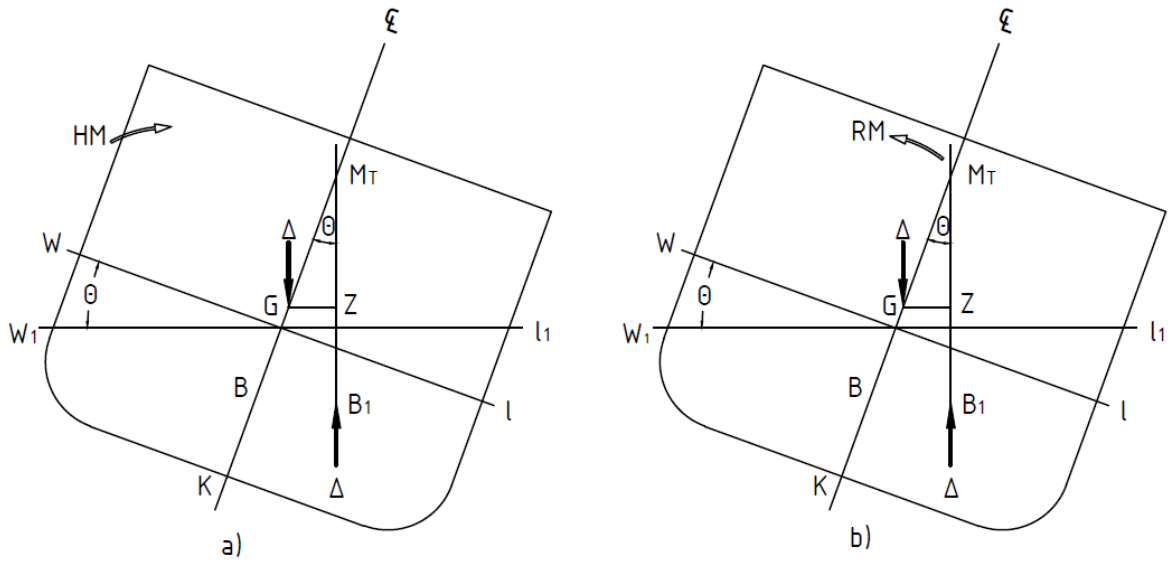


Figure 7.3 - Stable Equilibrium

GM_T is used as the measure of stability and is related to GZ by:

$$GZ = GM_T sin\Theta$$

The RM or Moment of Statical Stability, expressed in mass terms is:

$$\Delta GZ = \Delta GM_T sin\Theta$$

When expressed in force terms:

$$\Delta gGZ = \Delta gGM_T sin\Theta$$

Finally from Figure 7.3(b) it can be deduced that:

$$KM_T = KB + BM_T = KG + GM_T$$

7.3.2 Unstable Equilibrium

In the event that loading of a ship results in the position of C of G being higher than the position of M_T, then as in Figure 7.4, the RM will end up acting in concert with the HM, so that when even the external force is removed the vessel will continue to heel and may capsize. Such a situation is referred to as *UNSTABLE EQUILIBRIUM*, and GZ and GM_T are considered to be negative.

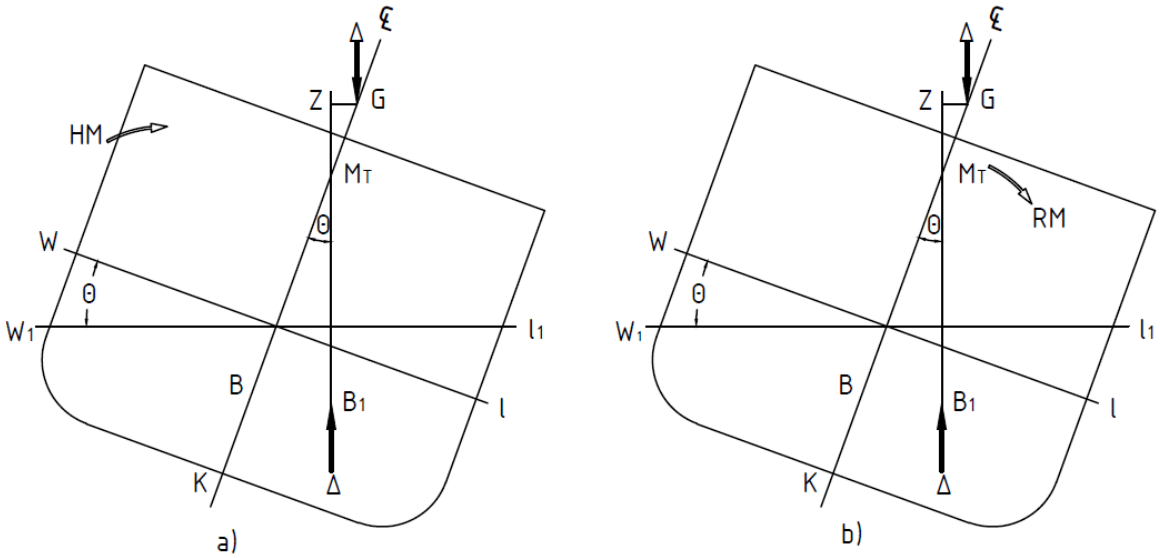


Figure 7.4 – Unstable Equilibrium

From Figure 7.4(b):

$$KM_T = KB + BM_T = KG - GM_T$$

7.3.3 Neutral Equilibrium

In situations where loading makes the position of the C of G to coincide with that of the transverse metacentre, then after the HM has been removed the ship will have no RM to bring it back and will remain at the heeled angle until another force again acts. The ship under this condition is said to be in *NEUTRAL* EQUILIBRIUM ant the angle at which this occurs is referred to as the *ANGLE OF LOLL*.

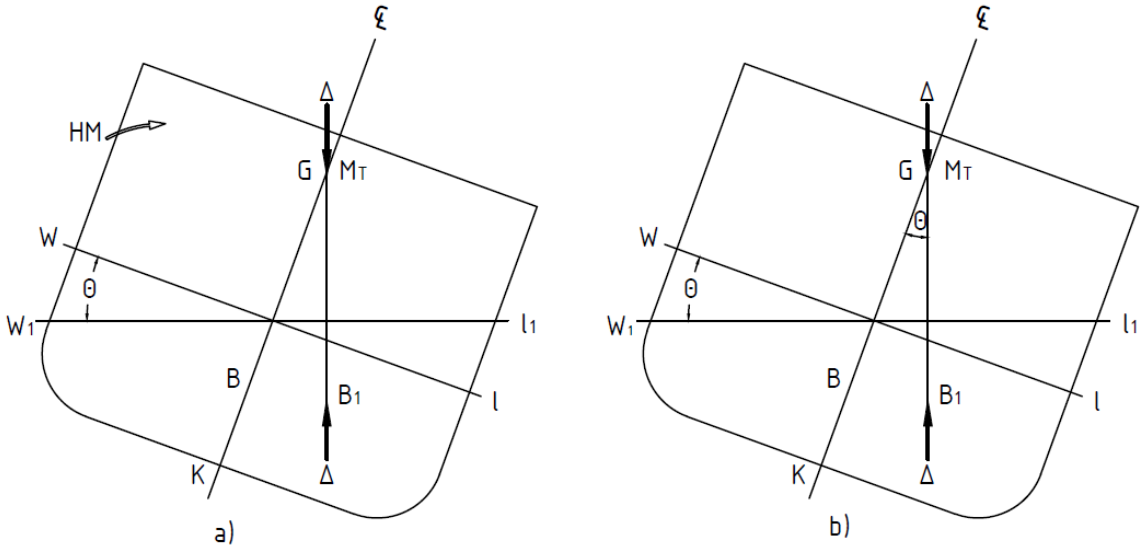


Figure 7.5 – Neutral Equilibrium

From Figure 7.5(b) the metacentric height is zero and:

$$KM_T = KB + BM_T = KG$$

7.4 CONCEPT OF INITIAL STABILITY

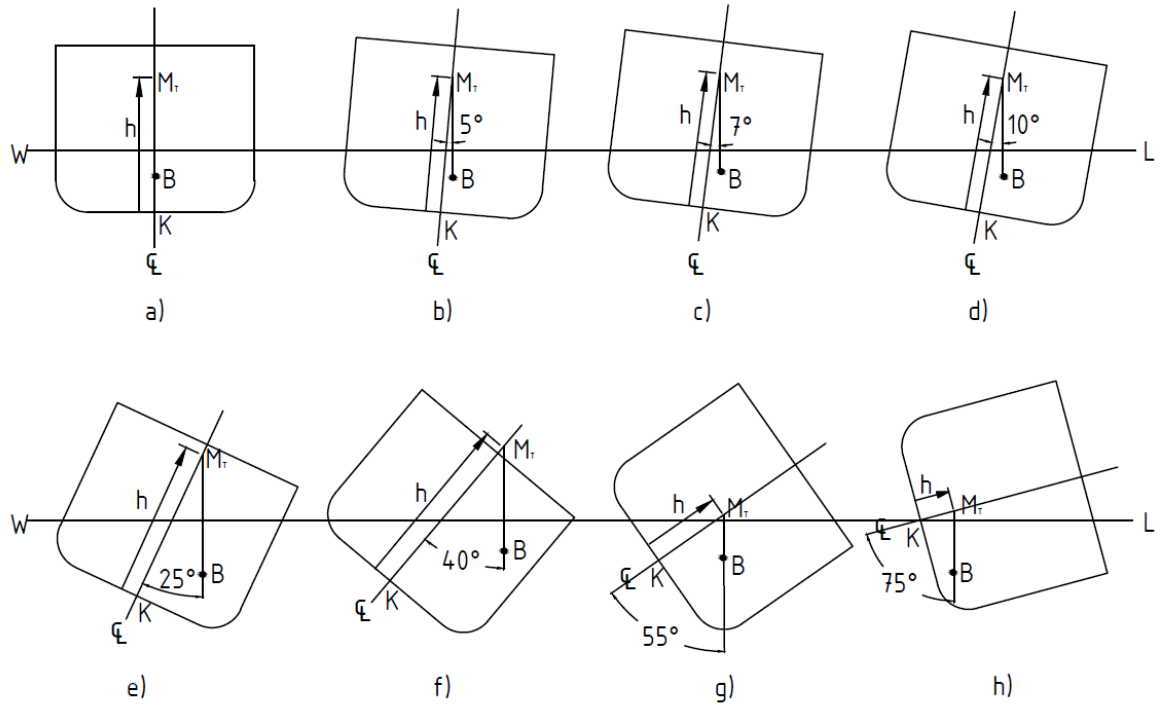


Figure 7.7 – Effect of Angle of Heel on Metacentric Height

For a fixed displacement when KM_T is calculated for different angles of heel, the value increases negligibly from zero up to about 10º. Beyond 10º a higher rate of increase is registered until at the point when the deck edge becomes awash or the bilge becomes exposed, it decreases sharply to zero, by which time the vessel is almost capsizing as shown in Figures 7.7 and 7.8.

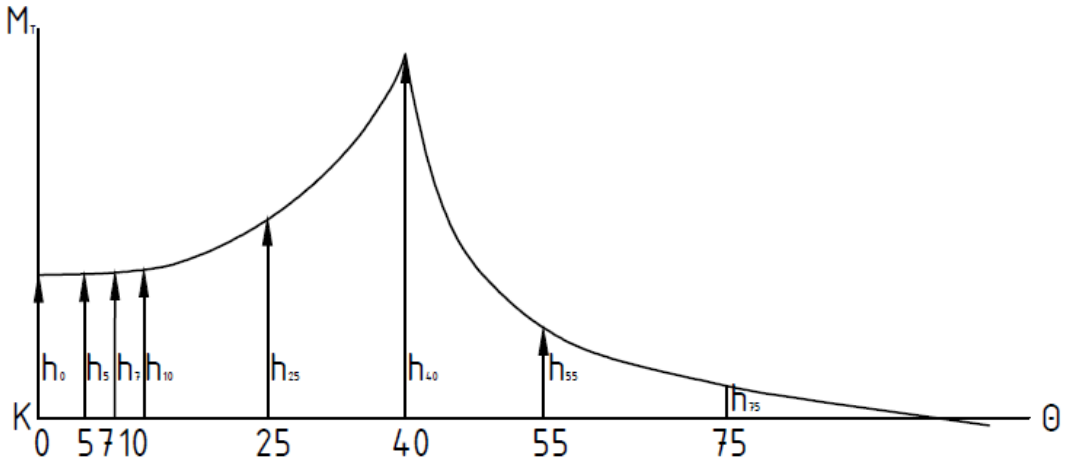


Figure 7.8 – Locus of M_T as Θ Increases

KM_T may therefore be considered to be constant for the initial stages of heel. The position of M_T can therefore be comfortably used as reference for measuring GM_T at small angles of heel up to 10º. This is what defines the concept of Initial Stability or Stability at Small Angles of Heel.

7.5 DETERMINATION OF THE VALUE OF KM_T

When a ship is heeled to a small angle less than TEN degrees from upright, the area carved out by the inclined waterline W_1L_1 will not be symmetrical about the centreline. Its centroid will hence lie on a line (i.e. neutral axis (NA)) which will be parallel to the centreline in the upright condition thereby causing the two waterlines WL and W_1L_1 to intersect at S as shown in Figure 7.9 (a).

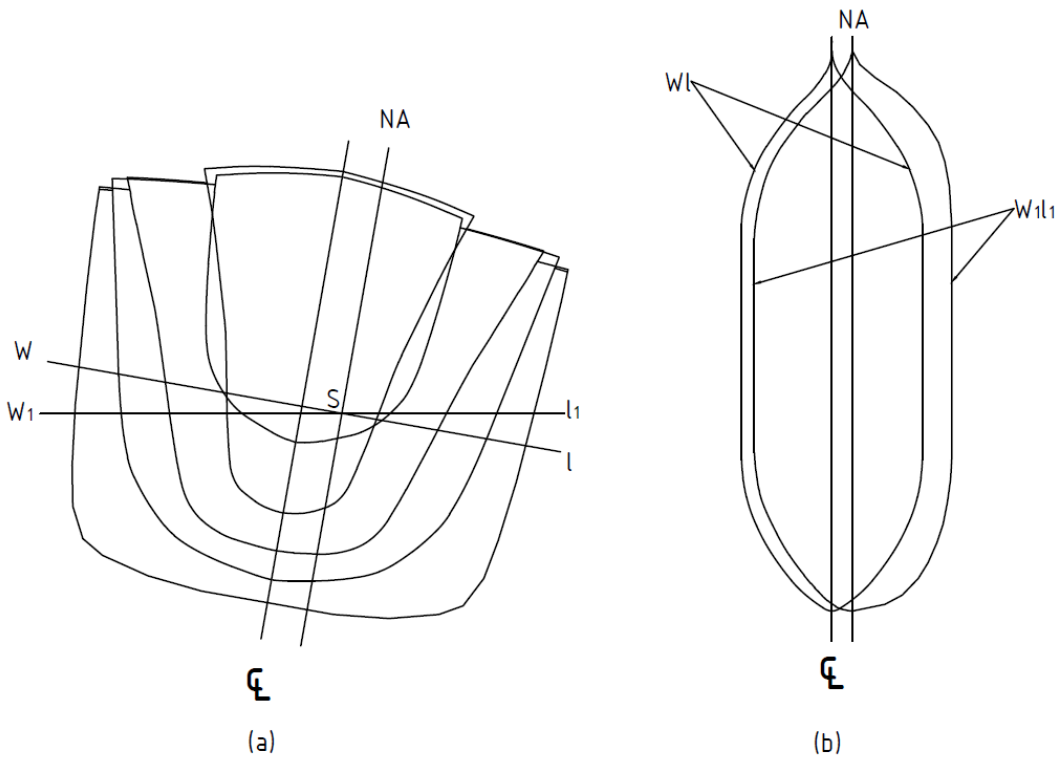


Figure 7.9

If however the ship is now assumed to be WALL-SIDED in the region of the waterline then for all practical purposes the two neutral axes will coincide making the waterlines to INTERSECT at O. On the basis of these assumptions the metacentric height (GM_T) is used as the measure of Initial Stability.

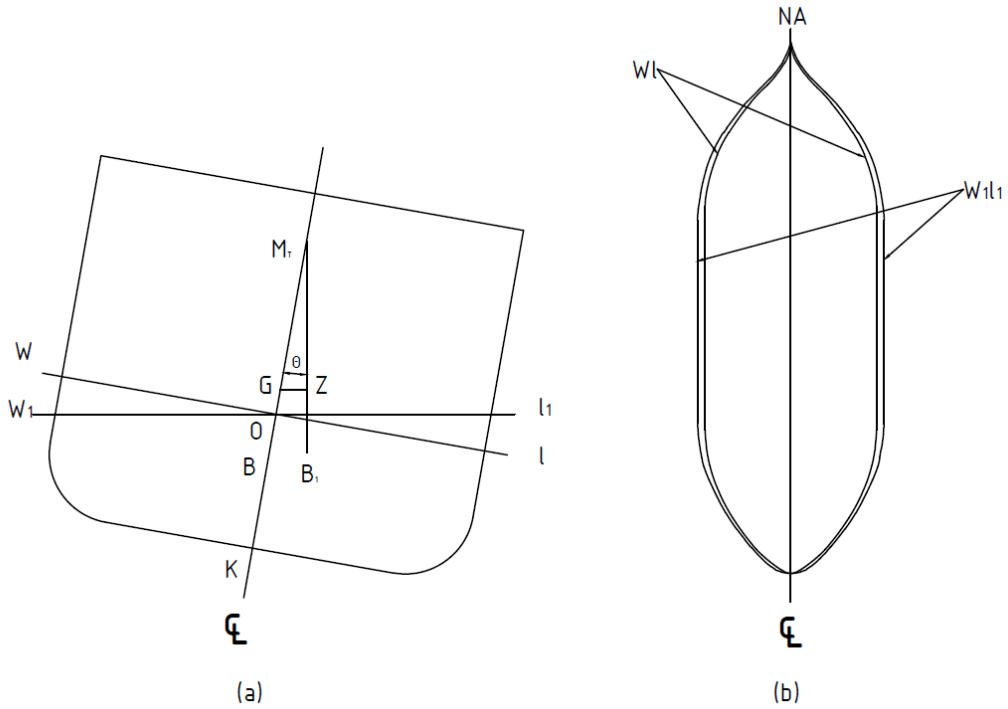


Figure 7.10

Considering a small length dx of the hull when heeled to a small angle Θ shown in Figure 7.11(a), it can be said that a volume WW_1Odx has emerged from the water and has been replaced an immersed volume LL_1Odx of similar dimensions. A small volume with centre at b has therefore shifted to b_1.

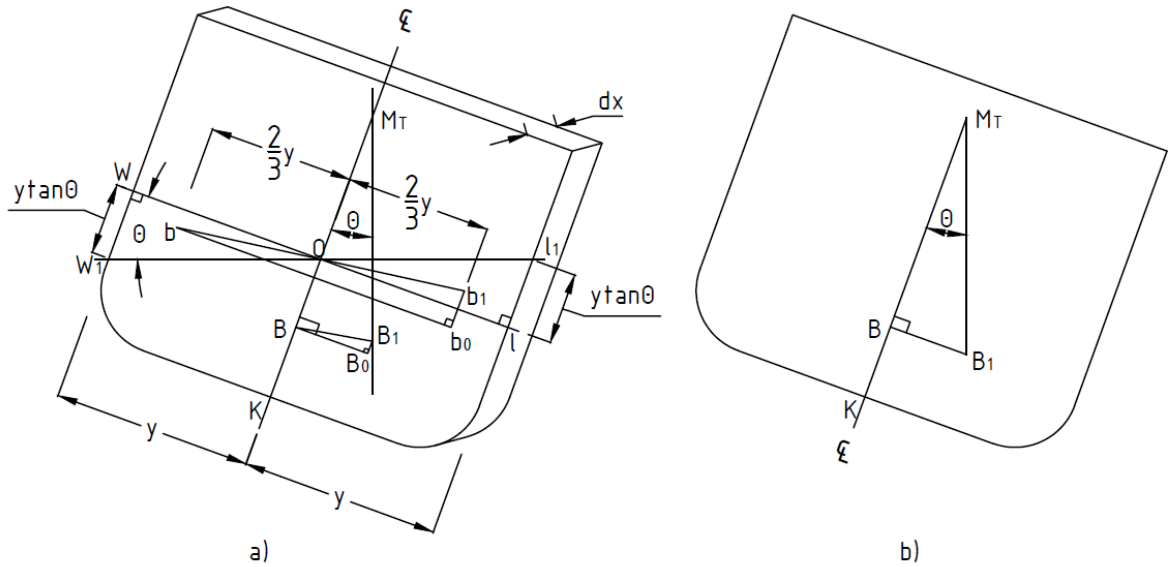


Figure 7.11 – Metacentric Height

The cross-sections of these volumes are right-angled triangles hence if y is the half-breadth of the vessel on the waterline WL, then for the whole length of ship L,

Transverse shift in centre of buoyancy of ship:

$$BB_0 = \frac{\text{Volume shifted} \times \text{Distance shifted}}{\text{Ship's Volume of Displacement}} = \frac{\int_0^L WW_1 dx \times bb_0}{\nabla}$$

$$BB_0 = \frac{\int_0^L \frac{1}{2} y . y\tan\Theta . dx . 2 . \frac{2}{3} y}{\nabla} = \frac{\int_0^L \frac{2}{3} y^3 . dx . \tan\Theta}{\nabla}$$

But

$$\int_0^L \frac{2}{3} y^3 . dx = I_{CL}$$

Therefore

$$BB_0 = \frac{I_{CL}\tan\Theta}{\nabla}$$

Vertical shift in centre of buoyancy of ship:

$$B_0B_1 = \frac{\text{Volume shifted} \times \text{Distance shifted}}{\text{Ship's Volume of Displacement}} = \frac{\int_0^L LL_1 dx \times b_0b_1}{\nabla}$$

$$B_0B_1 = \frac{\int_0^L \frac{1}{2} y . y\tan\Theta . dx . 2 . \frac{1}{3} y\tan\Theta}{\nabla} = \frac{\frac{1}{2}\int_0^L \frac{2}{3} y^3 . dx . \tan^2\Theta}{\nabla}$$

Therefore

$$B_0B_1 = \frac{\frac{1}{2} I_{CL}\tan^2\Theta}{\nabla}$$

Now for all practical purposes ½$\tan^2\Theta$ approximates to zero, therefore B_0B_1 equals zero and B_0 will merge with B_1 to reduce BB_0B_1 to BB_1 as shown in Figure 7.11(b).

From triangle BB_1M_T

$$BM_T\tan\Theta = BB_1 = BB_0$$

$$BM_T\tan\Theta = \frac{I_{CL}\tan\Theta}{\nabla}$$

Therefore

$$BM_T = \frac{I_{CL}}{\nabla}$$

From which

$$KM_T = KB + BM_T$$